47. Serial Sound Interface Enhanced (SSIE)

47.1 Overview

The Serial Sound Interface Enhanced (SSIE) can transmit and receive audio data to and from various devices that support any of audio data formats, such as I^2S, monaural, and TDM.

Table 47.1 lists the SSIE specifications, Figure 47.3, Figure 47.4 and Figure 47.5 show a block diagram of SSIE, and Table 47.4 lists the I/O pins.

Table 47.1 SSIE specifications

Item		Description
Number of channels		Two channels, SSIE0 and SSIE1
Communication mode		• Master/slave • Transmission/reception (SSIE0:full duplex communication, SSIE1:half-duplex communication)
Communication format		• I^2S format • Monaural format • TDM format
Serial data		• MSB first • Data can be left-justified or right-justified. • Data delay (1 clock cycle) or no delay selectable for the period from SSILRCKn/SSIFSn (n = 0, 1) to SSITXD0/SSIRXD0/SSIDATA1 • System word length: 8, 16, 24, 32, 48, 64, 128, or 256 bits • Data word length: 8, 16, 18, 20, 22, 24, or 32 bits • Padding polarity: Low or high
Bit clock (SSIBCKn (n = 0, 1))	In master mode	• Two clock sources available (AUDIO_CLK/GPT output (GTIOC2A)) • Clock source division ratio: 1/1, 1/2, 1/4, 1/6, 1/8, 1/12, 1/16, 1/24, 1/32, 1/48, 1/64, 1/96, and 1/128. • Supply/stop is selectable while communication is halted.
	In master/slave mode	• Polarity (rising edge or falling edge) selectable
LR clock/frame synchronization (SSILRCKn/SSIFSn (n = 0, 1))	In master mode	• Polarity (low level or high level) selectable • Supply/stop is selectable while communication is halted.
Transmit data (SSITXD0/SSIDATA1) and receive data (SSIRXD0/SSIDATA1)	Transmission	• Muting method (transmission of transmit FIFO data or transmission of data fixed to 0) selectable
FIFO	Capacity	• Transmit FIFO/receive FIFO: 4 bytes × 32 stages
	Data alignment	• Data alignment method (left-justification or right-justification) selectable for the data transfer between FIFO and shift register
Interrupt	Interrupt output	• Communication error/idle mode • Receive data full • Transmit data empty
Low power consumption function		• Whether to supply the audio clock selectable in master mode
Module stop function		Module stop state can be set for each channel to reduce power consumption.
TrustZone Filter		Security and Privilege attribution can be set for each channel.

The following table lists and defines the terms used for the communication formats SSIE can use:

Table 47.2 Definition of terms (1 of 2)

Term	Definition
Start trigger	First edge of the signal on the SSILRCKn/SSIFSn (n = 0, 1) pin when the signal is set to the value specified in LRCKP to enable communication

Table 47.2 Definition of terms (2 of 2)

Term	Definition
Frame boundary	Point where SSIE starts transferring the first data of a frame or the point where SSIE ends transferring the last data of the frame
Frame word number	Number of sound channels per frame
System word length	Number of bits per channel
Data word length	Number of significant bits per channel
Control bits for communication formats	• SSICR register: FRM, DWL, SWL, LRCKP, SPDP, SDTA, PDTA, and DEL bits • SSIFCR register: BSW bit • SSIOFR register: OMOD bit • SSISCR register: TDES[4:0] and RDFS[4:0] bits

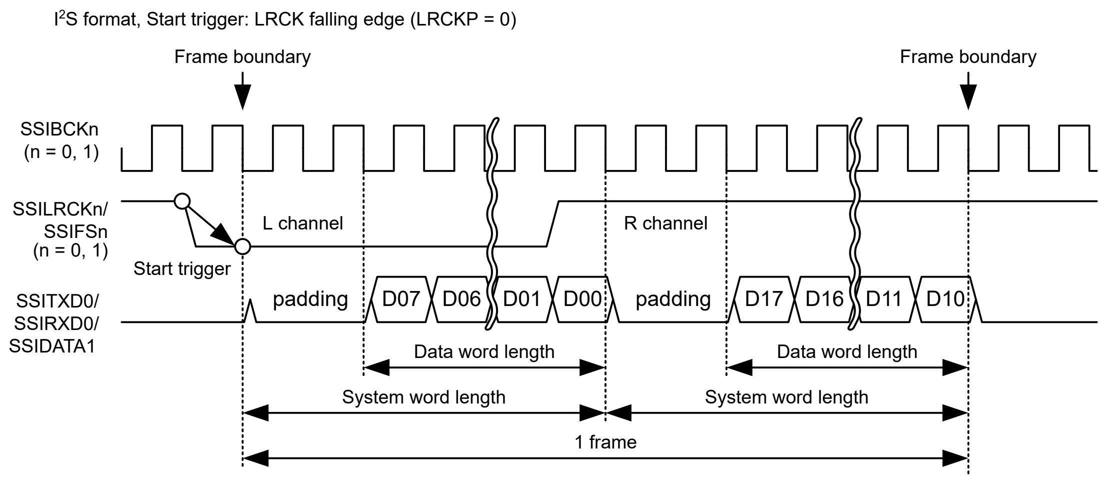
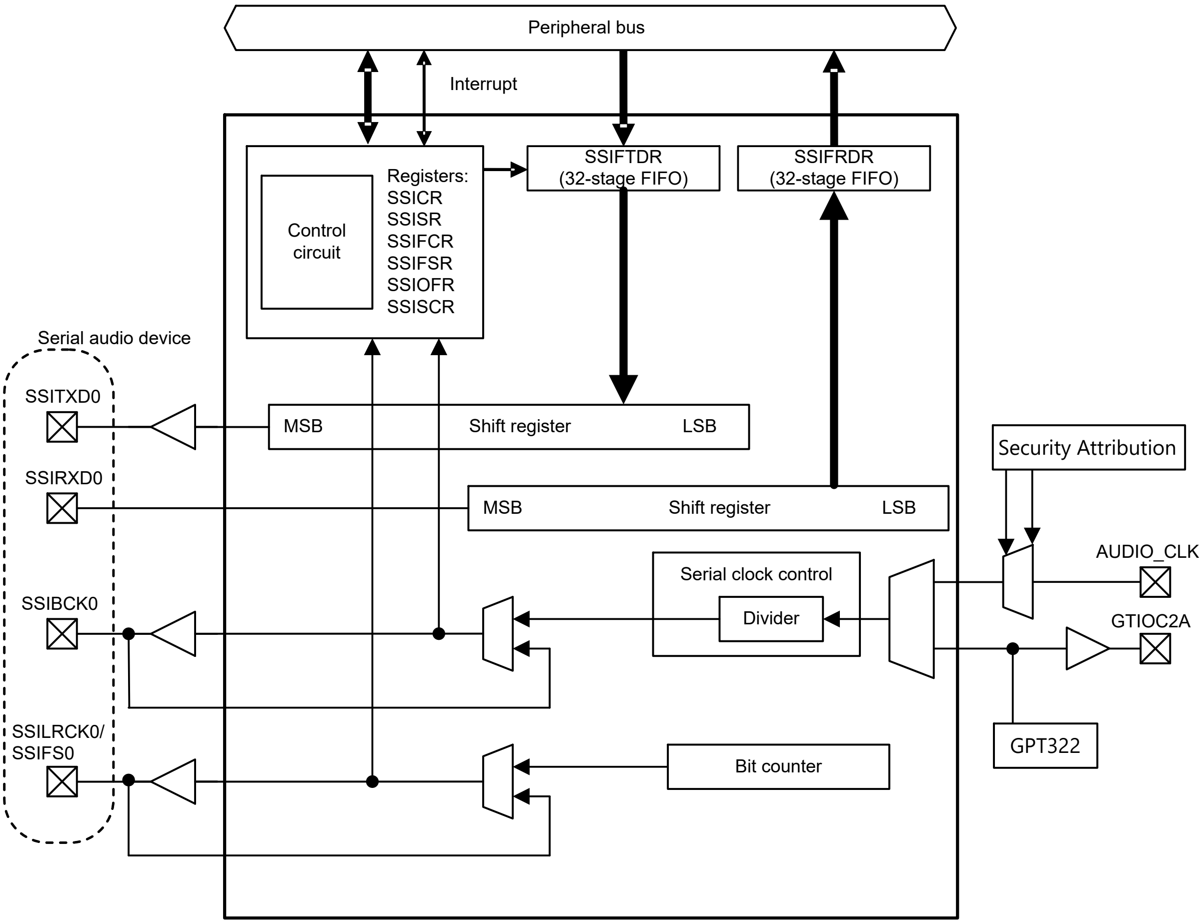


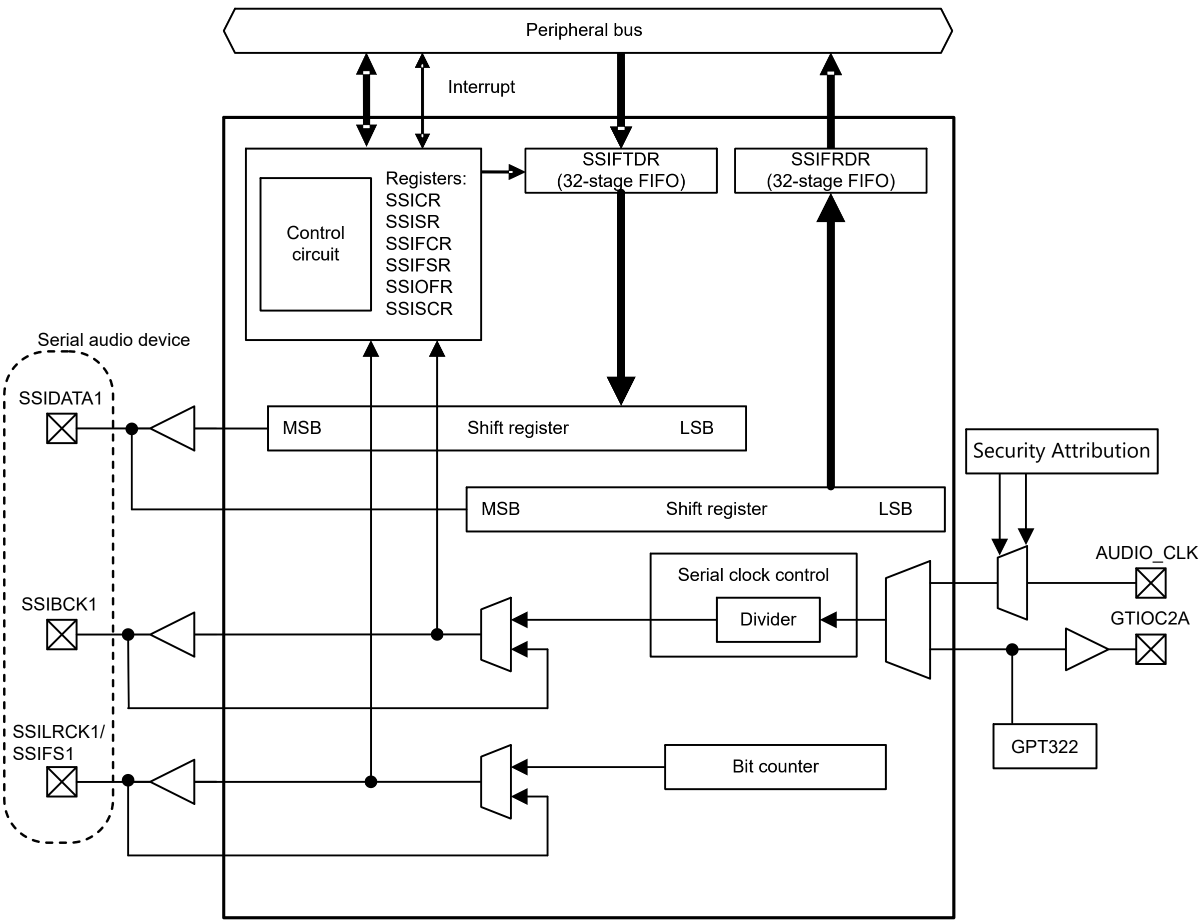
Figure 47.1 Definition of communication format

Figure 47.2 and Figure 47.3 shows a block diagram of SSIE.



[Legend]
SSICR: Control register
SSISR: Status register
SSIFCR: FIFO control register
SSIFSR: FIFO status register
SSIOFR: Audio format register
SSIFTDR: Transmit FIFO data register
SSIFRDR: Receive FIFO data register
SSISCR: Status control register

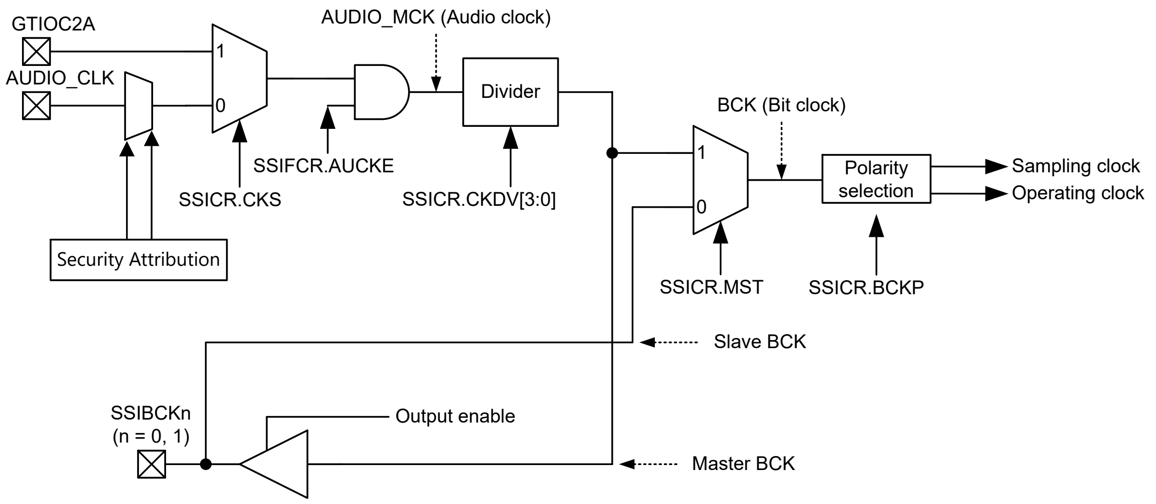
Figure 47.2 SSIE block diagram (SSIE0)



[Legend]
SSICR: Control Register
SSISR: Status Register
SSIFCR: FIFO Control Register
SSIFSR: FIFO Status Register
SSIOFR: Audio Format Register
SSIFTDR: Transmit FIFO Data Register
SSIFRDR: Receive FIFO Data Register
SSISCR: Status Control Register

Figure 47.3 SSIE block diagram (SSIE1)

Figure 47.4 shows the clock configuration of SSIE.



[Legend]
SSICR: Control Register
SSISR: Status Register
SSIFCR: FIFO Control Register
SSIFSR: FIFO Status Register
SSIOFR: Audio Format Register
SSIFTDR: Transmit FIFO Data Register
SSIFRDR: Receive FIFO Data Register
SSISCR: Status Control Register

Figure 47.4 SSIE clock configuration

The AUDIO_CLK can be input according to the security attribution of Port and SSIE and SSICR.CKS bit setting as shown in the Table 47.3.

Table 47.3 The input condition of AUDIO_CLK by the security setting

<table>
<tr><th>PORT Security Attribution (PmSAR (m = 0 - 9, A, B))</th><th>SSIE0 Security Attribution (PSARm (B - E))</th><th>SSIE0's SSICR.CKS</th><th>SSIE1 Security Attribution (PSARm (B - E))</th><th>SSIE1's SSICR.CKS</th><th>AUDIO_CLK pin input</th></tr>
<tr><td rowspan="2">0 (secure)</td><td>0 (secure)</td><td>0 (AUDIO_CLK)</td><td colspan="2">Don't care</td><td rowspan="6">AUDIO_CLK input enable</td></tr>
<tr><td colspan="2">Don't care</td><td>0 (secure)</td><td>0 (AUDIO_CLK)</td></tr>
<tr><td rowspan="4">1 (non-secure)</td><td rowspan="2">1 (non-secure)</td><td rowspan="2">Don't care</td><td>0 (secure)</td><td>1 (GTIOC2A)</td></tr>
<tr><td>1 (non-secure)</td><td>Don't care</td></tr>
<tr><td>0 (secure)</td><td>1 (GTIOC2A)</td><td rowspan="2">1 (non-secure)</td><td rowspan="2">Don't care</td></tr>
<tr><td>1 (non-secure)</td><td>Don't care</td></tr>
<tr><td colspan="5">other than above</td><td>AUDIO_CLK input disable</td></tr>
</table>

Table 47.4 lists the I/O pins.

Table 47.4 SSIE I/O pins

Channel	Pin name	I/O	Function
Common	SSIBLKn	Input	Serial bit clock pins
	AUDIO_CLK	Input	External clock pin for audio (input oversampling clock)
	SSILRCKn/SSIFSn	Output	LR clock/frame synchronization pins
SSIE0	SSIRXD0	Input	Serial data input pin
	SSITXD0	Output	Serial data output pin
SSIE1	SSIDATA1	I/O	Serial data input/output pin

Note: n = 0, 1

47.2 Register Descriptions

47.2.1 SSICR : Control Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x00

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	CKS	TUIEN	TOIEN	RUIEN	ROIEN	IIEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL	CKDV[3:0]				MUEN	—	TEN	REN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	REN	Reception Enable*2 0: Disables reception 1: Enables reception (starts reception)	R/W
1	TEN	Transmission Enable*2 0: Disables transmission 1: Enables transmission (starts transmission)	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
3	MUEN	Mute Enable 0: Disables muting on the next frame boundary 1: Enables muting on the next frame boundary	R/W
7:4	CKDV[3:0]	Selects Bit Clock Division Ratio*1 0x0: AUDIO_MCK 0x1: AUDIO_MCK/2 0x2: AUDIO_MCK/4 0x3: AUDIO_MCK/8 0x4: AUDIO_MCK/16 0x5: AUDIO_MCK/32 0x6: AUDIO_MCK/64 0x7: AUDIO_MCK/128 0x8: AUDIO_MCK/6 0x9: AUDIO_MCK/12 0xA: AUDIO_MCK/24 0xB: AUDIO_MCK/48 0xC: AUDIO_MCK/96 Others: Setting prohibited	R/W
8	DEL	Selects Serial Data Delay*1 In the monaural format, this bit controls the waveform of SSILRCKn/SSIFSn (n = 0, 1). For details, see section 47.3.2. Monaural Format. 0: Delay of 1 cycle of SSIBCKn (n = 0, 1) between SSILRCKn/SSIFSn (n = 0, 1) and SSITXD0/SSIRXD0/SSIDATA1 1: No delay between SSILRCKn/SSIFSn (n = 0, 1) and SSITXD0/SSIRXD0/SSIDATA1	R/W
9	PDTA	Selects Placement Data Alignment*1 0: Left-justifies placement data (SSIFTDR, SSIFRDR) 1: Right-justifies placement data (SSIFTDR, SSIFRDR)	R/W
10	SDTA	Selects Serial Data Alignment*1 0: Transmits and receives serial data first and then padding bits 1: Transmit and receives padding bits first and then serial data	R/W

<table>
<tr><th>Bit</th><th>Symbol</th><th>Function</th><th>R/W</th></tr>
<tr><td>11</td><td>SPDP</td><td>Selects Serial Padding Polarity[*1]
0: Padding data is at a low level
1: Padding data is at a high level</td><td>R/W</td></tr>
<tr><td>12</td><td>LRCKP</td><td>Selects the Initial Value and Polarity of LR Clock/Frame Synchronization Signal[*1]
0: The initial value is at a high level.
The start trigger for a frame is synchronized with a falling edge of SSILRCKn/SSIFSn (n = 0, 1).
1: The initial value is at a low level.
The start trigger for a frame is synchronized with a rising edge of SSILRCKn/SSIFSn (n = 0, 1).</td><td>R/W</td></tr>
<tr><td>13</td><td>BCKP</td><td>Selects Bit Clock Polarity[*1]
0: SSILRCKn/SSIFSn (n = 0, 1) and SSITXD0/SSIRXD0/SSIDATA1 change at a falling edge (SSILRCKn/SSIFSn (n = 0, 1) and SSIRXD0/SSIDATA1 are sampled at a rising edge of SSIBCKn (n = 0, 1)).
1: SSILRCKn/SSIFSn (n = 0, 1) and SSITXD0/SSIRXD0/SSIDATA1 change at a rising edge (SSILRCKn/SSIFSn (n = 0, 1) and SSIRXD0/SSIDATA1 are sampled at a falling edge of SSIBCKn (n = 0, 1)).</td><td>R/W</td></tr>
<tr><td>14</td><td>MST</td><td>Master Enable[*1]
0: Slave-mode communication
1: Master-mode communication</td><td>R/W</td></tr>
<tr><td>15</td><td>—</td><td>This bit is read as 0. The write value should be 0.</td><td>R/W</td></tr>
<tr><td>18:16</td><td>SWL[2:0]</td><td>Selects System Word Length[*1]
0 0 0: 8 bits
0 0 1: 16 bits
0 1 0: 24 bits
0 1 1: 32 bits
1 0 0: 48 bits
1 0 1: 64 bits
1 1 0: 128 bits
1 1 1: 256 bits</td><td>R/W</td></tr>
<tr><td>21:19</td><td>DWL[2:0]</td><td>Selects Data Word Length[*1]
0 0 0: 8 bits
0 0 1: 16 bits
0 1 0: 18 bits
0 1 1: 20 bits
1 0 0: 22 bits
1 0 1: 24 bits
1 1 0: 32 bits
1 1 1: Setting prohibited</td><td>R/W</td></tr>
<tr><td>23:22</td><td>FRM[1:0]</td><td>Selects Frame Word Number[*1]
<table>
<tr><th></th><th colspan="3">Communication format (SSIOFR.OMOD[1:0])</th></tr>
<tr><td>FRM[1:0]</td><td>I^{2}S (00b)</td><td>Monaural (10b)</td><td>TDM (01b)</td></tr>
<tr><td>00b</td><td>2</td><td>1</td><td>Setting prohibited</td></tr>
<tr><td>01b</td><td rowspan="3">Setting prohibited</td><td rowspan="3">Setting prohibited</td><td>4</td></tr>
<tr><td>10b</td><td>6</td></tr>
<tr><td>11b</td><td>8</td></tr>
</table></td><td>R/W</td></tr>
<tr><td>24</td><td>—</td><td>This bit is read as 0. The write value should be 0.</td><td>R/W</td></tr>
<tr><td>25</td><td>IIEN</td><td>Idle Mode Interrupt Output Enable
0: Disables idle mode interrupt output
1: Enables idle mode interrupt output</td><td>R/W</td></tr>
<tr><td>26</td><td>ROIEN</td><td>Receive Overflow Interrupt Output Enable
0: Disables receive overflow interrupt output
1: Enables receive overflow interrupt output</td><td>R/W</td></tr>
</table>

Bit	Symbol	Function	R/W
27	RUIEN	Receive Underflow Interrupt Output Enable 0: Disables receive underflow interrupt output 1: Enables receive underflow interrupt output	R/W
28	TOIEN	Transmit Overflow Interrupt Output Enable 0: Disables transmit overflow interrupt output 1: Enables transmit overflow interrupt output	R/W
29	TUIEN	Transmit Underflow Interrupt Output Enable 0: Disables transmit underflow interrupt output 1: Enables transmit underflow interrupt output	R/W
30	CKS	Selects an Audio Clock for Master-mode Communication*1 0: Selects the AUDIO_CLK input 1: Selects the GTIOC2A (GPT output)	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Writing to these bits while SSIE is in a communication state (SSISR.IIRQ = 0) is prohibited. If the value of these bits is changed by rewriting, subsequent operation is unpredictable.

Note 2. If the TEN bit or REN bit is rewritten, make sure that the SSISR.IIRQ bit is in the desired status. If the value of the TEN or REN bit is changed by rewriting, subsequent operation is unpredictable. For example, when transmission or reception is enabled, check that SSISR.IIRQ is 0; when transmission or reception is disabled, check that SSISR.IIRQ is 1.

With this register, select an audio clock, control interrupt requests, select data formats, and set an operation mode.

TEN and REN bits (Transmission and Reception Enable)

The TEN and REN bits enable/disable transmission and reception. When 1 is written to one of these bits, the corresponding communication operation starts in synchronization with a start trigger by the SSILRCKn/SSIFSn (n = 0, 1) signal. For details, see section 47.6.2. Transmission to section 47.6.4. Transmission and Reception. When 0 is written to this bit, the current communication operation stops at the next frame boundary. To use SSIE for both transmission and reception, always write 1 to these bits together. When stopping the communication using SSIE, always disable both transmission and reception (write 0 to the TEN and REN bits).

If you want to stop SSIE before a frame boundary is reached, perform a software reset procedure.

MUEN bit (Mute Enable)

The MUEN bit sets/clears the mute function for the data output from the SSITXD0/SSIDATA1 pin. When this bit is set to 1 in the middle of a frame, the SSITXD0/SSIDATA1 output changes to 0 at the next frame boundary. When this bit is set to 0 in the middle of a frame, the SSITXD0/SSIDATA1 output changes to the data of transmit FIFO data register at the next frame boundary. Note that this bit controls data only. Status flags and interrupt signals are normally generated.

Changing the value of this bit must be performed only after setting the communication format to be used.

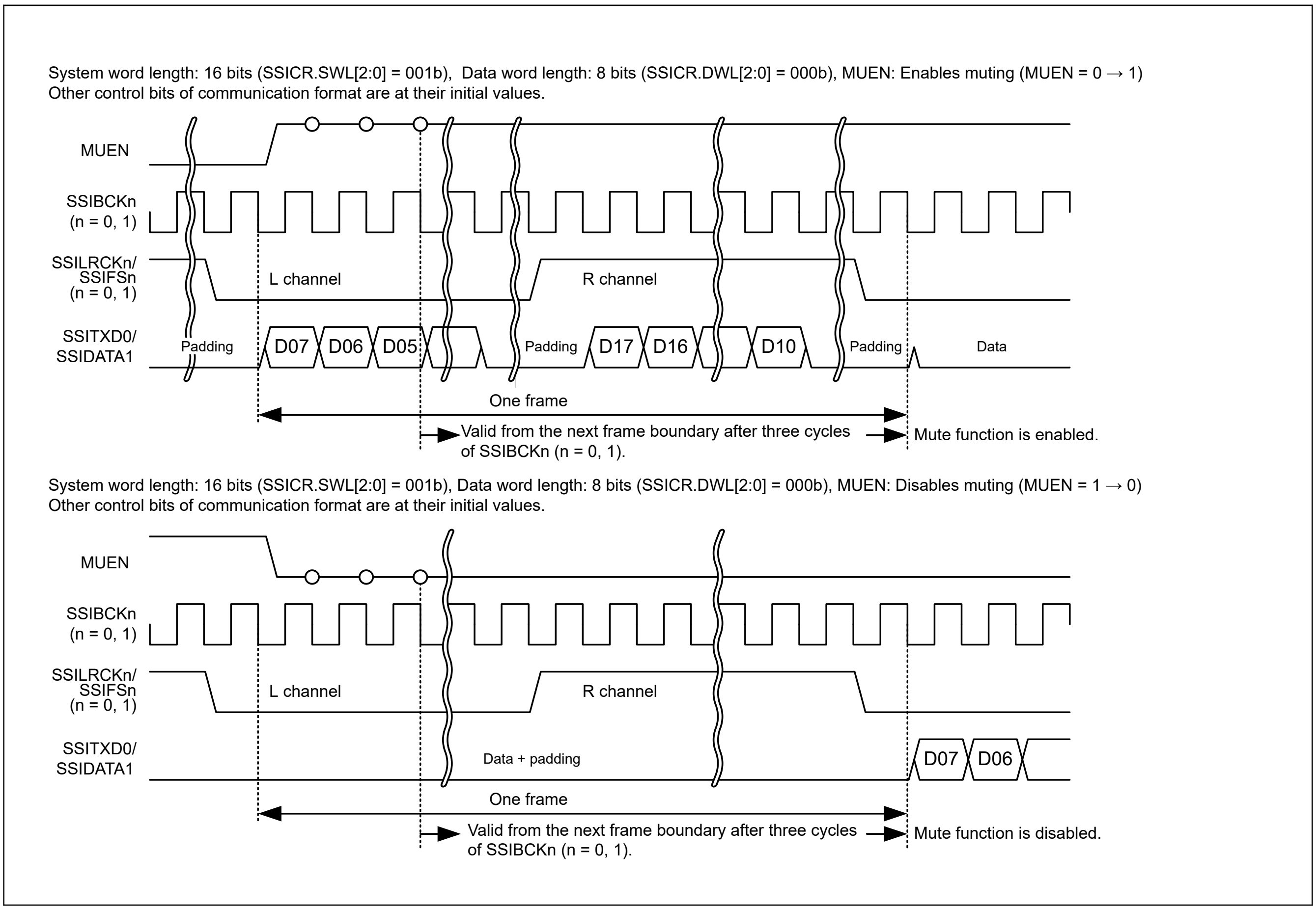


Figure 47.5 Transmit data with the mute function set

CKDV[3:0] bits (Selects Bit Clock Division Ratio)

The CKDV[3:0] bits set the division ratio of the bit clock based on AUDIO_MCK in master-mode communication (MST= 1). In slave-mode communication (MST = 0), setting of these bits are invalid.

Writing to this bit must be performed when the supply of AUDIO_MCK is stopped. For details about the timing, see the detailed description of the AUCKE bit in section 47.2.3. SSIFCR : FIFO Control Register.

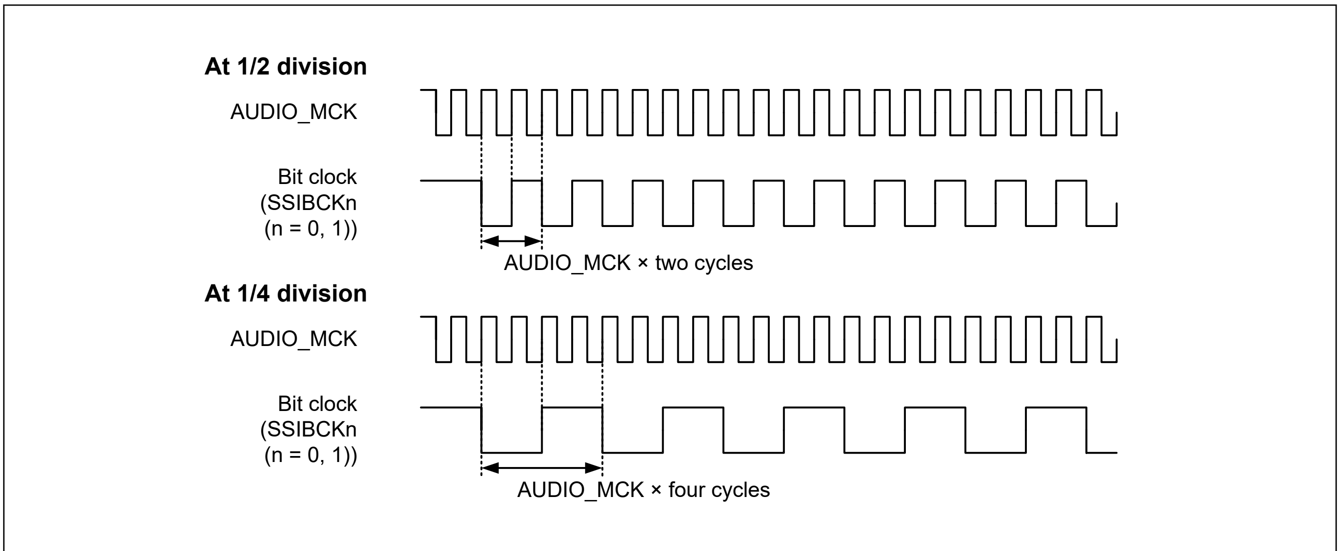


Figure 47.6 Sampling frequencies in master-mode communication

DEL bit (Selects Serial Data Delay)

The DEL bit sets whether or not there will be a delay between SSILRCKn/SSIFSn (n = 0, 1) and SSITXD0/SSIRXD0/SSIDATA1.

For the I^2S or TDM format, set the DEL bit to 0. When the monaural format is used, setting of this bit changes the high period width of SSILRCKn/SSIFSn (n = 0, 1). For details, see section 47.3.2. Monaural Format. When using a compatible communication format, specify a setting of this bit that enables communication.

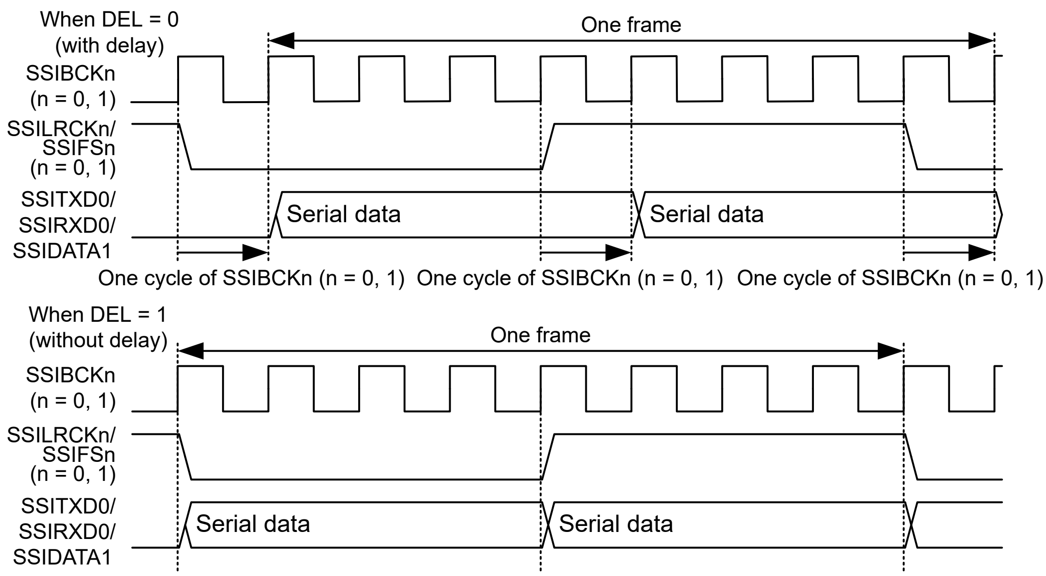


Figure 47.7 Setting of delay in serial data

PDTA bit (Selects Placement Data Alignment)

The PDTA bit sets how to align placement data. With the setting of data word length as 32 bits (SSICR.DWL[2:0] = 110b), this bit is invalid.

At transmission, see Figure 47.8.

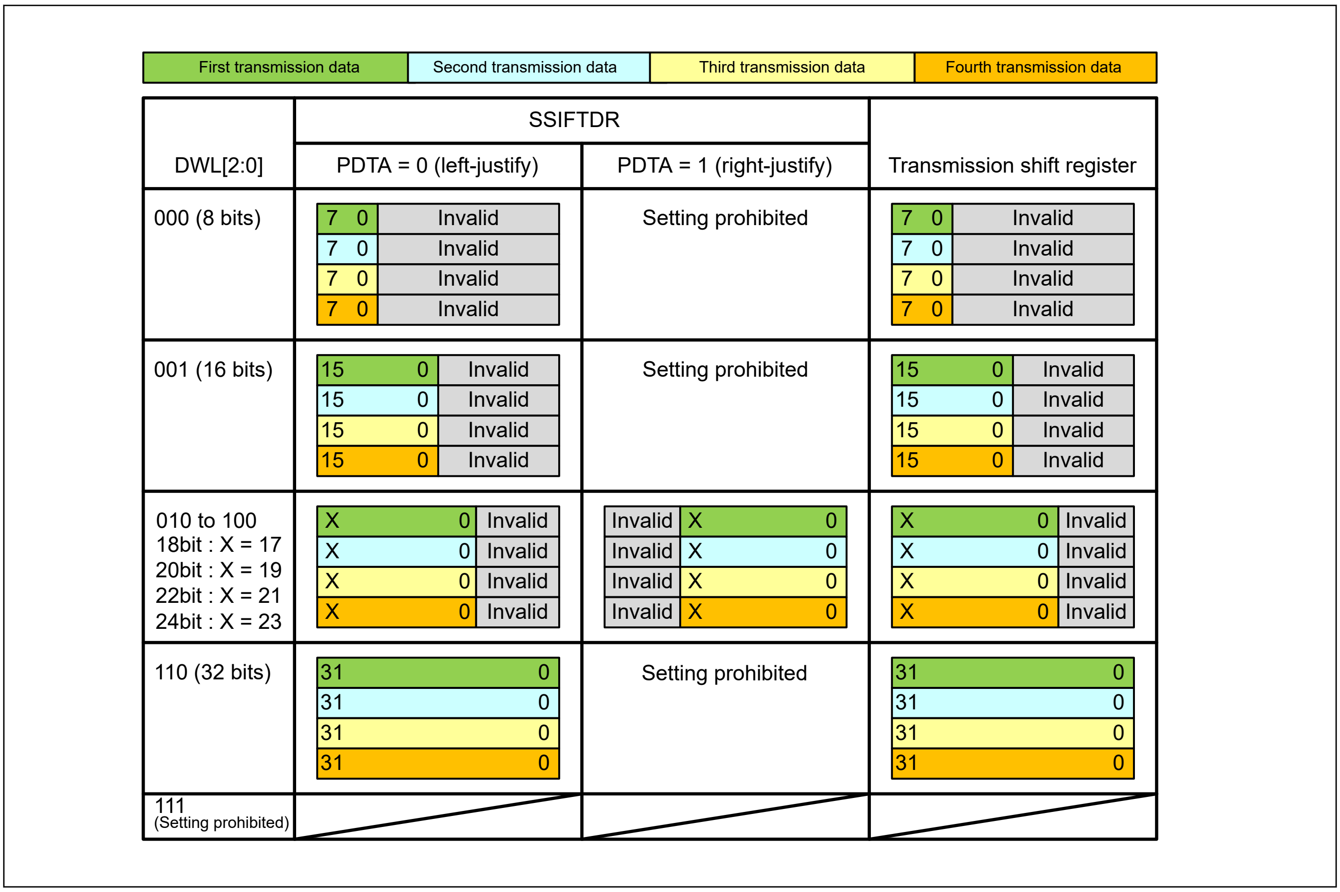


Figure 47.8 Alignment of placement data at transmission

At reception, see Figure 47.9.

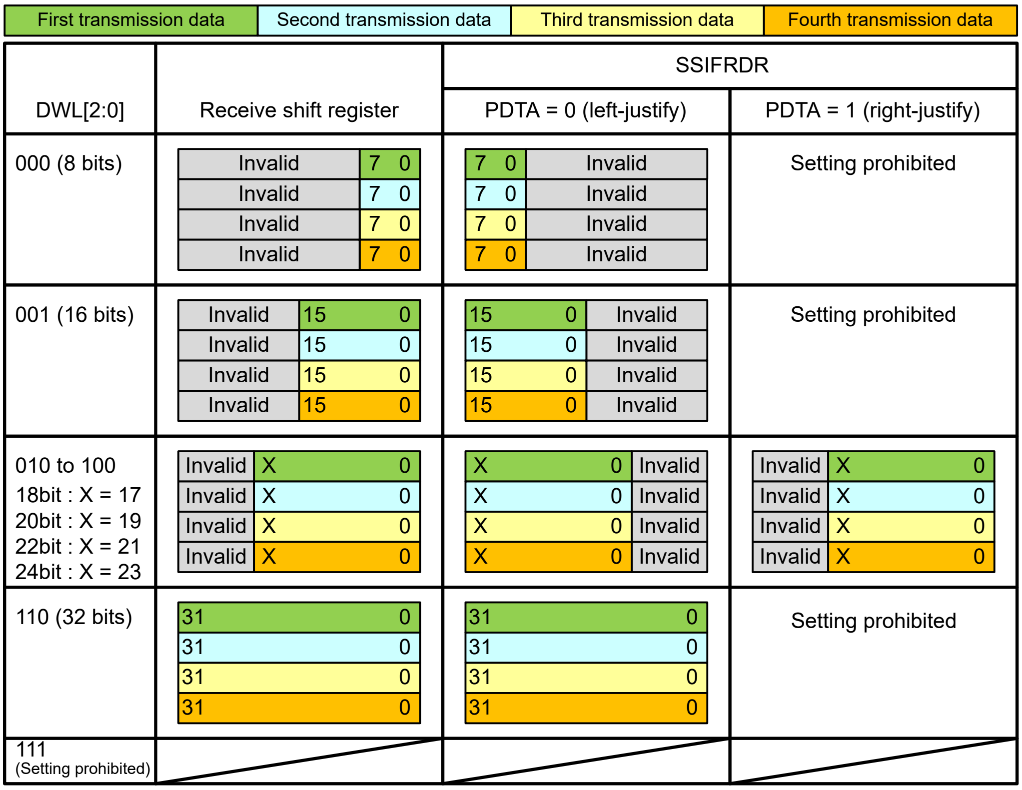


Figure 47.9 Alignment of placement data at reception

SDTA bit (Selects Serial Data Delay)

The SDTA bit sets how to align serial data and padding bits. For communication without padding bits, this bit is invalid.

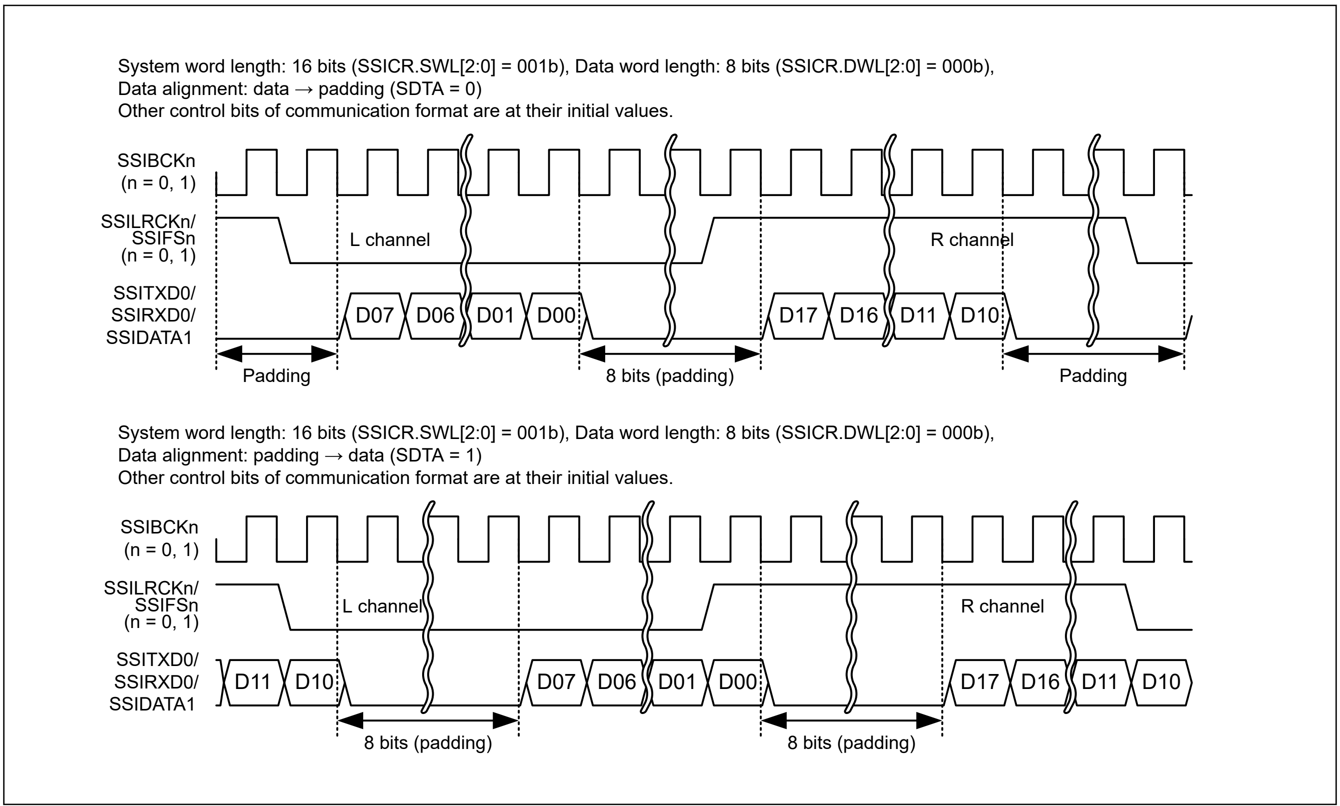


Figure 47.10 Alignment setting of serial data with padding bits

SPDP bit (Selects Serial Padding Polarity)

The SPDP bit sets polarity of padding bits.

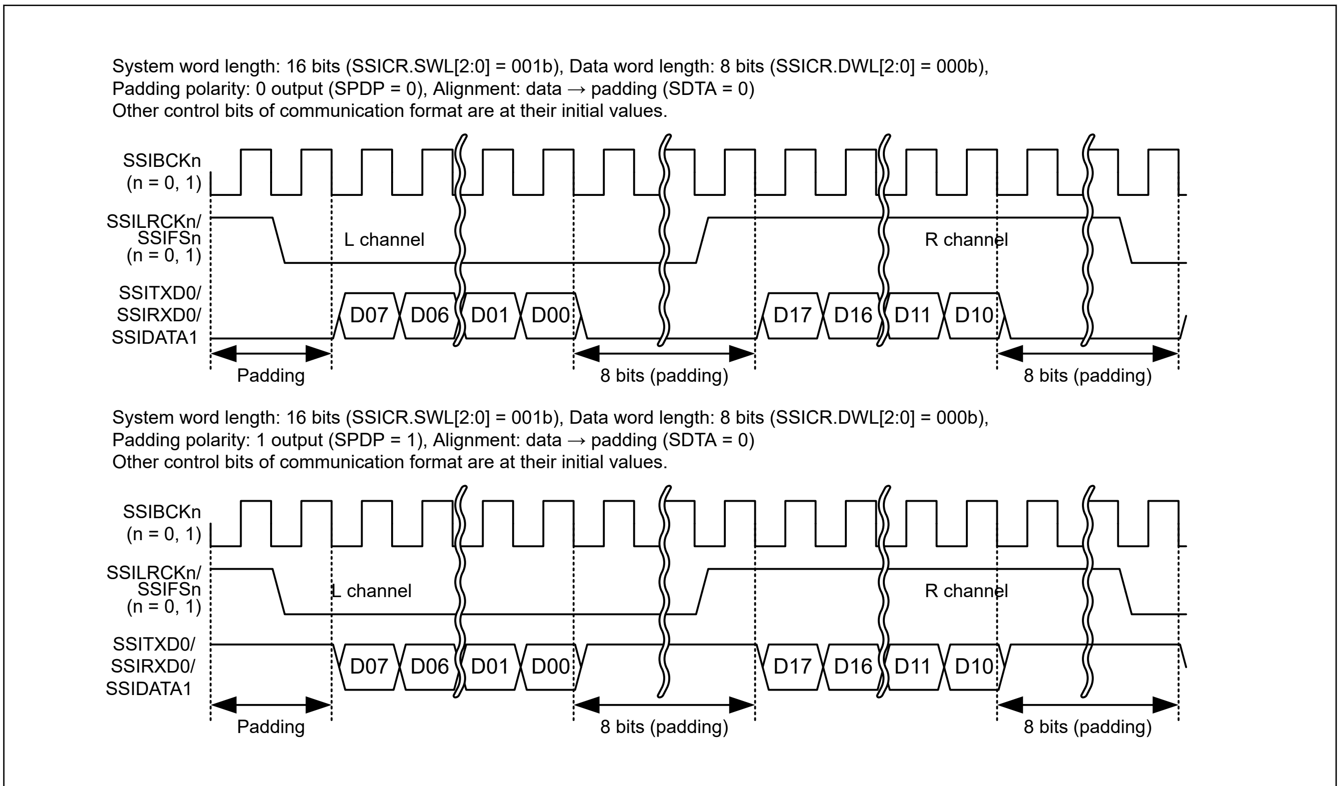


Figure 47.11 Padding bit polarity

LRCKP bit (Selects the Initial Value and Polarity of LR Clock/Frame Synchronization Signal)

The LRCKP bit sets the initial value and polarity of SSILRCKn/SSIFSn (n = 0, 1). Set this bit according to the communication format to be used in SSIE. See Table 47.5 Initial output value and polarity of SSILRCKn/SSIFSn (n = 0, 1) pin. For the slave-mode communication (MST = 0), only the start trigger is used.

Writing to these bits must be performed when the LR clock supply to the SSILRCKn/SSIFSn (n = 0, 1) pin is stopped. For details about the output of LR clock, see the detailed description of the LRCONT bit in section 47.2.7. SSIOFR : Audio Format Register.

Table 47.5 Initial output value and polarity of SSILRCKn/SSIFSn (n = 0, 1) pin

Communication Format	Expected Initial State	Setting Value of LRCKP
I^2S	High	0
Monaural	Low	1
TDM	Low	1

Note: When the format to be used is compatible with the I^2S, monaural, or TDM format, specify settings to enable communication with the respective formats.

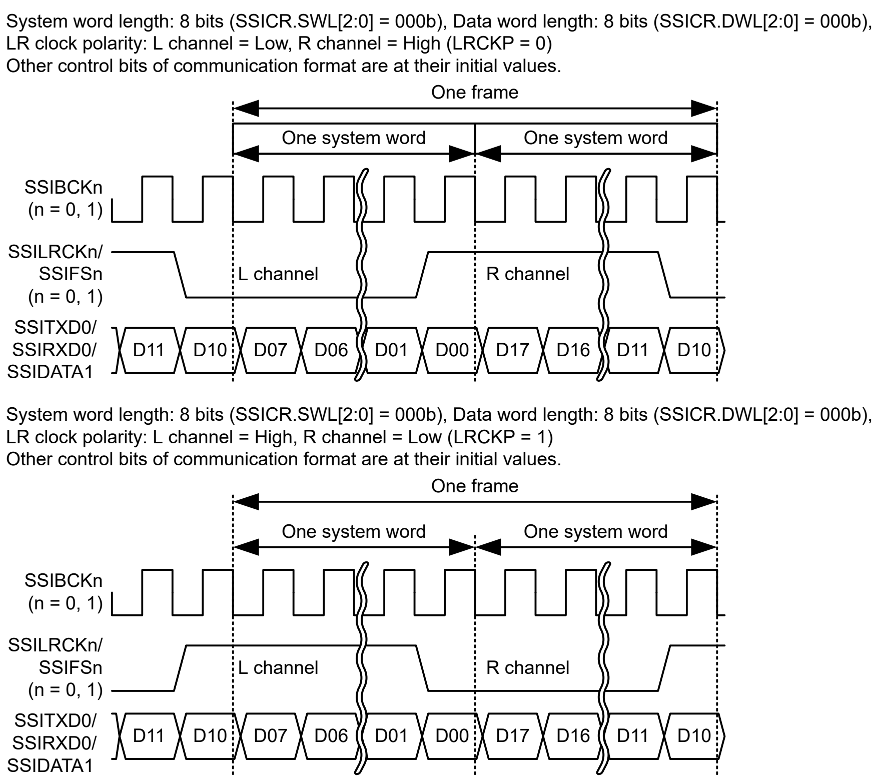


Figure 47.12 LR clock/frame synchronization polarity setting

BCKP bit (Selects Bit Clock Polarity)

The BCKP bit sets the bit clock polarity.

Writing to this bit must be performed when the supply of AUDIO_MCK is stopped. For details about the timing, see the detailed description of the AUCKE bit in section 47.2.3. SSIFCR : FIFO Control Register.

Table 47.6 Bit clock polarity

Communication	Master/Slave	Timing	BCKP = 0	BCKP = 1
Reception	Slave	At SSILRCKn/SSIFSn (n = 0, 1) sampling	SSIBCKn (n = 0, 1) rising edge	SSIBCKn (n = 0, 1) falling edge
	Master/slave	At SSIRXD0/SSIDATA1 sampling	SSIBCKn (n = 0, 1) rising edge	SSIBCKn (n = 0, 1) falling edge
Transmission	Master	At change of SSILRCKn/SSIFSn (n = 0, 1) output	SSIBCKn (n = 0, 1) falling edge	SSIBCKn (n = 0, 1) rising edge
	Master/slave	At change of SSITXD0/SSIDATA1 output	SSIBCKn (n = 0, 1) falling edge	SSIBCKn (n = 0, 1) rising edge

MST bit (Master Enable)

The MST bit sets master-/slave-mode communication.

Writing to this bit must be performed when the supply of AUDIO_MCK is stopped. For details about the timing, see the detailed description of the AUCKE bit in section 47.2.3. SSIFCR : FIFO Control Register.

SWL[2:0] bits (Selects System Word Length)

The SWL[2:0] bits set the number of bits in one system word. Padding bits are sent and received in relation with one data word set with DWL[2:0]. See Table 47.13 for details.

Writing to these bits must be performed when the LR clock supply to the SSILRCKn/SSIFSn (n = 0, 1) pin is stopped. For details about the output of LR clock, see the detailed description of the LRCONT bit in section 47.2.7. SSIOFR : Audio Format Register.

DWL[2:0] bits (Selects Data Word Length)

The DWL[2:0] bits set the number of bits in one data word. The data word length (number of bits per data word) must not exceed the system word length (number of bits per system word). For details, see Table 47.13.

FRM[1:0] bits (Selects Frame Word Number)

The FRM[1:0] bits set the frame word number in individual communication formats.

Writing to these bits must be performed when the LR clock supply to the SSILRCKn/SSIFSn (n = 0, 1) pin is stopped. For details about the output of LR clock, see the detailed description of the LRCONT bit in section 47.2.7. SSIOFR : Audio Format Register.

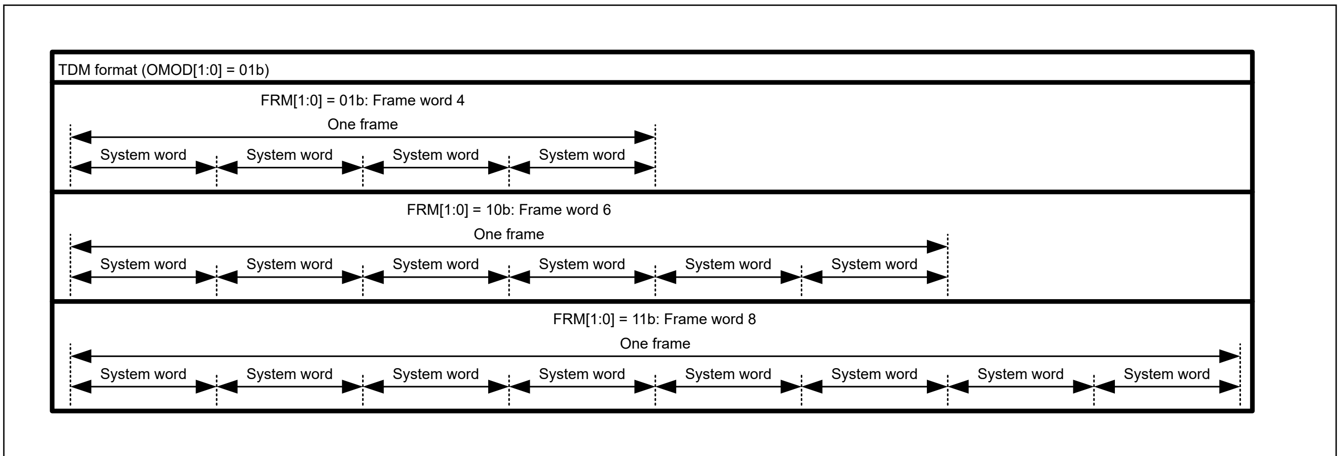


Figure 47.13 Frame word number

IIEN bit (Idle Mode Interrupt Output Enable)

The IIEN bit enables/disables output of idle mode interrupts. By enabling this bit (set it to 1), an interrupt is output at a rising edge of SSISR.IIRQ = 1. An interrupt is also output when this bit is changed from 0 to 1 while SSISR.IIRQ = 1.

ROIEN bit (Receive Overflow Interrupt Output Enable)

The ROIEN bit enables/disables output of receive overflow interrupts. By enabling this bit (set it to 1), an interrupt is output at a rising edge of SSISR.ROIRQ = 1. An interrupt is also output when this bit is changed from 0 to 1 while SSISR.ROIRQ = 1.

RUIEN bit (Receive Underflow Interrupt Output Enable)

The RUIEN bit enables/disables output of receive underflow interrupts. By enabling this bit (set it to 1), an interrupt is output at a rising edge of SSISR.RUIRQ = 1. An interrupt is also output when this bit is changed from 0 to 1 while SSISR.RUIRQ = 1.

TOIEN bit (Transmit Overflow Interrupt Output Enable)

The TOIEN bit enables/disables output of transmit overflow interrupts. By enabling this bit (set it to 1), an interrupt is output at a rising edge of SSISR.TOIRQ = 1. An interrupt is also output when this bit is changed from 0 to 1 while SSISR.TOIRQ = 1.

TUIEN bit (Transmit Underflow Interrupt Output Enable)

The TUIEN bit enables/disables output of transmit underflow interrupts. By enabling this bit (set it to 1), an interrupt is output at a rising edge of SSISR.TUIRQ = 1. An interrupt is also output when this bit is changed from 0 to 1 while SSISR.TUIRQ = 1.

CKS bit (Selects an Audio Clock for Master-mode Communication)

The CKS bit sets the audio clock in master-mode communication (MST = 1). In slave-mode communication (MST = 0), setting of this bit is invalid.

Writing to this bit must be performed when the supply of AUDIO_MCK is stopped. For details about the timing, see the detailed description of the AUCKE bit in section 47.2.3. SSIFCR : FIFO Control Register.

47.2.2 SSISR : Status Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x04

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
24:0	—	These bits are read as 0. The write value should be 0.	R/W
25	IIRQ	Idle Mode Status Flag 0: In the communication state 1: In the idle state	R
26	ROIRQ	Receive Overflow Error Status Flag 0: No receive overflow error is generated. 1: A receive overflow error is generated.	R/W
27	RUIRQ	Receive Underflow Error Status Flag 0: No receive underflow error is generated. 1: A receive underflow error is generated.	R/W
28	TOIRQ	Transmit Overflow Error Status Flag 0: No transmit overflow error is generated. 1: A transmit overflow error is generated.	R/W

Bit	Symbol	Function	R/W
29	TUIRQ	Transmit Underflow Error Status flag 0: No transmit underflow error is generated. 1: A transmit underflow error is generated.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register is configured with status flags that indicate SSIE operational state.

IIRQ flag (Idle Mode Status Flag)

The IIRQ flag is a status flag that indicates the idle state. It indicates whether SSIE is in the idle state or communication state.

For details, see Figure 47.14 and Figure 47.15.

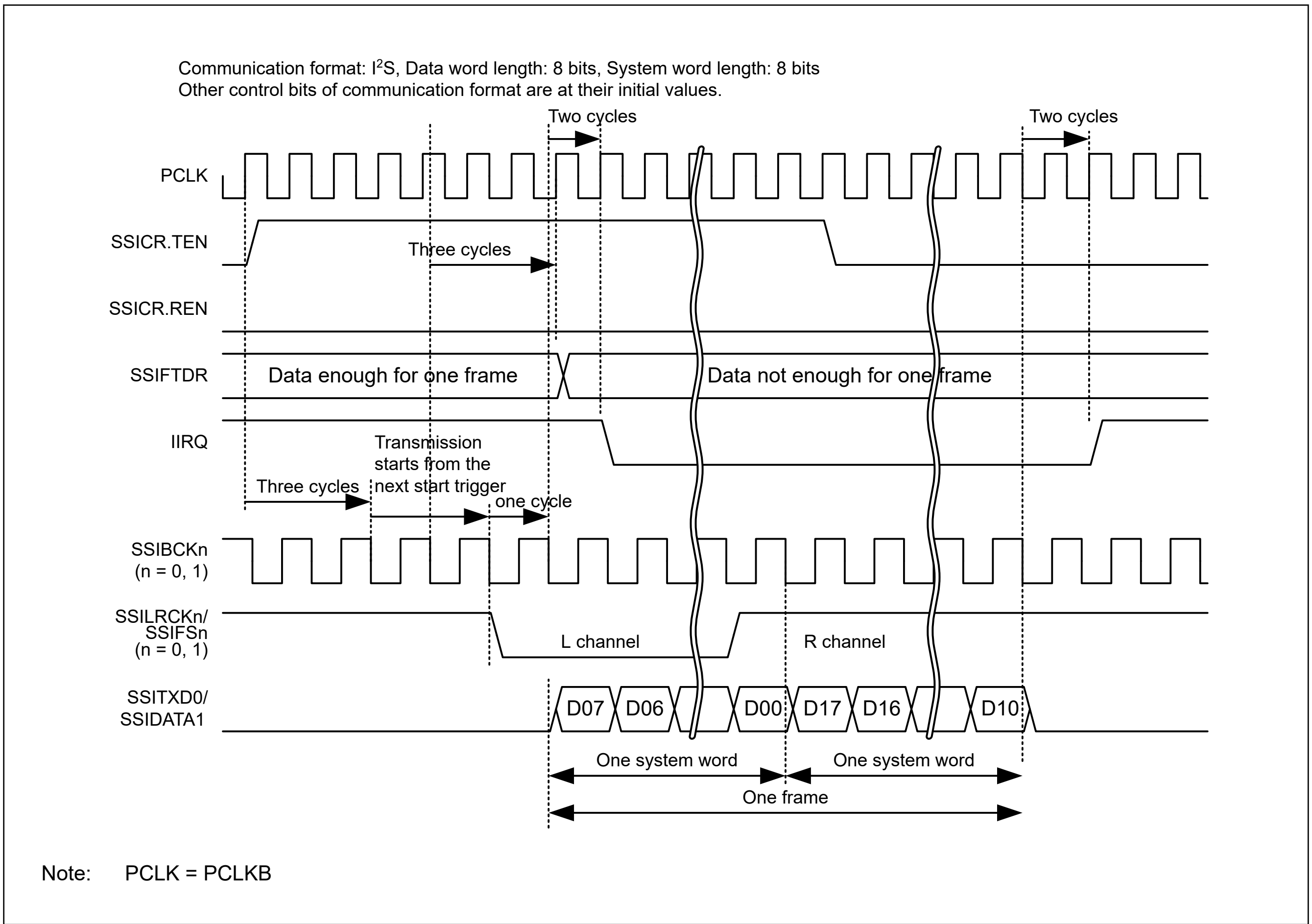


Figure 47.14 IIRQ setting timing (transmission)

- Transmitter (dedicated to transmission)

[Clearing condition]

While transmission was enabled (SSICR.TEN = 1 and SSICR.REN = 0), the transmit data for a transmission frame was written to the SSIFTDR register, and a start trigger was generated by the SSILRCKn/SSIFSn (n = 0, 1) signal.

[Clearing timing]

1 SSIBCKn (n = 0, 1) cycle + 2 PCLKB cycles after generation of the start trigger that is the clearing condition.

[Setting condition]

While transmission and reception were disabled (SSICR.TEN = 0 and SSICR.REN = 0), transmission of one frame was complete.

[Setting timing]

2 PCLKB cycles after the end of transmission (at a frame boundary) that is the setting condition.

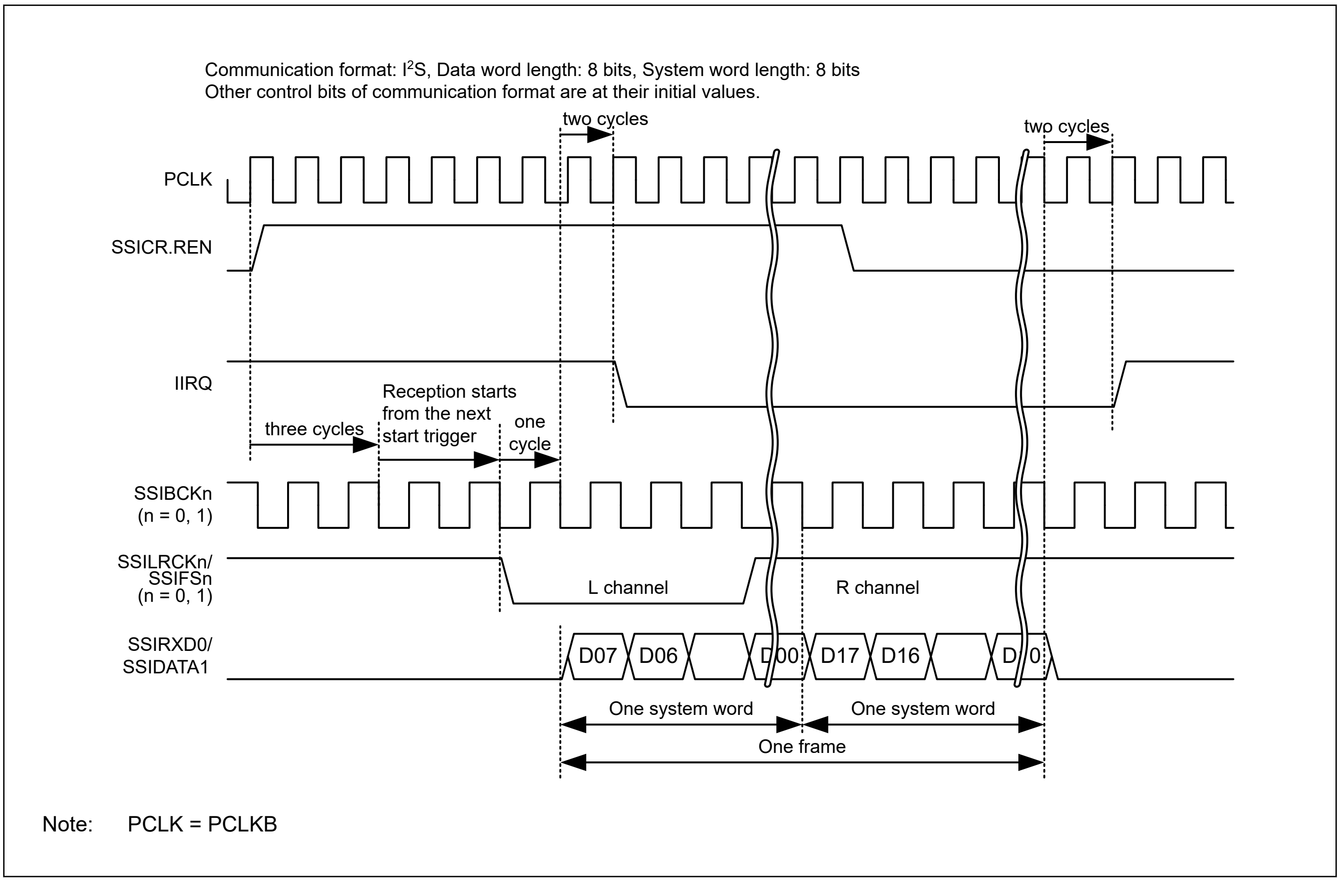


Figure 47.15 IIRQ setting timing (reception)

- Receiver (dedicated to reception)

[Clearing condition]

While reception was enabled (SSICR.TEN = 0 and SSICR.REN = 1, a start trigger was generated by the SSILRCKn/SSIFSn (n = 0, 1) signal.

[Clearing timing]

1 SSIBCKn (n = 0, 1) cycle + 2 PCLKB cycles after generation of the start trigger that is the clearing condition.

[Setting condition]

While transmission and reception were disabled (SSICR.TEN = 0 and SSICR.REN = 0), reception of one frame was complete.

[Setting timing]

2 PCLKB cycles after the end of reception (at a frame boundary) that is the setting condition.

- Transceiver (transmission and reception)

[Clearing condition]

While transmission and reception were enabled (SSICR.TEN = 1 and SSICR.REN = 1), the transmit data for a transmission frame was written to the SSIFTDR register, and a start trigger is generated by the SSILRCKn/SSIFSn (n = 0, 1) signal.

[Clearing timing]

1 SSIBCKn (n = 0, 1) cycle + 2 PCLKB cycles after generation of the start trigger that is the clearing condition.

[Setting condition]

While transmission and reception were disabled (SSICR.TEN = 0 and SSICR.REN = 0), transmission of one frame was complete.

[Setting timing]

2 PCLKB cycles after the end of transmission (at a frame boundary) that is the setting condition.

ROIRQ flag (Receive Overflow Error Status Flag)

The ROIRQ flag is a status flag that indicates a receive overflow error. This flags is set by automatic determination, but it must be cleared by register access. This flag indicates that received data is supplied at a higher rate than requested. Data is not transferred from the receive shift register to SSIFRDR where a receive overflow error is generated. For the procedure to recover from the overflow error, see section 47.6.6. Error Handling. This flag is not cleared by a receive FIFO data register reset (SSIFCR.RFRST).

[Priority order for setting and clearing]

Setting is prioritized.*1

[Clearing condition]

When either of the following operations is done:

1. Writing 0 to this bit after reading 1 from this bit*2
2. Enabling communication (changing SSICR.REN from 0 to 1).

[Clearing timing]

Clearing timing corresponding to the above clearing condition:

1. When 0 is written to this bit after reading 1 from this bit (same as the timing in Figure 47.19)
2. 1 PCLKB cycle after writing 1 to SSICR.REN.*3

[Setting condition]
At completion of receiving new data while SSIFRDR is full.

[Setting timing]
3 cycles of PCLKB after reception is completed.

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

- A software reset (SSIFCR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.REN.

Note 3. After communication is enabled (by changing the value of SSICR.REN bit from 0 to 1), the reception error flags (RUIRQ and ROIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the reception error flags might be unable to be read.

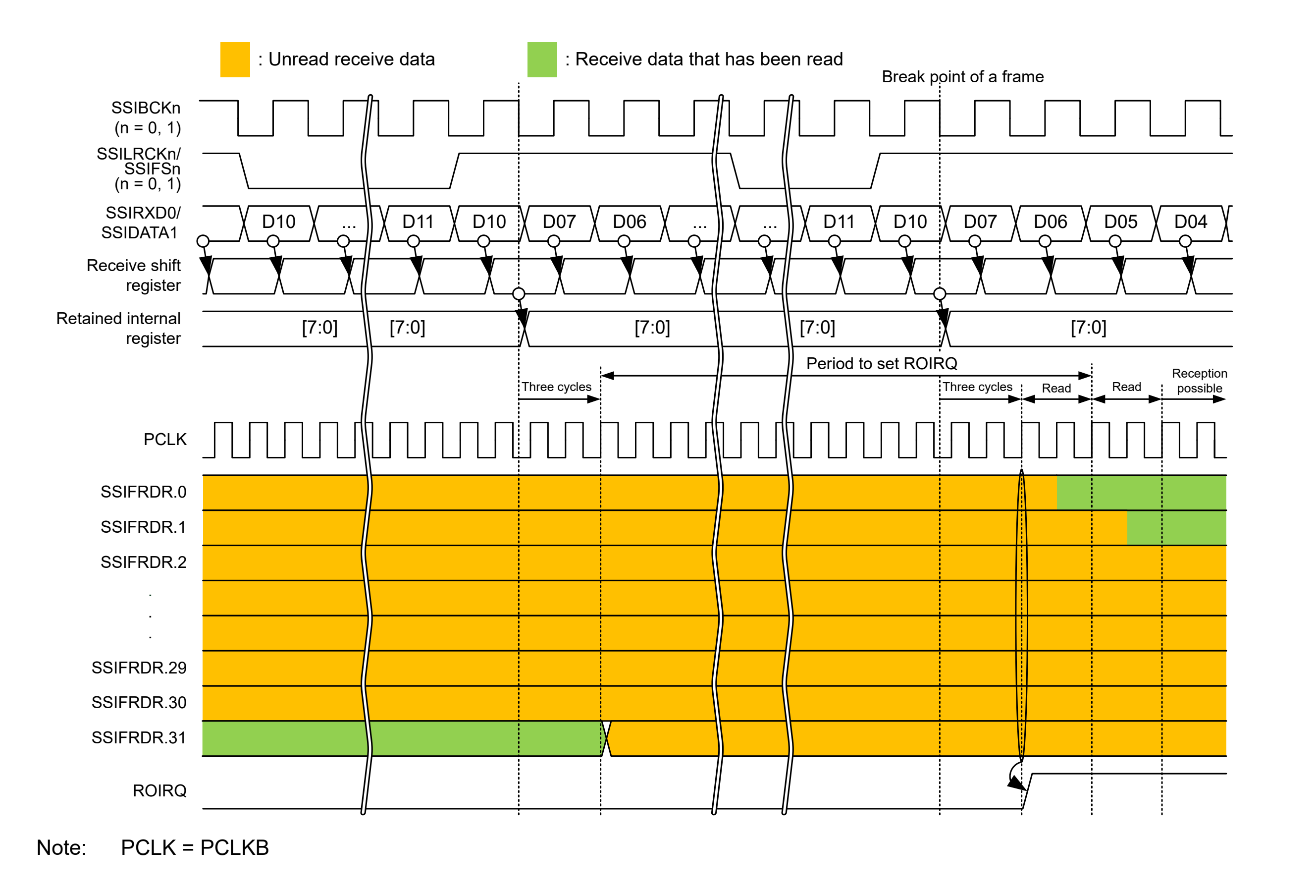


Figure 47.16 ROIRQ setting timing

RUIRQ flag (Receive Underflow Error Status Flag)

The RUIRQ flag is a status flag that indicates a receive underflow error. This flags is set by automatic determination, but it must be cleared by register access. This flag indicates that SSIFRDR is read while it is empty. Data read from SSIFRDR where a receive underflow error is generated is invalid. See section 47.6.6. Error Handling for the error recovery procedure. This flag is not cleared by a receive FIFO data register reset (SSIFCR.RFRST). Note, however, that this flag is not set even if the SSIFRDR register is read while the receive FIFO data register is reset (by setting SSIFCR.RFRST to 1).

[Priority order for setting and clearing]

Setting is prioritized.*1

[Clearing condition]

When either of the following operations is done:

1. Writing 0 to this bit after reading 1 from this bit*2
2. Enabling communication (changing SSICR.REN from 0 to 1).

[Clearing timing]
Clearing timing corresponding to the above clearing condition

1. When 0 is written to this bit after reading 1 from this bit (same as the timing in Figure 47.19)
2. 1 PCLKB cycle after writing 1 to SSICR.REN.*3

[Setting condition]

Reading from SSIFRDR while it is empty.

[Setting timing]

At completion of reading from SSIFRDR. See Figure 47.17.

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

- A software reset (SSIFCR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.REN.

Note 3. After communication is enabled (by changing the value of SSICR.REN bit from 0 to 1), the reception error flags (RUIRQ and ROIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the reception error flags might be unable to be read.

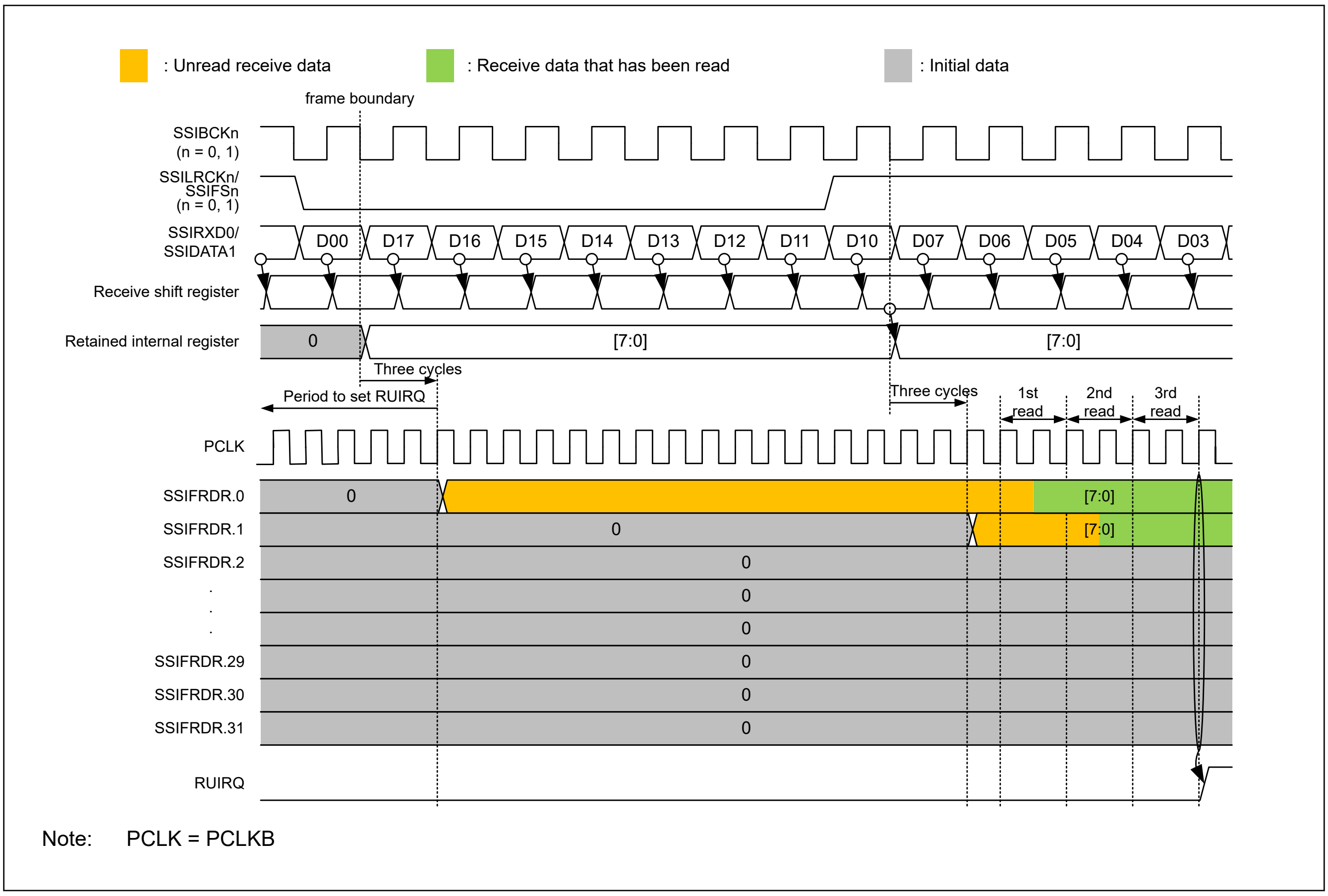


Figure 47.17 RUIRQ setting timing

TOIRQ flag (Transmit Overflow Error Status Flag)

The TOIRQ flag is a status flag that indicates a transmit overflow error. This flag is set by automatic determination, but it must be cleared by register access. This flag indicates that an attempt has been made to write data to the SSIFTDR register when the register is full of data. The data writing that causes a transmit overflow is ignored. For the procedure to recover from the overflow error, see section 47.6.6. Error Handling. This flag is not cleared by a transmit FIFO data register reset (SSIFCR.TFRST).

[Priority order for setting and clearing]

Setting is prioritized.[*1]

[Clearing condition]

When either of the following operations is done:

1. Writing 0 to this bit after reading 1 from this bit[*2]
2. Enabling communication (changing SSICR.TEN from 0 to 1).

[Clearing timing]

Clearing timing corresponding to the above clearing condition

1. When 0 is written to this bit after reading 1 from this bit (same as the timing in Figure 47.19)
2. 1 PCLKB cycle after writing 1 to SSICR.TEN.*3

[Setting condition]

An attempt was made to write data to the SSIFTDR register when the register is full of data.

[Setting timing]

At completion of writing to SSIFTDR. For details, see Figure 47.18.

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

- A software reset (SSIFCR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.TEN.

Note 3. After communication is enabled (by changing the value of SSICR.TEN bit from 0 to 1), the transmission error flags (TOIRQ and TUIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the transmission error flags might be unable to be read.

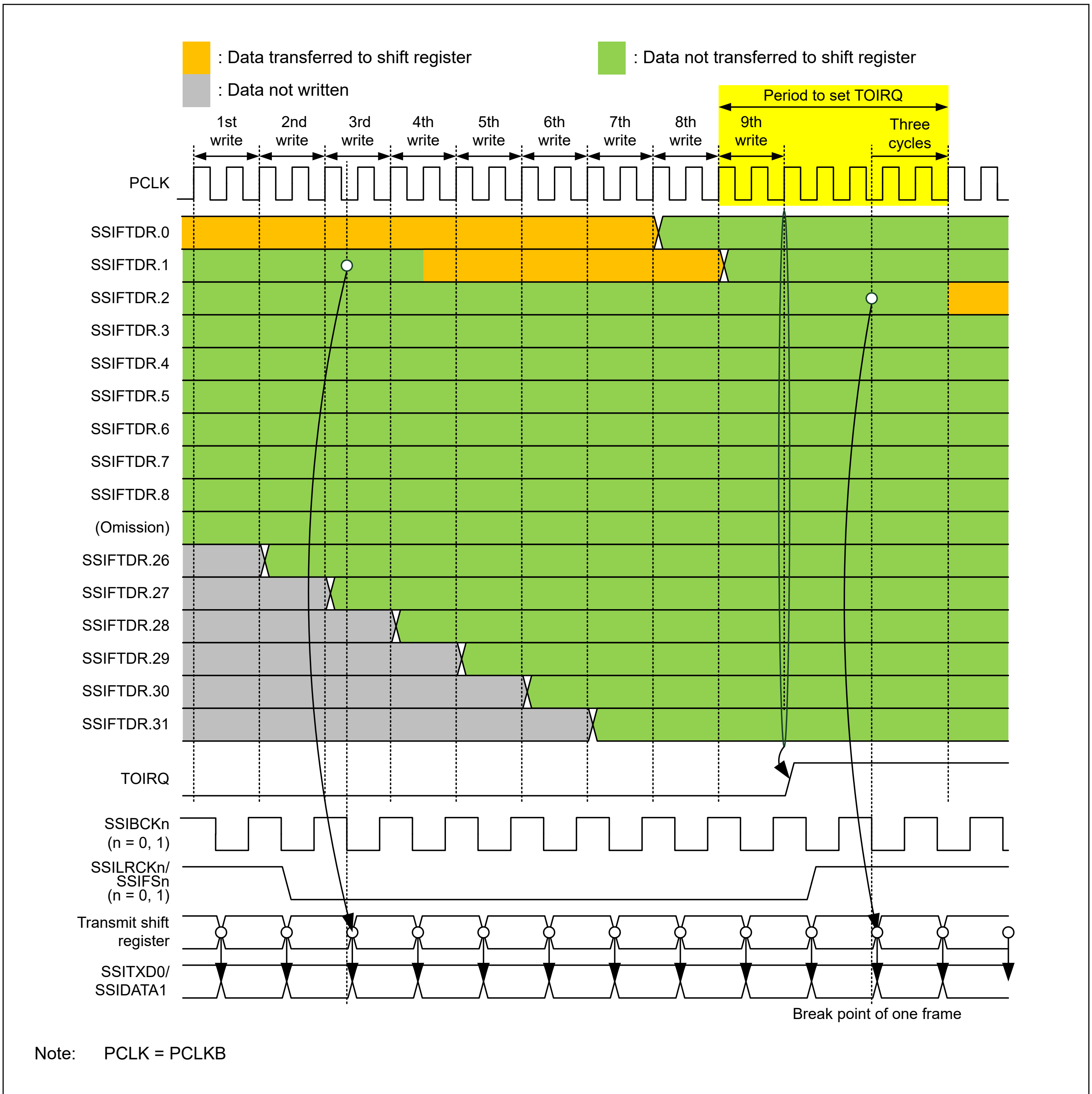


Figure 47.18 TOIRQ setting timing

TUIRQ flag (Transmit Underflow Error Status flag)

The TUIRQ flag is a status flag that indicates a transmit underflow error. This flag is set by automatic determination, but it must be cleared by register access. This flag indicates that writing the serial data required for a frame to SSIFTDR did not catch up with transmission of the frame. Even if this flag is cleared after it has been set, the SSITXD0/SSIDATA1 output remains to be 0. To output the data written to the transmit FIFO data register (SSIFTDR) to the SSITXD0/SSIDATA1 pin, follow the communication stop procedure in Figure 47.56 and error-handling procedure in Figure 47.57. For the procedure to recover from an error, see section 47.6.6. Error Handling. This flag is not cleared by a reset of transmit FIFO data register (by the SSIFCR.TFRST signal).

[Priority order for setting and clearing]

Setting is prioritized.*1

[Clearing condition]

When either of the following operations is done:

1. Writing 0 to this bit after reading 1 from this bit*2
2. Enabling communication (changing SSICR.TEN from 0 to 1).

[Clearing timing]

Clearing timing corresponding to the above clearing condition

1. When 0 is written to this bit after reading 1 from this bit
2. 1 PCLKB cycle after writing 1 to SSICR.TEN.*3

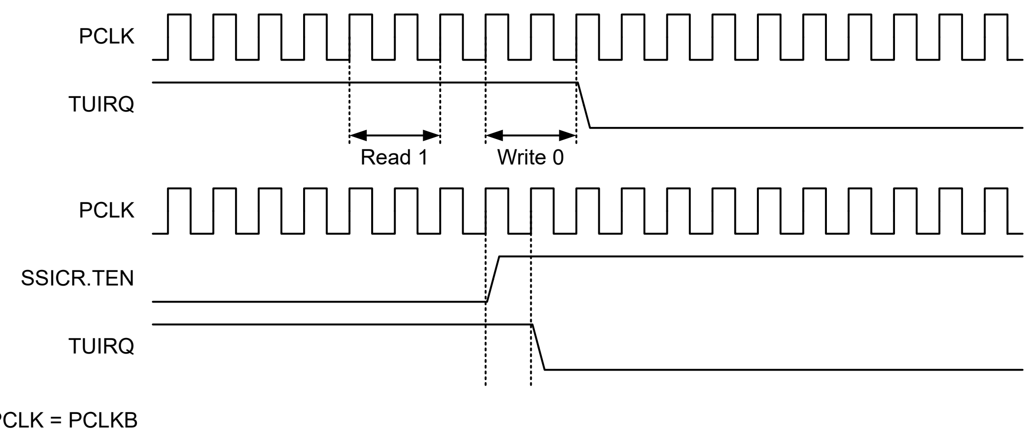


Figure 47.19 TUIRQ clearing timing

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1). The software reset has priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following three conditions is met:

- A software reset (SSIFCR.SSIRST = 1) is done.
- After 1 has been read, writing of 0 is complete.
- 1 PCLKB cycle passes after 1 has been written to SSICR.TEN.

Note 3. After communication is enabled (by changing the value of SSICR.TEN bit from 0 to 1), the transmission error flags (TOIRQ and TUIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the transmission error flags might be unable to be read.

[Setting condition]

When communication continues over a frame boundary, the transmit data required for the next frame has not been written to SSIFTDR. For details, see Figure 47.20 and Figure 47.21.

[Setting timing]

3 PCLKB cycles after the frame boundary. For details, see Figure 47.20.

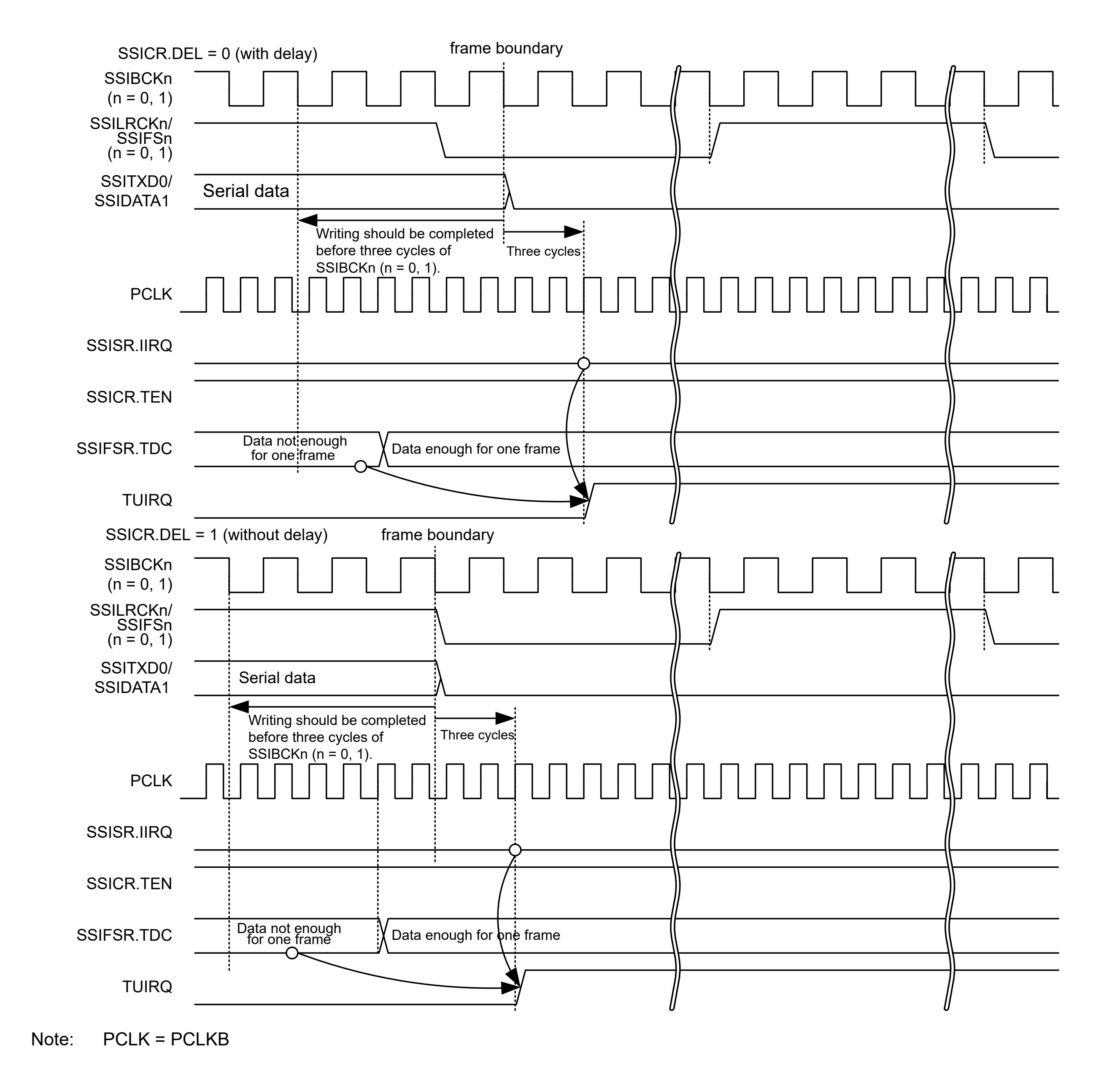


Figure 47.20 TUIRQ setting timing (when communication continues)

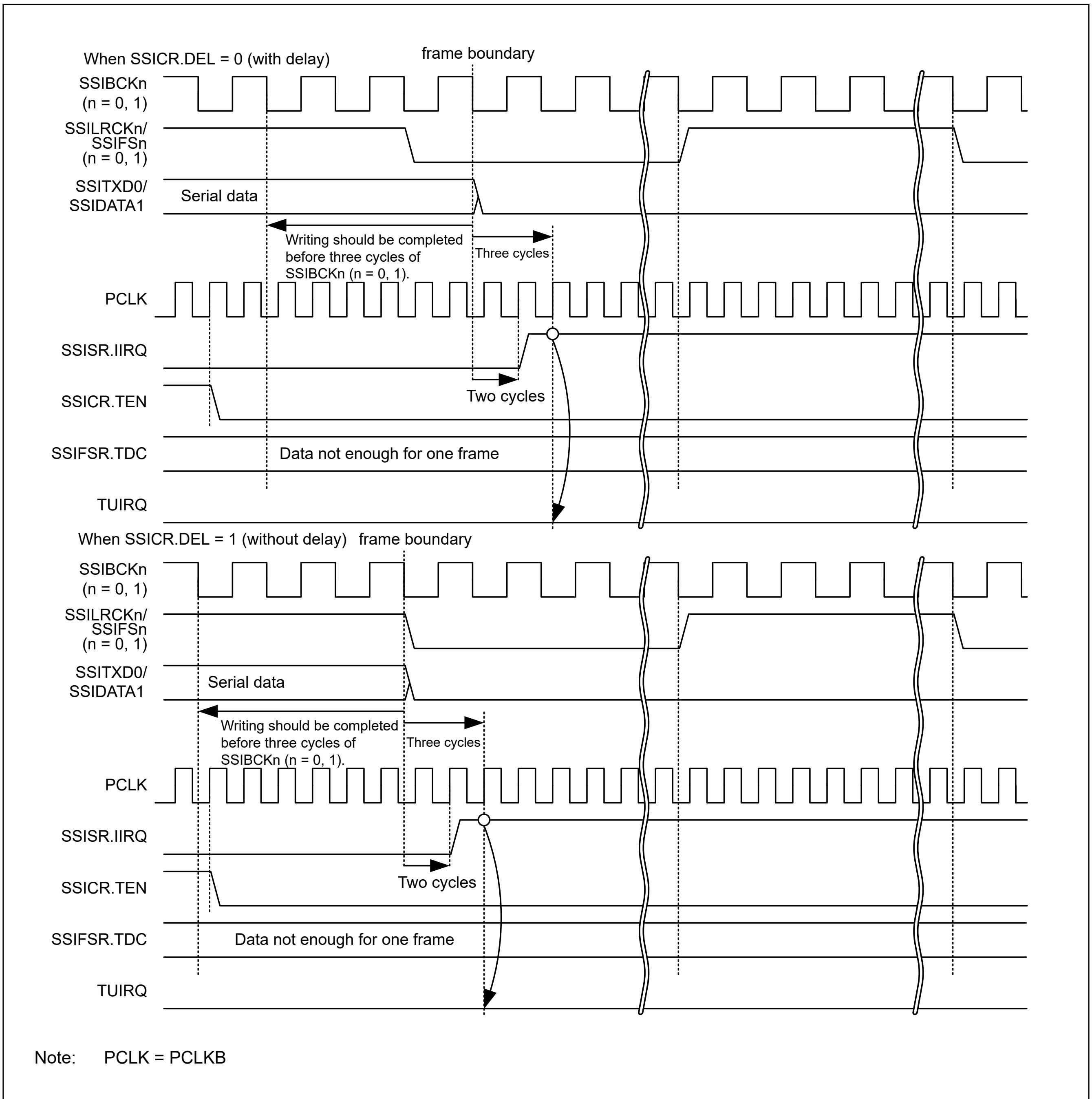


Figure 47.21 TUIRQ setting timing (when communication stops)

47.2.3 SSIFCR : FIFO Control Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x10

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RFRST	Receive FIFO Data Register Reset*1 0: Clears a receive data FIFO reset condition 1: Sets a receive data FIFO reset condition	R/W
1	TFRST	Transmit FIFO Data Register Reset*1 0: Clears a transmit data FIFO reset condition 1: Sets a transmit data FIFO reset condition	R/W
2	RIE	Receive Data Full Interrupt Output Enable 0: Disables receive data full interrupts 1: Enables receive data full interrupts	R/W
3	TIE	Transmit Data Empty Interrupt Output Enable 0: Disables transmit data empty interrupts 1: Enables transmit data empty interrupts	R/W
10:4	—	These bits are read as 0. The write value should be 0.	R/W
11	BSW	Byte Swap Enable*1 0: Disables byte swap 1: Enables byte swap	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	SSIRST	Software Reset 0: Clears a software reset condition 1: Sets a software reset condition	R/W
30:17	—	These bits are read as 0. The write value should be 0.	R/W
31	AUCKE	AUDIO_MCK Enable in Master mode Communication*1 0: Disables supply of AUDIO_MCK 1: Enables supply of AUDIO_MCK	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Writing to these bits while SSIE is in a communication state (SSISR.IIRQ = 0) is prohibited. If the value of these bits is changed by rewriting, subsequent operation is unpredictable.

This register sets a software reset, byte swap, and enable/disable of interrupt requests.

RFRST bit (Receive FIFO Data Register Reset)

The RFRST bit sets a software reset of the receive FIFO data register (SSIFRDR). Writing 1 to this bit initializes the internal state related to SSIFRDR. The register bits subject to the software reset triggered by this bit are indicated by shading in Table 47.7. Because this bit is not automatically cleared after it has been set, write 0 to this bit to release the register bits from the software reset. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedural step.

This bit is subject to the software reset by the SSIRST bit. Because the software reset by the SSIRST bit has priority over the reset by this bit, setting this bit is ignored when the SSIRST bit is set.

Table 47.7 Bits subject to software reset by the RFRST bit

Symbol	Address (BASE+)		+0								+1							
			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSICR	0x00	+0	—	CKS	TUIEN	TOIEN	RUIEN	ROIEN	IIEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
		+2	—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL	CKDV[3:0]				MUEN	—	TEN	REN
SSISR	0x04	+0	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
SSIFCR	0x10	+0	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
		+2	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
SSIFSR	0x14	+0	—	—	TDC[5:0]						—	—	—	—	—	—	—	TDE
		+2	—	—	RDC[5:0]						—	—	—	—	—	—	—	RDF
SSIFTDR	0x18	+0	FTDR[31:16]															
		+2	FTDR[15:0]															
SSIFRDR	0x1C	+0	FRDR[31:16]															
		+2	FRDR[15:0]															
SSIOFR	0x20	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	
SSISCR	0x24	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	TDES[4:0]					—	—	—	RDFS[4:0]				

TFRST bit (Transmit FIFO Data Register Reset)

The TFRST bit sets a software reset of the transmit FIFO data register (SSIFTDR). Writing 1 to this bit initializes the internal state related to SSIFTDR. The register bits subject to the software reset triggered by this bit are indicated by shading in Table 47.8. Because this bit is not automatically cleared after it has been set, write 0 to this bit to release the register bits from the software reset. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedural step.

This bit is subject to the software reset by the SSIRST bit. Because the software reset by the SSIRST bit has priority over the reset by this bit, setting this bit is ignored when the SSIRST bit is set.

Table 47.8 Bits subject to software reset by the TFRST bit (1 of 2)

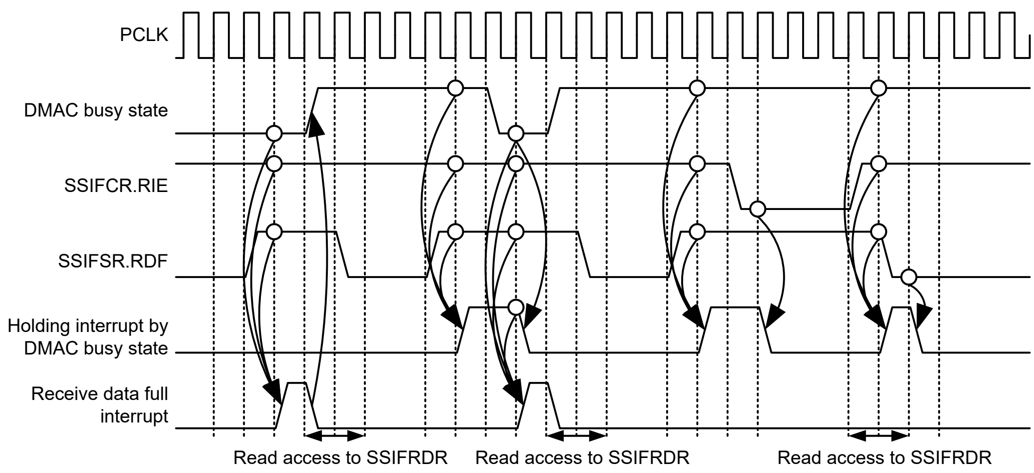
Symbol	Address (BASE+)		+0								+1							
			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSICR	0x00	+0	—	CKS	TUIEN	TOIEN	RUIEN	ROIEN	IIEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
		+2	—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL	CKDV[3:0]				MUEN	—	TEN	REN
SSISR	0x04	+0	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 47.8 Bits subject to software reset by the TFRST bit (2 of 2)

Symbol	Address (BASE+)		+0								+1							
			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSIFCR	0x10	+0	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
		+2	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
SSIFSR	0x14	+0	—	—	TDC[5:0]						—	—	—	—	—	—	—	TDE
		+2	—	—	RDC[5:0]						—	—	—	—	—	—	—	RDF
SSIFTDR	0x18	+0	FTDR[31:16]															
		+2	FTDR[15:0]															
SSIFRDR	0x1C	+0	FRDR[31:16]															
		+2	FRDR[15:0]															
SSIOFR	0x20	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	
SSISCR	0x24	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	TDES[4:0]					—	—	—	RDFS[4:0]				

RIE bit (Receive Data Full Interrupt Output Enable)

The RIE bit enables/disables output of receive data full interrupts. Use a receive data full interrupt as an interrupt to trigger data reading from the receive FIFO data register. Write 1 to this bit after specifying the setting condition for receive data full interrupt (by using the SSISCR.RDFS bit). Figure 47.22 shows the timing of generating the receive data full interrupt.



Note: PCLK = PCLKB

Figure 47.22 Timing of receive data full interrupt

TIE bit (Transmit Data Empty Interrupt Output Enable)

The TIE bit enables/disables output of transmit data empty interrupts. Use a transmit data empty interrupt as an interrupt to trigger data writing to the transmit FIFO data register. Write 1 to this bit after specifying the setting condition for transmit data empty interrupt (by using the SSISCR.TDES bit). Figure 47.23 shows the timing of generating the transmit data empty interrupt.

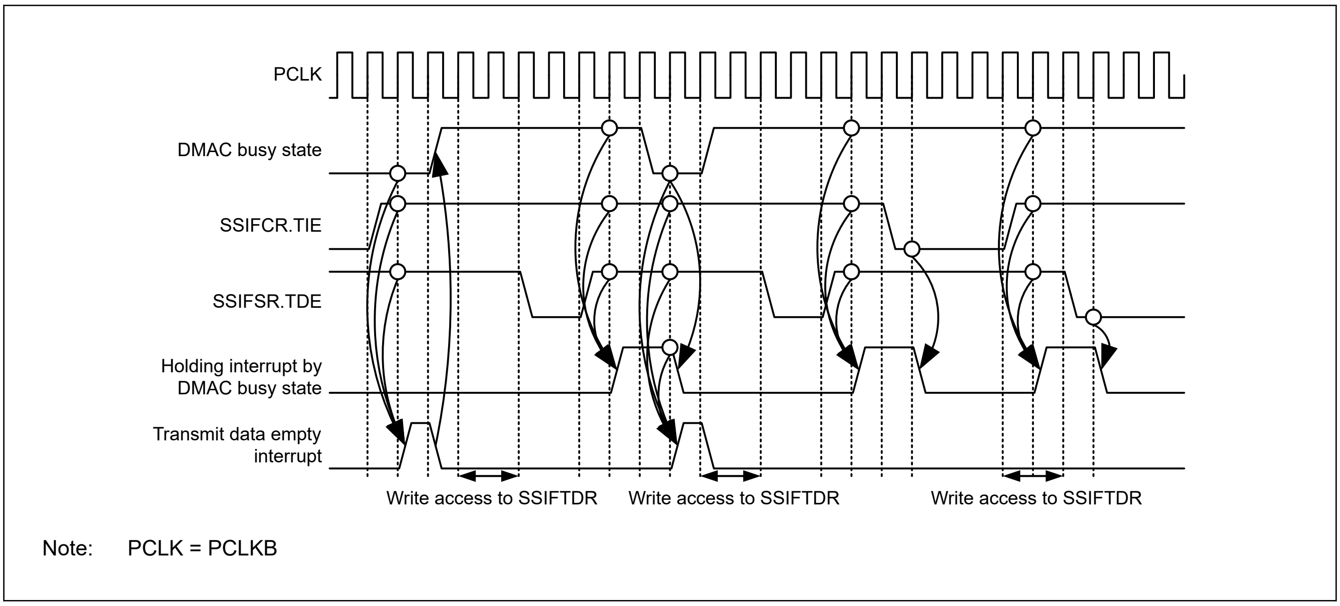


Figure 47.23 Timing of transmit data empty interrupt

BSW bit (Byte Swap Enable)

The BSW bit enables/disables byte swap of register access for the transmit FIFO data register (SSIFTDR) and the receive FIFO data register (SSIFRDR). This bit is valid only with 16-bit access or 32-bit access to SSIFTDR and SSIFRDR. For details, see Figure 47.24.

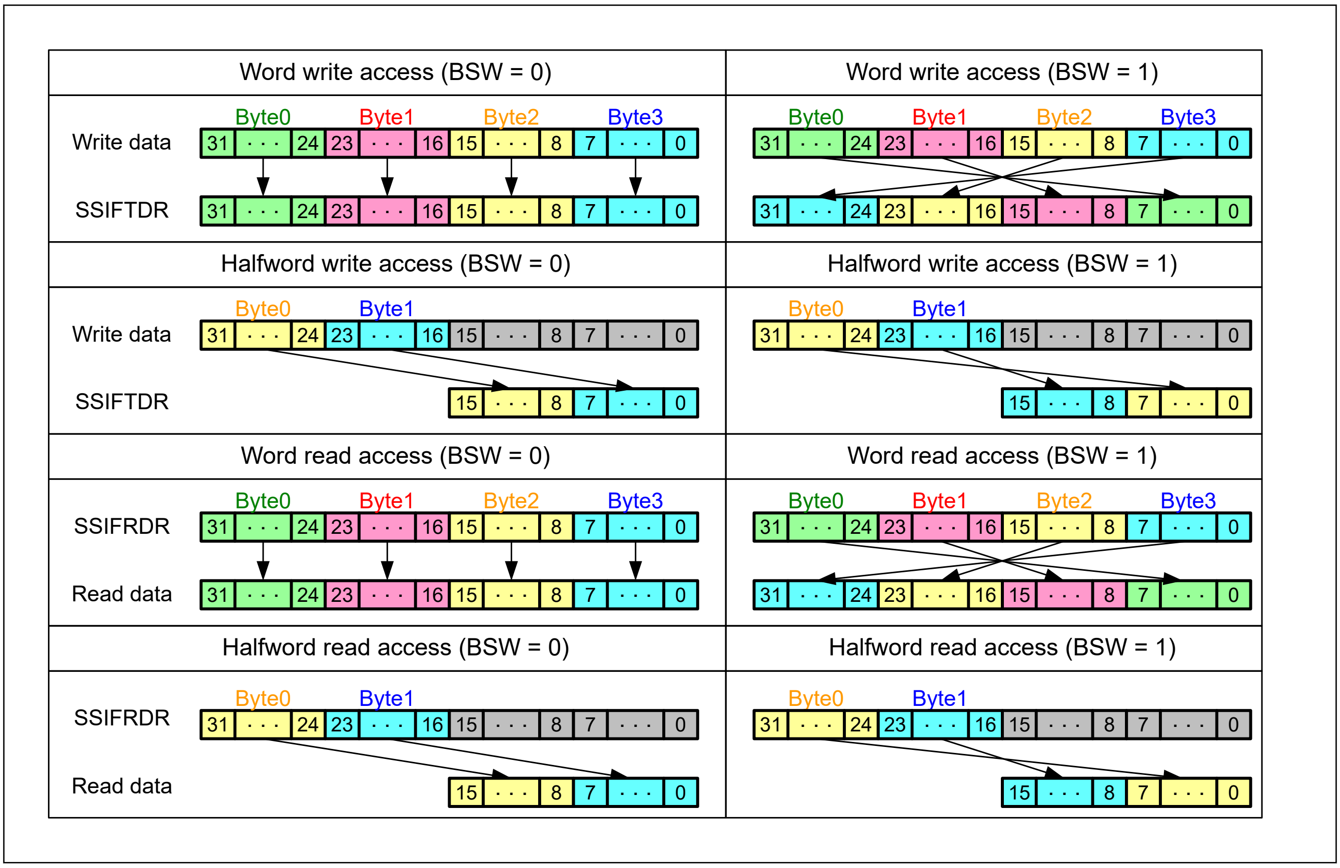


Figure 47.24 Operation example of byte swap

SSIRST bit (Software Reset)

The SSIRST bit sets a software reset of SSIE. Writing 1 to this bit initializes the internal state of SSIE. The register bits subject to the software reset triggered by this bit are indicated by shading in Table 47.9. Because this bit is not automatically cleared after it has been set, write 0 to this bit to release the register bits from the software reset. After writing 0 to this bit, be sure to check that this bit is 0 before starting the next procedural step.

To stop communication of SSIE immediately, after turning off the peripheral functions, write 1 to this bit. Initialization by a software reset is performed without any relation with the bit clock.

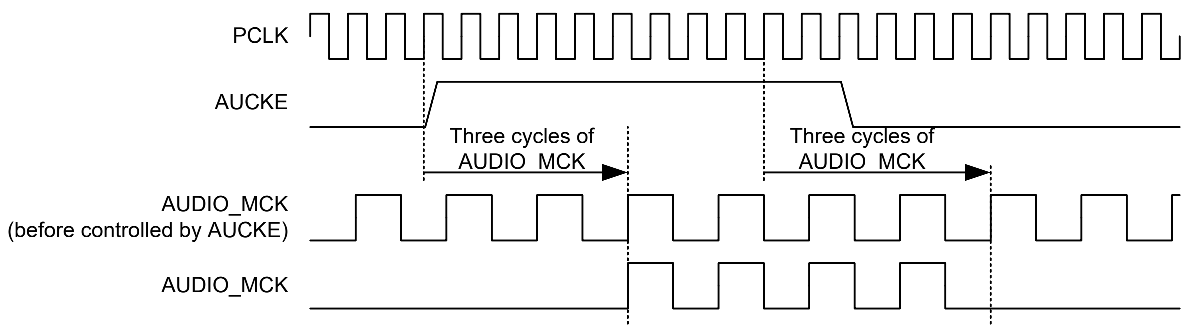
Table 47.9 Bits subject to software reset by the SSIRST bit

Symbol	Address (BASE+)		+0								+1							
			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSICR	0x00	+0	—	CKS	TUIEN	TOIEN	RUIEN	ROIEN	IIEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
		+2	—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL	CKDV[3:0]				MUEN	—	TEN	REN
SSISR	0x04	+0	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
SSIFCR	0x10	+0	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
		+2	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
SSIFSR	0x14	+0	—	—	TDC[5:0]						—	—	—	—	—	—	—	TDE
		+2	—	—	RDC[5:0]						—	—	—	—	—	—	—	RDF
SSIFTDR	0x18	+0	FTDR[31:16]															
		+2	FTDR[15:0]															
SSIFRDR	0x1C	+0	FRDR[31:16]															
		+2	FRDR[15:0]															
SSIOFR	0x20	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	
SSISCR	0x24	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	TDES[4:0]					—	—	—	RDFS[4:0]				

AUCKE bit (AUDIO_MCK Enable in Master mode Communication)

The AUCKE bit enables/disables supply to AUDIO_MCK while in master-mode communication (MST = 1).

Changing the value of this bit must be performed only after specifying the settings related to AUDIO_MCK (by using the CKS, MST, BCKP, and CKDV bits in the SSICR register).



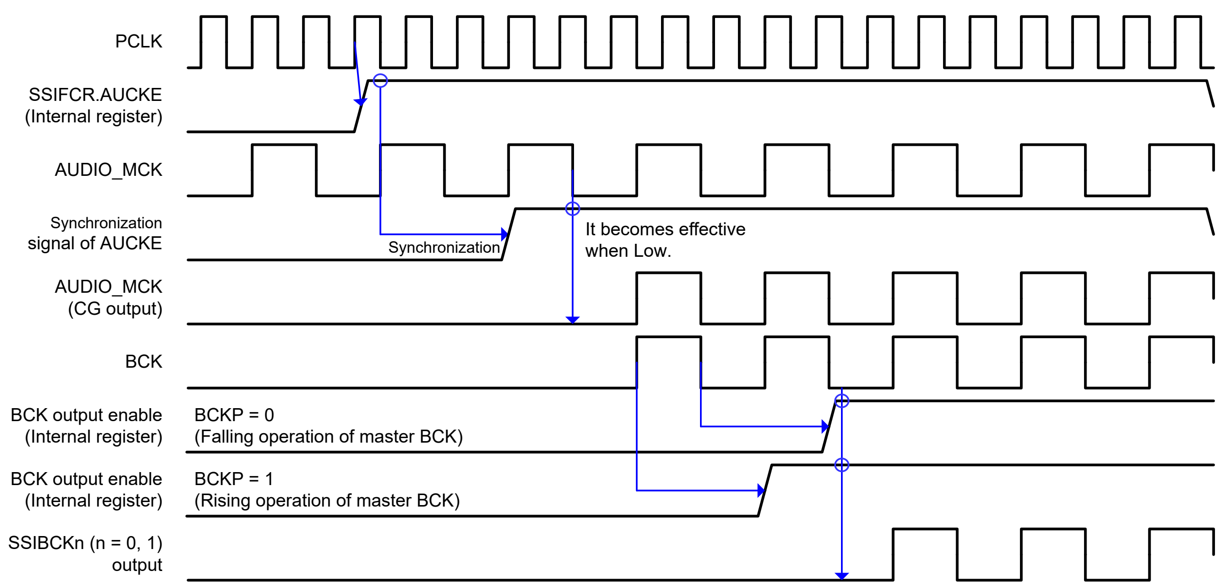
Note: PCLK = PCLKB

Figure 47.25 Stop/resume of AUDIO_MCK

Note: In slave-mode communication (SSICR.MST = 0), SSIE needs supply of SSIBCKn (n = 0, 1). To stop BCK on the master side, make sure that SSIE is in the idle state (SSISR.IIRQ = 1). If BCK is stopped before SSIE becomes idle, take the procedure to start communication in Figure 47.52 or wait for an idle state by taking the procedure to resume communication in Figure 47.58.

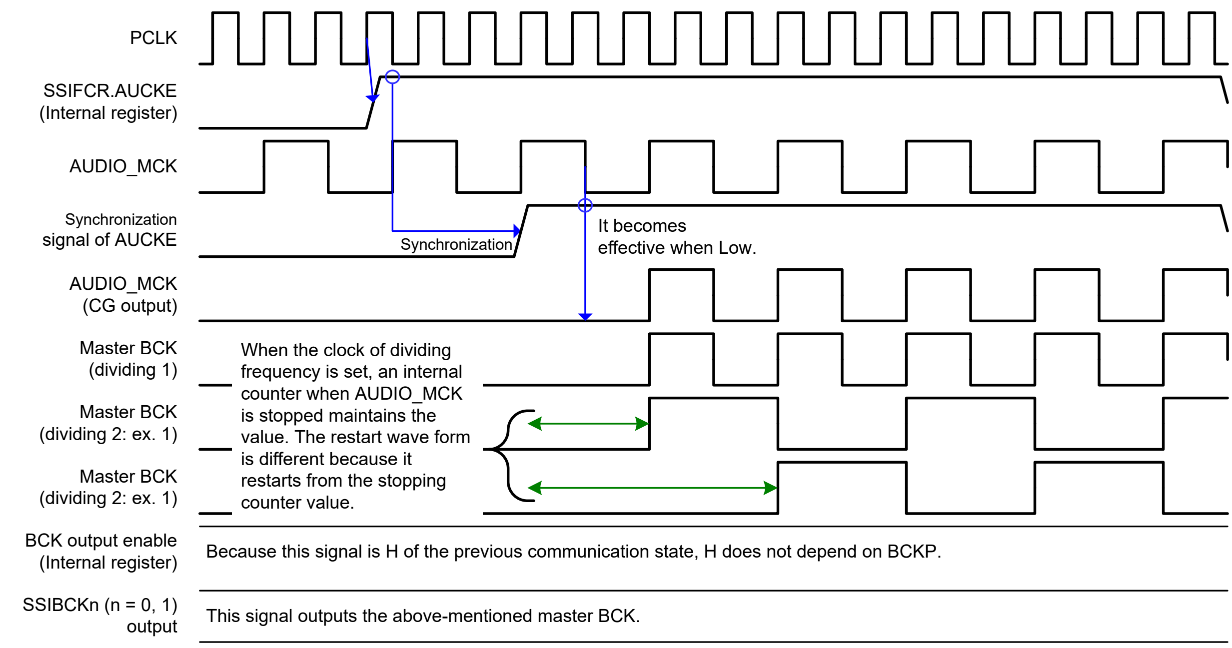
In master-mode communication (SSICR.MST = 1), SSIE operates with the audio clock (AUDIO_MCK). To stop SSIE completely, make sure that SSIE is in the idle state (SSISR.IIRQ = 1) and then write 0 to SSIFCR.AUCKE. If 0 is written to SSIFCR.AUCKE before SSIE becomes idle, take the procedure to start communication in Figure 47.52.

Figure 47.26 and Figure 47.27 show the timings of signal operation in the period from setting this bit to 1 to the output to the SSIBCKn (n = 0, 1) pin.



Note: PCLK = PCLKB

Figure 47.26 Timing diagram for the operation from system reset to start of master-mode communication



Note: PCLK = PCLKB

Figure 47.27 Timing diagram for the operation from stop of communication to start of master-mode communication

Note: If the supply of AUDIO_MCK stops, the value of the SSIBCKn (n = 0, 1) pin is held. Therefore, the SSIBCKn (n = 0, 1) signal might stop in the H (high level) state.

47.2.4 SSIFSR : FIFO Status Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x14

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	TDC[5:0]						—	—	—	—	—	—	—	TDE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	RDC[5:0]						—	—	—	—	—	—	—	RDF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RDF	Receive Data Full Flag 0: The size of received data in SSIFRDR is not more than the value of SSISCR.RDFS. 1: The size of received data in SSIFRDR is not less than the value of SSISCR.RDFS plus one.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
13:8	RDC[5:0]	Number of Receive FIFO Data Indication Flag Number of receive FIFO data indication flag	R
15:14	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
16	TDE	Transmit Data Empty Flag 0: The free space of SSIFTDR is not more than the value of SSISCR.TDES. 1: The free space of SSIFTDR is not less than the value of SSISCR.TDES plus one.	R/W
23:17	—	These bits are read as 0. The write value should be 0.	R/W
29:24	TDC[5:0]	Number of Transmit FIFO Data Indication Flag Number of transmit FIFO data indication flag	R
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

This register is configured with status flags that indicate the status of the transmit FIFO data register and the receive FIFO data register.

RDF flag (Receive Data Full Flag)

The RDF flag indicates that the receive FIFO data register (SSIFRDR) has unread received data not less than the amount set with the SSISCR.RDFS bit plus one. This flag is set by automatic determination, but it must be cleared by register access.

[Priority order for setting and clearing]

Clearing is prioritized.

[Clearing condition]

Either of the following two:*1

1. Writing 0 to this bit after reading 1 from this bit (CPU operation)*2
2. Last access (DTC/DMAC operation) to read data from SSIFRDR by an interrupt routine using the DTC and DMAC.

[Clearing timing]

Clearing timing corresponding to the above clearing condition

1. When 0 is written to this bit after reading 1 from this bit (same as the timing in Figure 47.19)
2. After the PCLKB cycle in which the last access instruction is issued to read data from SSIFRDR by an interrupt routine using the DTC and DMAC.

[Setting condition]

SSIFTDR has free space not less than the amount set with the SSIFCR.TTRG bit plus one.

[Setting timing]

At completion of transfer from the shift register that results in SSIFRDR having data not less than the amount set with the SSISCR.RDFS bit plus one.

Note 1. These bits are cleared by a software reset (SSIFCR.SSIRST = 1) and receive FIFO data register reset (SSIFCR.RFRST = 1). Reset conditions available for these bits are the software reset and receive FIFO data register reset as well as the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following four conditions is met:
- A software reset is done (SSIFCR.SSIRST = 1).
- A receive FIFO data register reset is done (SSIFCR.RFRST = 1).
- After 1 has been read, writing of 0 is complete.
- Last access is performed to read data from SSIFRDR by an interrupt routine using the DTC and DMAC.

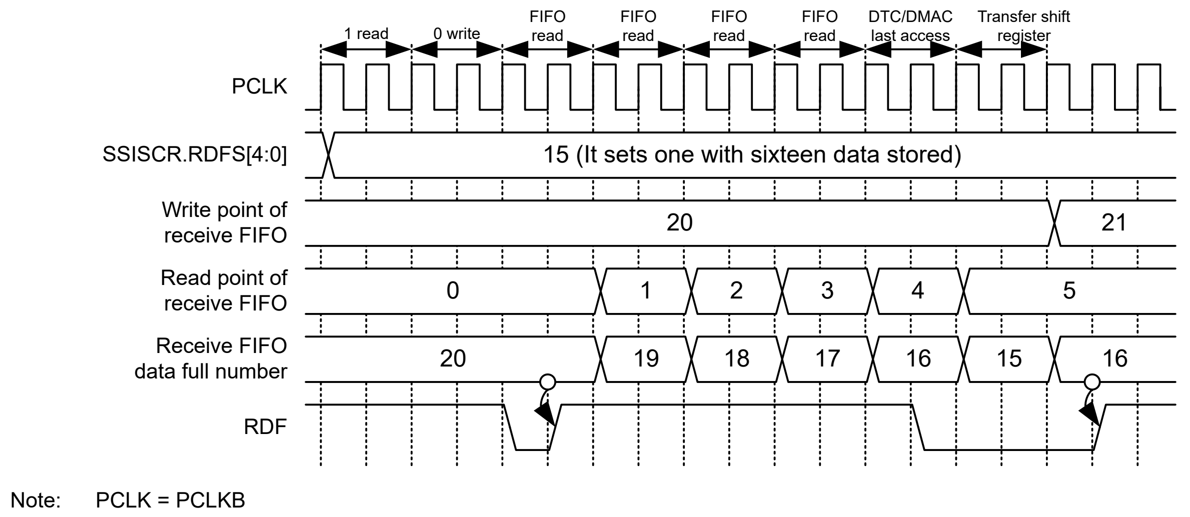


Figure 47.28 Timing diagram for setting and clearing RDF

RDC[5:0] flags (Number of Receive FIFO Data Indication Flag)

The RDC[5:0] flags indicate the number of valid data that are stored in the receive FIFO data register (SSIFRDR). With this flag as 0x00, there is no received data. With 0x20, the register is filled with received data and there is no free space.

TDE flag (Transmit Data Empty Flag)

The TDE flag indicates that the transmit FIFO data register (SSIFTDR) has free space not less than the amount set with the SSIFCR.TTRG bit plus one. This flag is set by automatic determination, but it must be cleared by register access.

[Priority order for setting and clearing]

Clearing is prioritized.*1

[Clearing condition]

Either of the following two:

1. Writing 0 to this bit after reading 1 from this bit (CPU operation)*2
2. Last access (DTC/DMAC operation) to write data to SSIFTDR by an interrupt routine using the DTC and DMAC.

[Clearing timing]

Clearing timing corresponding to the above clearing condition

1. When 0 is written to this bit after reading 1 from this bit (same as the timing in Figure 47.19)
2. Last access (DTC/DMAC operation) to write data to SSIFTDR by an interrupt routine using the DTC and DMAC.

[Setting condition]

SSIFTDR has free space not less than the amount set with the SSIFCR.TTRG bit plus one.

[Setting timing]

While operating on PCLKB, SSIFTDR is found to have free space not less than “size set in the SSISCR.TDES bits + 1.”

Note 1. This bit is cleared by a software reset (SSIFCR.SSIRST = 1) and transmit FIFO data register reset (SSIFCR.TFRST = 1). The software reset and transmit FIFO data register reset have priority over all the clearing conditions described above.

Note 2. After reading 1 from this bit, this bit is cleared when one of the following four conditions is met:
- A software reset is done (SSIFCR.SSIRST = 1).
- A transmit FIFO data register reset is done (SSIFCR.TFRST = 1).
- After 1 has been read, writing of 0 is complete.
- Last access is performed to write data to SSIFTDR by an interrupt routine using the DTC and DMAC.

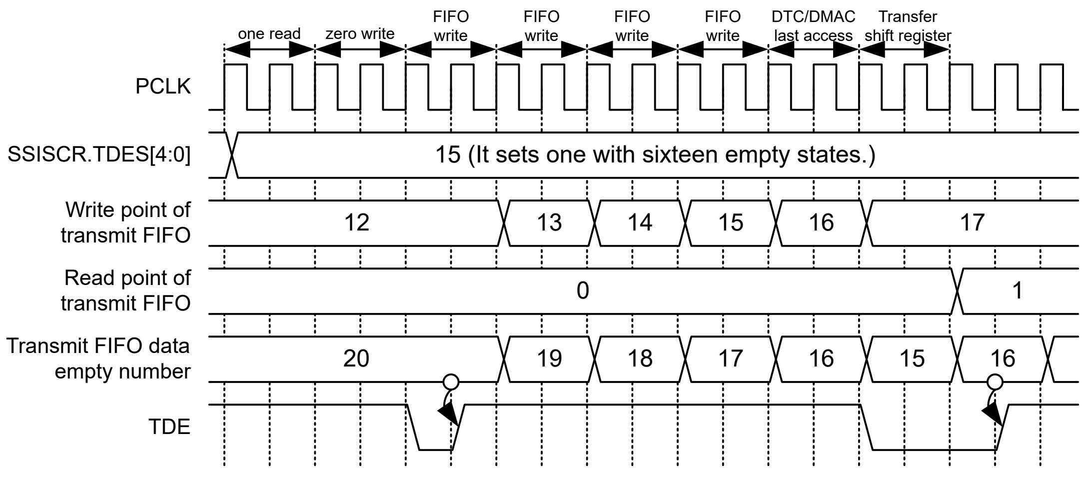


Figure 47.29 Timing diagram for setting and clearing TDE

TDC[5:0] flags (Number of Transmit FIFO Data Indication Flag)

The TDC[5:0] flags indicate the number of valid data that are stored in the transmit FIFO data register (SSIFTDR). With this flag as 0x00, there is no data to be transmitted. With 0x20, there is no space to write data.

47.2.5 SSIFTDR : Transmit FIFO Data Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x18

Bit position:	31 … 0
Bit field:	SSIFTDR[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
31:0	SSIFTDR[31:0]	Transmit FIFO Data	W

Note: S-TYPE-3, P-TYPE-3

This register stores data to be serially transmitted. 0 is returned when this register is read.

When you use this register for transmission, specify data writing to this register as the DTC/DMAC operation that is triggered by a transmit data empty interrupt. Determine the access size to this register according to the data word length to be communicated in Table 47.10.

Table 47.10 Register access restriction to FIFOs (1 of 2)

Access Size				
SSICR.DWL[2:0]	Data Word Length	Byte	Halfword	Word
000b	8	✓	—	—
001b	16	—	✓	—
010b	18	—	—	✓
011b	20	—	—	✓
100b	22	—	—	✓
101b	24	—	—	✓

Table 47.10 Register access restriction to FIFOs (2 of 2)

Access Size				
SSICR.DWL[2:0]	Data Word Length	Byte	Halfword	Word
110b	32	—	—	✓
111b	Setting prohibited	—	—	—

Figure 47.30 shows register access to the transmit FIFO data register.

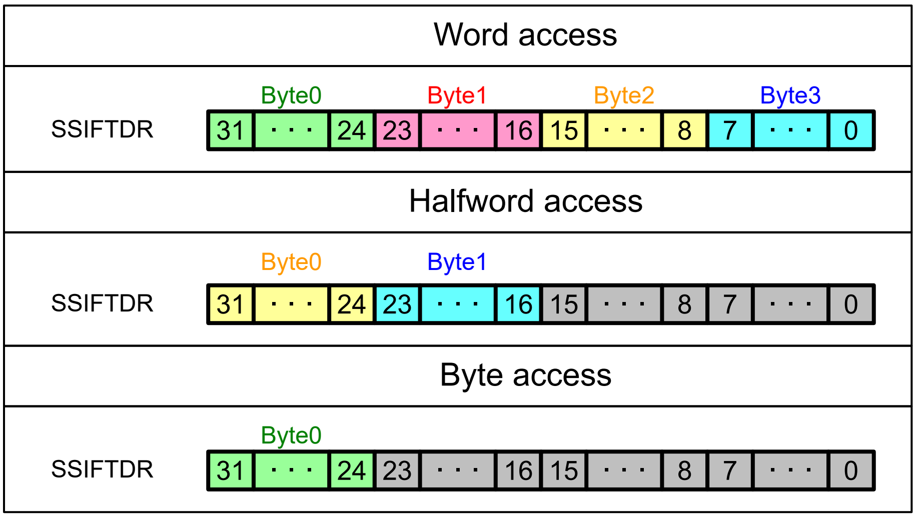


Figure 47.30 Example of register access to the transmit FIFO data register

Figure 47.31 shows the configurations and operation examples of the transmit FIFO data register and transmit shift register. The configurations are for storing data to FIFO and not related with communication.

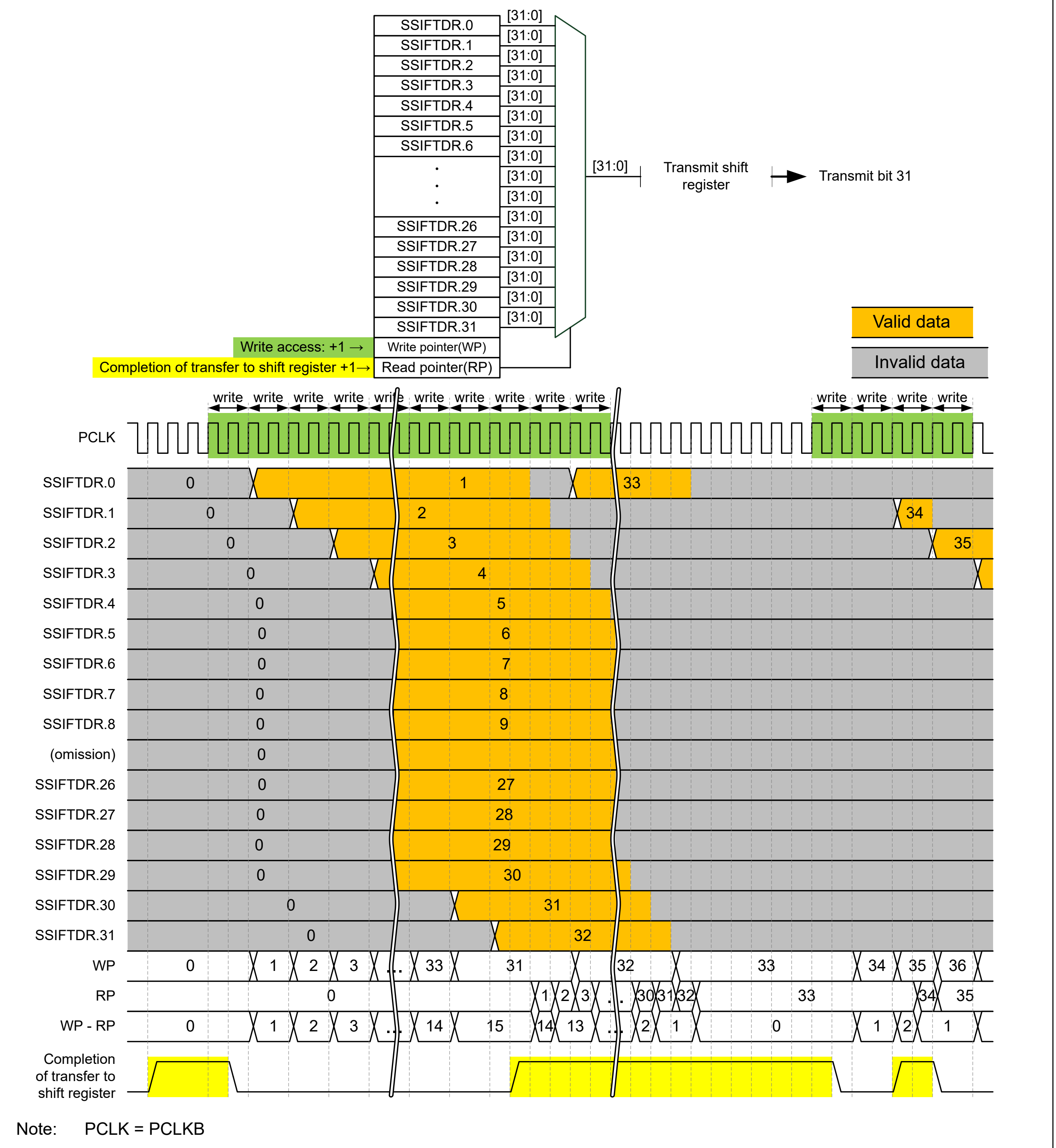


Figure 47.31 Configuration of the transmit FIFO data register and transmit shift register, and FIFO operation example

47.2.6 SSIFRDR : Receive FIFO Data Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x1C

Bit position:	31 … 0
Bit field:	SSIFRDR[31:0]
Value after reset:	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

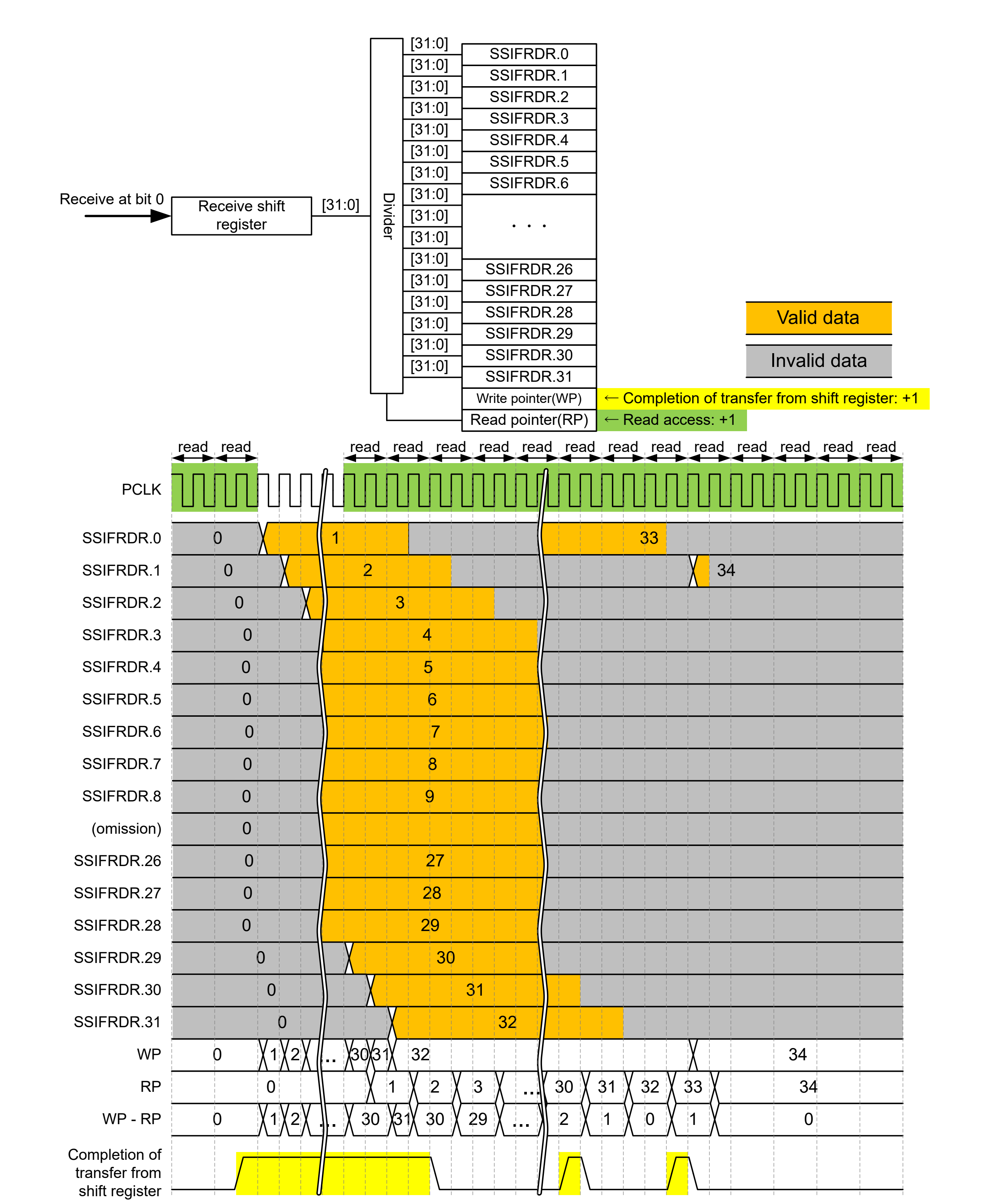
Bit	Symbol	Function	R/W
31:0	SSIFRDR[31:0]	Receive FIFO Data	R

Note: S-TYPE-3, P-TYPE-3

When you use this register for reception, specify data reading from this register as the DTC/DMAC operation that is triggered by a transmit data empty interrupt. Determine the access size to this register according to the data word length to be communicated in Table 47.10.

Register access to the receive FIFO data register is same as for the transmit FIFO data register.

Figure 47.31 shows the configurations and operation examples of the receive FIFO data register and receive shift register.



Note: PCLK = PCLKB

Figure 47.32 Configuration of the transmit FIFO data register and transmit shift register, and FIFO operation example

47.2.7 SSIOFR : Audio Format Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x20

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	OMOD[1:0]	Audio Format Select*3 *4 0 0: I²S format 0 1: TDM format 1 0: Monaural format 1 1: Setting prohibited	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
8	LRCONT	Whether to Enable LRCK/FS Continuation*1 *2 0: Disables LRCK/FS continuation 1: Enables LRCK/FS continuation	R/W
9	BCKASTP	Whether to Enable Stopping BCK Output When SSIE is in Idle Status*1 *2 0: Always outputs BCK to the SSIBCKn (n = 0, 1) pin 1: Automatically controls output of BCK to the SSIBCKn (n = 0, 1) pin	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. This bit is valid only in master-mode communication (SSICR.MST = 1). The setting is invalid in slave-mode communication (SSICR.MST = 0).
Note 2. The BCKASTP and LRCONT bits must not be set to 1 together.
Note 3. While SSIE is communicating (SSISR.IIRQ = 0), writing to these bits is prohibited. If the value of these bits is changed by writing, subsequent operation is unpredictable.
Note 4. If the communication format of other-party device is compatible with a communication format of SSIE, specify and use the communication format that enables communication with the other-party device.

This register is used to set an audio format (which involves the settings of communication format, LR clock/frame synchronization continuation mode, and BCK output stop).

OMOD[1:0] bits (Audio Format Select)

The OMOD[1:0] bits set an audio format. Writing to these bits must be performed when the LR clock supply to the SSILRCKn/SSIFSn (n = 0, 1) pin is stopped. For details about the output of LR clock, see the detailed description of the LRCONT bit in section 47.2.7. SSIOFR : Audio Format Register.

LRCONT bit (Whether to Enable LRCK/FS Continuation)

The LRCONT bit enables or disables the output from SSILRCKn/SSIFSn (n = 0, 1) pin when the communication mode is master-mode communication (SSICR.MST = 1) and SSIE is in the idle state (SSISR.IIRQ = 1).

Even in the idle state, a signal can output from the SSILRCKn/SSIFSn (n = 0, 1) pin when this bit is set to 1 (to enable LR clock/frame synchronization continuation) in master mode (SSICR.MST = 1).

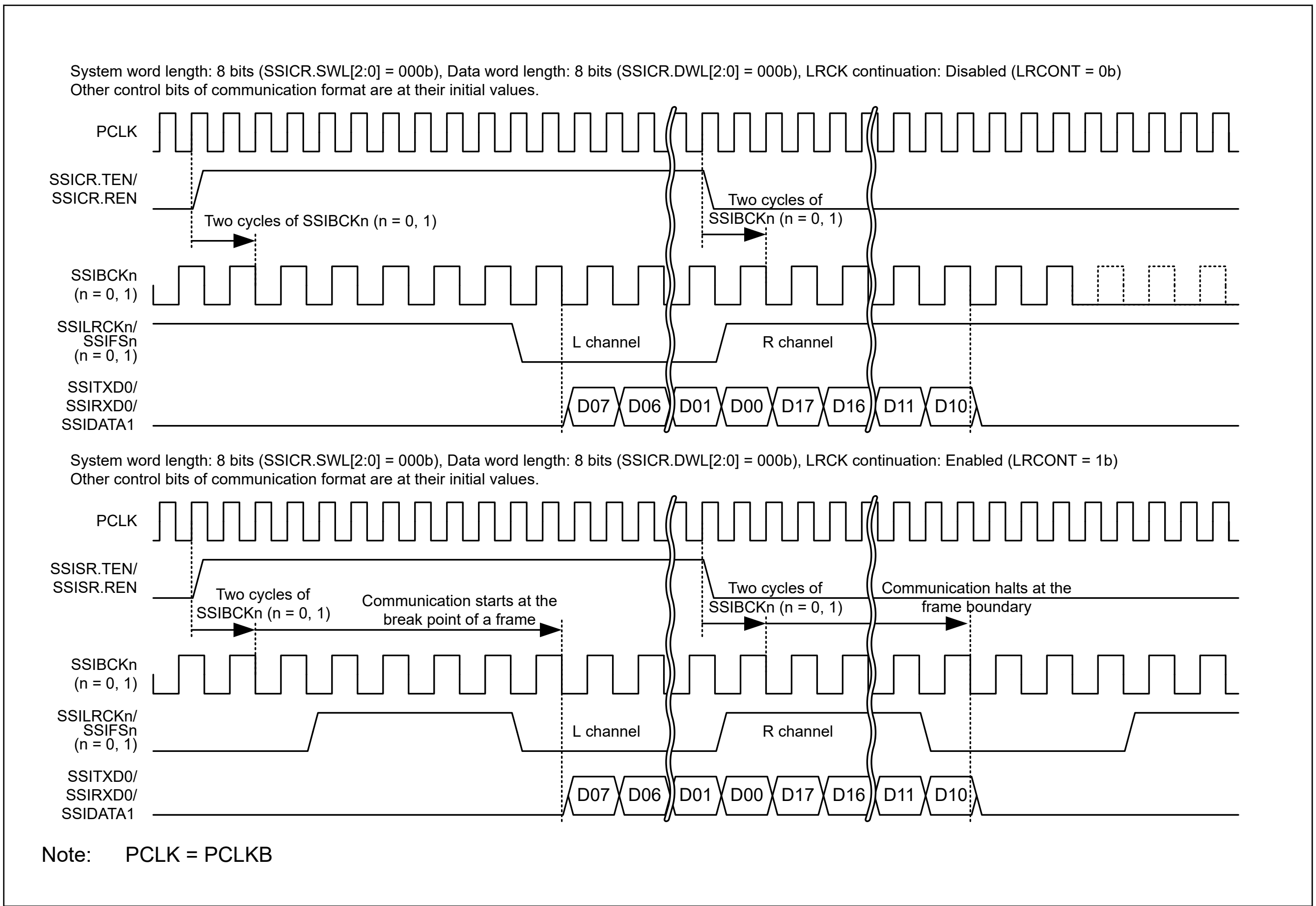


Figure 47.33 Example of LR clock/frame synchronization continuation operation

BCKASTP bit (Whether to Enable Stopping BCK Output When SSIE is in Idle Status)

The BCKASTP bit turns on or off the function to output BCK to the SSIBCKn (n = 0, 1) pin according to the communication shown in Figure 47.34 and Figure 47.35 in master-mode communication (SSICR.MST = 1).

Changing the value of this bit must be performed only after setting the communication format to be used.

This bit must be used in the following way:

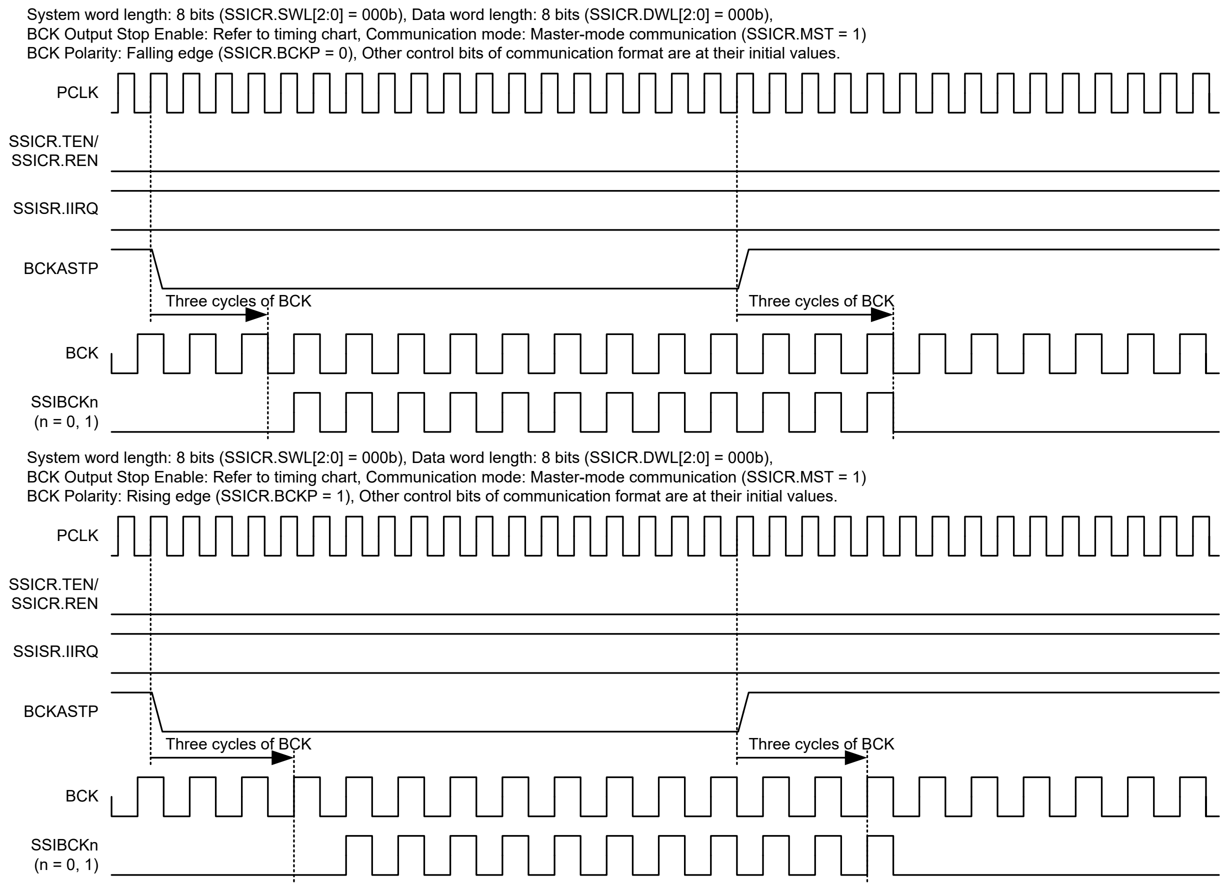
Write 0 to the BCKASTP bit, and then start communication. During the communication, write 1 to the BCKASTP bit. By this operation, the bit clock output to the SSIBCKn (n = 0, 1) pin stops automatically when the communication stops. To resume the communication, set SSIE to the idle state (SSICR.IIRQ = 1), enable the supply of AUDIO_MCK (SSIFCR.AUCKE = 1), and then write 0 to the BCKASTP bit.

When the communication mode is master-mode communication (SSICR.MST = 1) and SSIE is in the idle state (SSICR.IIRQ = 1):

Table 47.11 BCKASTP bit status and SSIBCKn (n = 0, 1) pin output

BCKASTP Bit	SSIBCKn (n = 0, 1) Pin Output Status
0	Output
1	Stopped

Note: The BCKASTP bit cannot be used when the other-party device (which is a slave) requires the clock output from the SSIBCKn (n = 0, 1) pin before and during communication. In such a case, use the BCKASTP bit to stop the clock only after communication. For the timing of enabling the clock stop function, see Figure 47.34.



Note: PCLK = PCLKB

Figure 47.34 Example operation of the BCKASTP bit (idle state)

When the communication mode is master-mode communication (SSICR.MST = 1) and the BCK output stop function is enabled (BCKASTP = 1):

Details of the BCK output to the SSIBCKn (n = 0, 1) pin are as follows:

Output start timing: BCK is output in appropriate timing so that a valid edge is generated when the LR clock/frame synchronization signal shifts to a valid value.

Output stop timing: 1 to 1.5 clock cycles after a frame boundary.

For details about the timings, see the timing diagram in Figure 47.35.

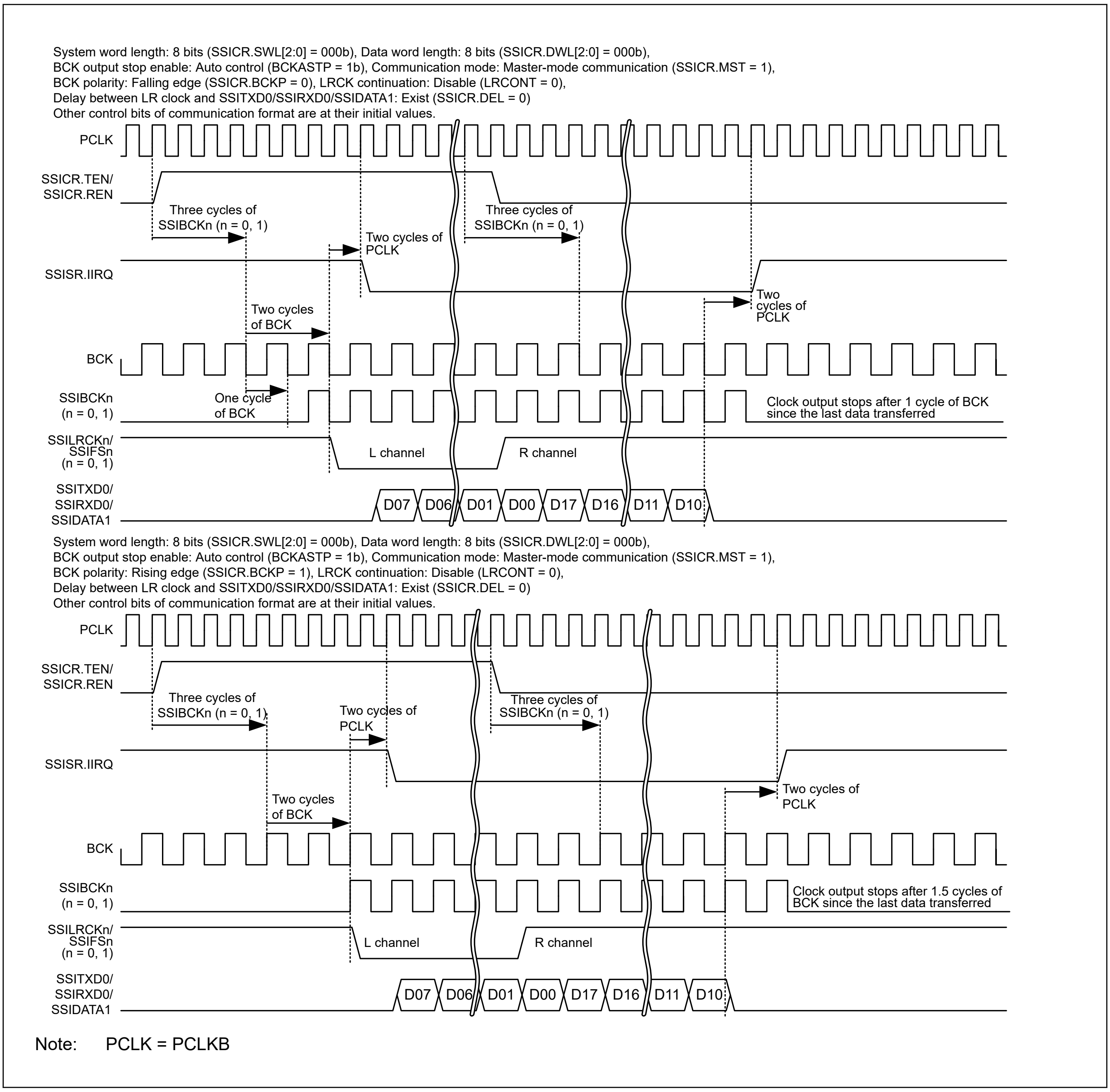


Figure 47.35 Example operation of the BCKASTP bit (communication operation with BCKASTP = 1)

47.2.8 SSISCR : Status Control Register

Base address: SSIEn = 0x4025_D000 + 0x0100 × n (n = 0, 1)
SSIEn_NS = 0x5025_D000 + 0x0100 × n (n = 0, 1)

Offset address: 0x24

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	TDES[4:0]					—	—	—	RDFS[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	RDFS[4:0]	RDF Setting Condition Select*1 0x00: SSIFRDR has one stage or more data size. 0x01: SSIFRDR has two stages or more data size. ⋮ 0x1E: SSIFRDR has thirty-one stages or more data size. 0x1F: SSIFRDR has thirty-two stages or more data size.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
12:8	TDES[4:0]	TDE Setting Condition Select*1 0x00: SSIFTDR has one stage or more free space. 0x01: SSIFTDR has two stages or more free space. ⋮ 0x1E: SSIFTDR has thirty-one stages or more free space. 0x1F: SSIFTDR has thirty-two stages or more free space.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Writing to these bits while SSIE is in a communication state (SSISR.IIRQ = 0) is prohibited. If written, the operation performed immediately after writing is not guaranteed.

RDFS[4:0] bits (RDF Setting Condition Select)

The RDFS[4:0] bits set the setting condition of the receive data full flag (RDF).

TDES[4:0] bits (TDE Setting Condition Select)

The TDES[4:0] bits set the setting condition of the transmit data empty flag (TDE).

47.3 Communication Formats

SSIE supports three communication formats. Table 47.12 shows supported communication formats.

Table 47.12 Supported communication formats

Communication Format	SSIOFR.OMOD[1:0]
I^2S format	00
TDM format	01
Monaural format	10

The following describes the serial data structure shared by communication formats. A serial data structure is defined by the system word length (set in SSICR.SWL[2:0]) and the data word length (set in SSICR.DWL[2:0]). If the data word length is shorter than the system word length, padding bits are transferred in the serial data. For details, see Figure 47.36.

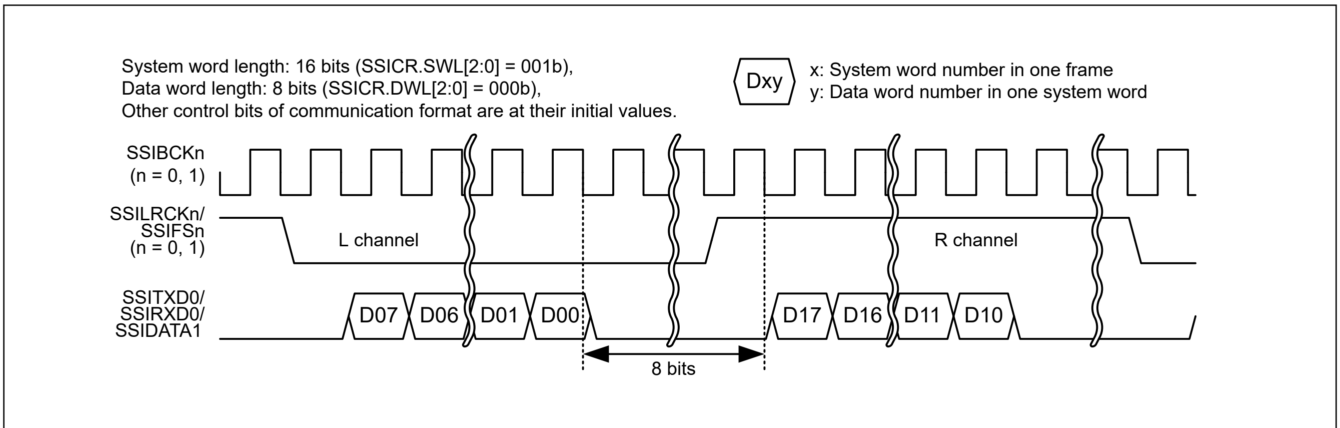


Figure 47.36 Example of padding bit transfer (I^2S format: system word length > data word length)

Table 47.13 lists the number of padding bits to be transferred with each combination of system word length (SSICR.SWL[2:0]) and data word length (SSICR.DWL[2:0]). "-" indicates that the setting is prohibited.

Table 47.13 Number of padding bits

	SSICR.DWL[2:0]	000b	001b	010b	011b	100b	101b	110b	111b
SSICR.SWL[2:0]	System Word Length	8	16	18	20	22	24	32	Setting prohibited
000b	8	0	—	—	—	—	—	—	—
001b	16	8	0	—	—	—	—	—	—
010b	24	16	8	6	4	2	0	—	—
011b	32	24	16	14	12	10	8	0	—
100b	48	40	32	30	28	26	24	16	—
101b	64	56	48	46	44	42	40	32	—
110b	128	120	112	110	108	106	104	96	—
111b	256	248	240	238	236	234	232	224	—

47.3.1 I²S Format

The I²S format is a communication format used for connection with I²S-compatible serial devices. With this format setting (SSIOFR.OMOD[1:0] = 00b), one frame is configured with two system words, one for the channel L and the other for channel R. The SSILRCKn/SSIFSn (n = 0, 1) signals are at a low level for the channel L and at a high level for the channel R. Set the polarity of the signals with the SSICR.LRCKP bit. Figure 47.37 shows the I²S format without padding. See Figure 47.36 for the format with padding.

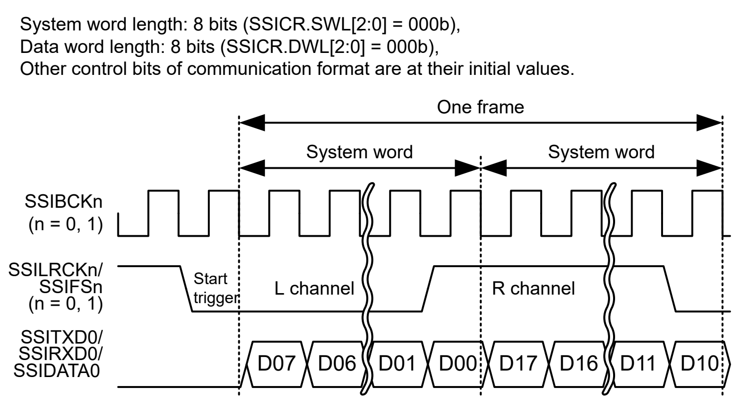


Figure 47.37 I²S format (without padding: system word length = data word length)

For the state of external pins when SSIE is in the idle state, see section 47.5.1. Idle State.

Note: SSIE has the SSILRCKn/SSIFSn (n = 0, 1) pin, which indicates the synchronization of communication. When SSIE is in slave mode (SSICR.MST = 0), the communication format SSIE uses must match that of the other-party device to communicate. SSIE uses the signal input by the SSILRCKn/SSIFSn (n = 0, 1) pin only as a trigger to start communication.

47.3.2 Monaural Format

The monaural format is a communication format used for connection with monaural-compatible serial devices. When the monaural format is specified (SSIOFR.OMOD[1:0] = 10b) for use, one frame consists of one system word. Also, a rising edge of the SSILRCKn/SSIFSn (n = 0, 1) signal indicates a communication start trigger. Figure 47.38 and Figure 47.39 respectively show the monaural formats without and with padding.

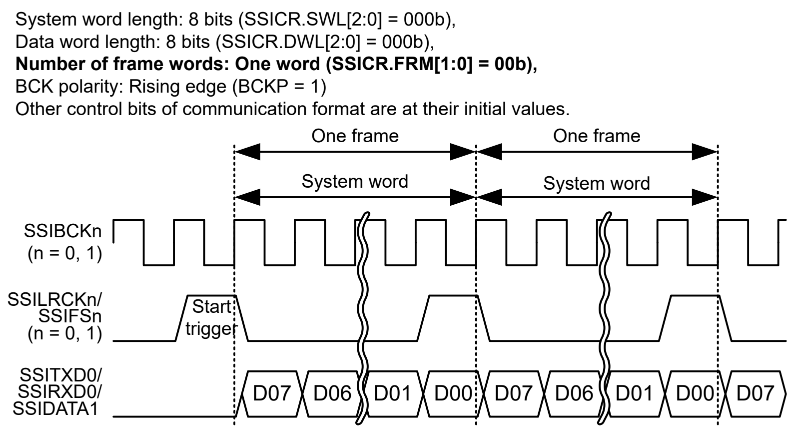


Figure 47.38 Short frame in monaural format (without padding: system word length = data word length)

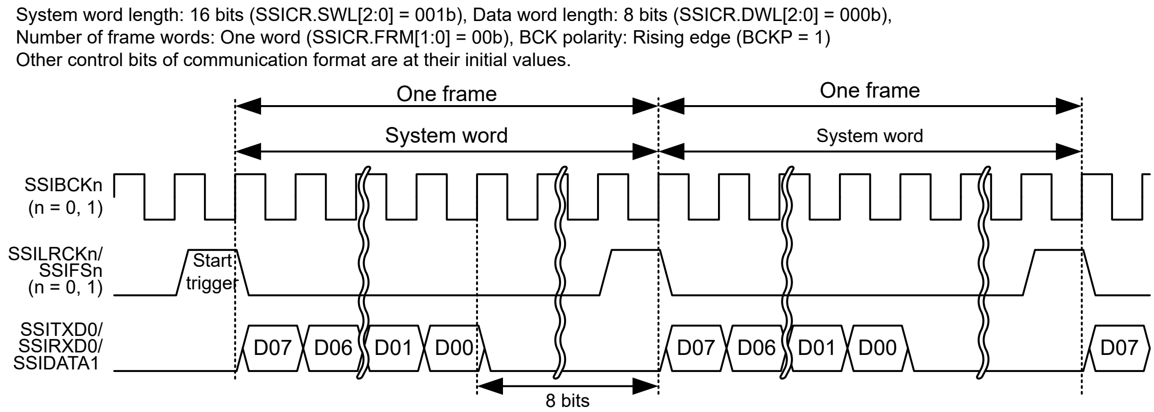


Figure 47.39 Short frame in monaural format (with padding: system word length > data word length)

The monaural formats supported by SSIE consist of short frames and long frames. See section 47.3.2.1. Short Frame and section 47.3.2.2. Long Frame for the difference between these two frames.

For the state of external pins state when SSIE is in the idle state, see section 47.5.1. Idle State.

Note: SSIE has the SSILRCKn/SSIFSn (n = 0, 1) pin, which indicates the synchronization of communication. When SSIE is in slave mode (SSICR.MST = 0), the communication format SSIE uses must match that of the other-party device to communicate. SSIE uses the signal input by the SSILRCKn/SSIFSn (n = 0, 1) pin only as a trigger to start communication.

47.3.2.1 Short Frame

When a short frame is used (SSICR.DEL = 0), the SSILRCKn/SSIFSn (n = 0, 1) signal indicating the start of serial data is set to high level only for 1 cycle of SSIBCKn (n = 0, 1). Data transfer starts at the falling edge of the signal.

47.3.2.2 Long Frame

When a long frame is used (SSICR.DEL = 1), the SSILRCKn/SSIFSn (n = 0, 1) signal indicating the start of serial data is set to high level only for 2 cycles of SSIBCKn (n = 0, 1). See Figure 47.40. Data transfer starts at the rising edge of the signal.

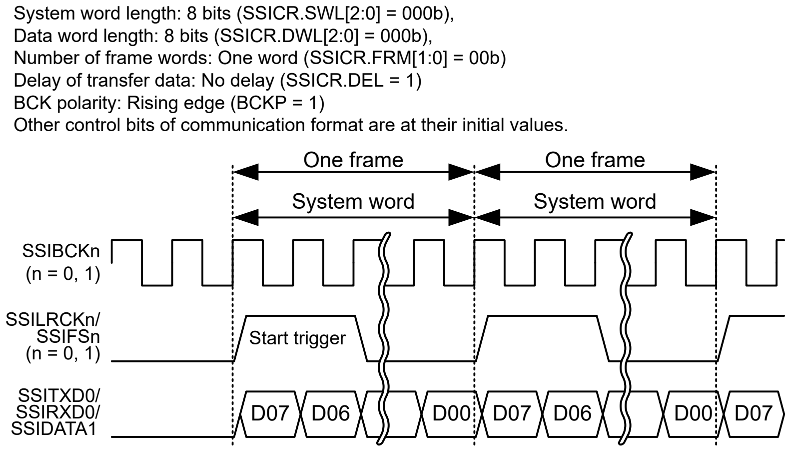


Figure 47.40 Long frame in monaural format (without padding)

47.3.3 TDM Format

The TDM format is a communication format used for connection with TDM-compatible multi-channel devices. With this format setting (SSIOFR.OMOD[1:0] = 01b), one frame is configured with four to eight system words set with the SSICR.FRM[1:0] bits. With this format, the SSILRCKn/SSIFSn (n = 0, 1) signal is at a high level for the first one system word and at a low level for the rest. The pulse generated on the SSILRCKn/SSIFSn (n = 0, 1) signal is defined as the SYNC pulse and its rising edge means a start of one frame. Figure 47.41 and Figure 47.42 respectively show the TDM formats without and with padding.

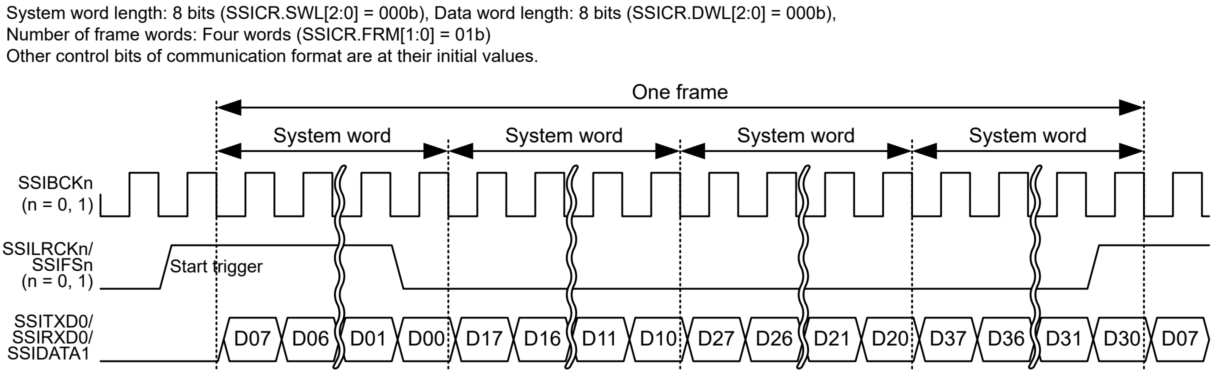


Figure 47.41 TDM format (without padding: system word length = data word length)

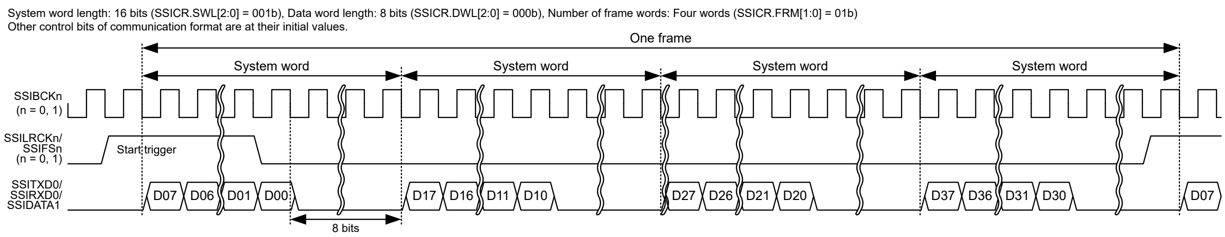


Figure 47.42 TDM format (with padding: system word length > data word length)

For the state of external pins when SSIE is in the idle state, see section 47.5.1. Idle State.

Note: SSIE has the SSILRCKn/SSIFSn (n = 0, 1) pin, which indicates the synchronization of communication. When SSIE is in slave mode (SSICR.MST = 0), the communication format SSIE uses must match that of the other-party device to communicate. SSIE uses the signal input by the SSILRCKn/SSIFSn (n = 0, 1) pin only as a trigger to start communication.

47.4 Communication Modes

SSIE supports the following communication modes. Table 47.15 lists the control bits that are not available with each communication mode. See section 47.4.1. Slave-mode Communication to section 47.4.5. Transmission and Reception for details of these communication modes.

Table 47.14 Communication modes

Communication Mode	SSICR.MST Bit	SSICR.REN Bit	SSICR.TEN Bit
Slave-mode transmission	0	0	1
Slave-mode reception	0	1	0
Slave-mode transmission and reception	0	1	1
Master-mode transmission	1	0	1
Master-mode reception	1	1	0
Master-mode transmission and reception	1	1	1

Table 47.15 Control bits that cannot be used in each communication mode

Control Bit	Slave-mode Reception	Slave-mode Transmission	Slave-mode Transmission and Reception	Master-mode Reception	Master-mode Transmission	Master-mode Transmission and Reception
SSICR.CKS	Invalid	Invalid	Invalid	Available	Available	Available
SSICR.CKDV	Invalid	Invalid	Invalid	Available	Available	Available
SSICR.MUEN	Invalid	Available	Available	Invalid	Available	Available
SSICR.TEN	Invalid	Available	Available	Invalid	Available	Available
SSICR.REN	Available	Invalid	Available	Available	Invalid	Available
SSIFCR.AUCKEN	Invalid	Invalid	Invalid	Available	Available	Available
SSIFCR.TIE	Invalid	Available	Available	Invalid	Available	Available
SSIFCR.RIE	Available	Invalid	Available	Available	Invalid	Available
SSIFCR.TFRST	Invalid	Available	Available	Invalid	Available	Available
SSIFCR.RFRST	Available	Invalid	Available	Available	Invalid	Available
SSIOFR.BCKASTP	Invalid	Invalid	Invalid	Available	Available	Available
SSIOFR.LRCONT	Invalid	Invalid	Invalid	Available	Available	Available
SSIOFR.OMOD	Available	Available	Available	Available	Available	Available
SSISCR.TDES	Invalid	Available	Available	Invalid	Available	Available
SSISCR.RDFS	Available	Invalid	Available	Available	Invalid	Available

“Invalid” means it has no effect on operation. Writing is possible.

47.4.1 Slave-mode Communication

SSIE operates in slave mode with SSICR.MST = 0. The SSIBCKn (n = 0, 1) and SSILRCKn/SSIFSn (n = 0, 1) signals to be used for serial-data communication must be supplied from an external device. If these signals do not match the communication format set for SSIE, operation is not guaranteed.

47.4.2 Master-mode Communication

SSIE operates in master mode with SSICR.MST = 1. The SSIBCKn (n = 0, 1) and SSILRCKn/SSIFSn (n = 0, 1) signals to be used for serial data communication must be internally generated from the audio clock. These signals use the format according to the setting of SSIE. If the communication format the slave device uses does not match the communication format set for SSIE, the operation is unpredictable.

47.4.3 Transmission

SSIE transmits serial data to the other-party device when the SSICR.TEN bit is 1 and the SSICR.REN bit is 0. If the communication format the other-party device uses does not match the communication format set for SSIE, the operation is unpredictable.

47.4.4 Reception

SSIE receives serial data from the other-party device when the SSICR.TEN bit is 0 and the SSICR.REN bit is 1. If the communication format the other-party device uses does not match the communication format set for SSIE, the operation is unpredictable.

47.4.5 Transmission and Reception

SSIE transmits and receives serial data to and from the other-party device when the SSICR.TEN bit is 1 and the SSICR.REN bit is 1. If the communication format the other-party device uses does not match the communication format set for SSIE, the operation is unpredictable.

47.5 Operation

SSIE has the following two main operation states Figure 47.43 shows SSIE state transition.

- Idle state (SSISR.IIRQ = 1)
- Communication state (SSISR.IIRQ = 0).

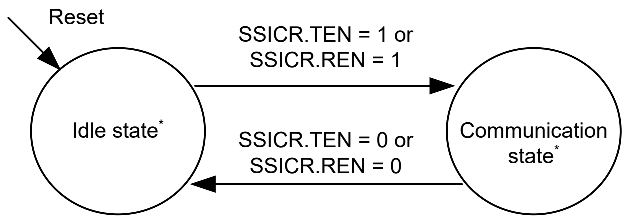


Figure 47.43 SSIE state transition

Note: See section 47.6.1. Start Communication for details of the idle state.
See section 47.6.2. Transmission for details of the communication state.

47.5.1 Idle State

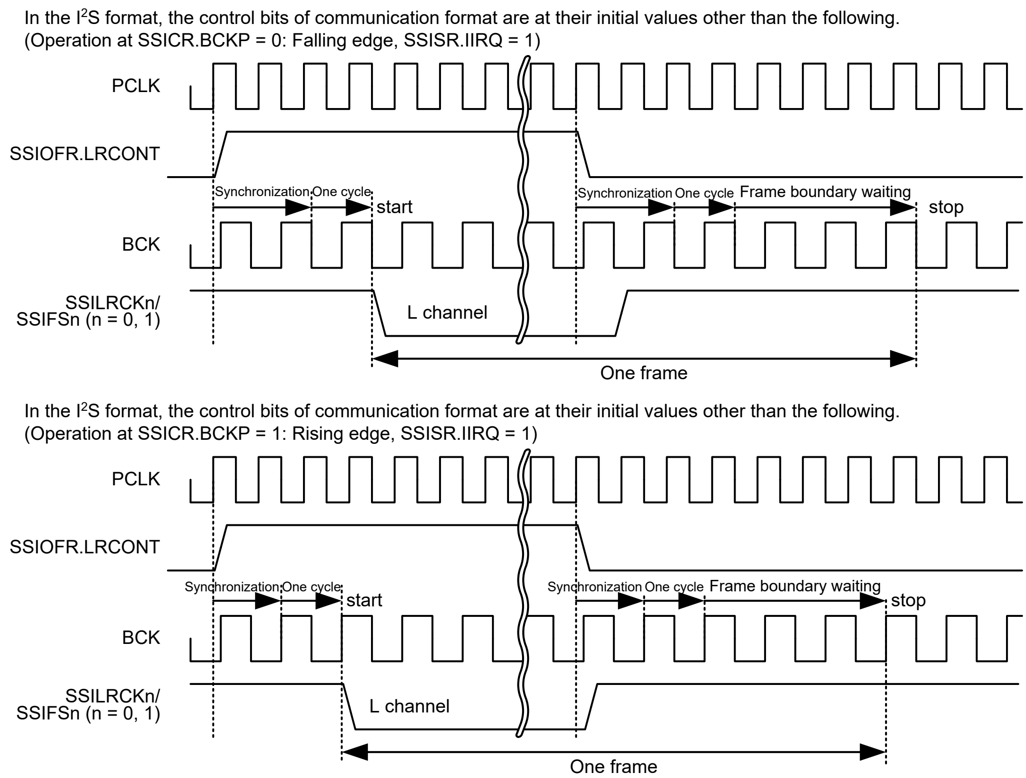
In this state, communication of SSIE is halted. If, however, the SSICR.MST bit is 1, output of the BCK and LR clock/ frame synchronization signals to external pins can be controlled according to the settings of SSIOFR.BCKASTP and SSIOFR.LRCONT bits. This function is common to all formats. For details, see Table 47.16.

Table 47.16 Output from external pins in the idle state (1 of 2)

SSICR.MST	SSIOFR.BCKASTP	SSIOFR.LRCONT	Output from Pins		
			SSIBCKn (n = 0, 1)	SSILRCKn/SSIFSn (n = 0, 1)	SSITXD0/SSIDATA1
0	—	—	Stop	Stop	Stop

Table 47.16 Output from external pins in the idle state (2 of 2)

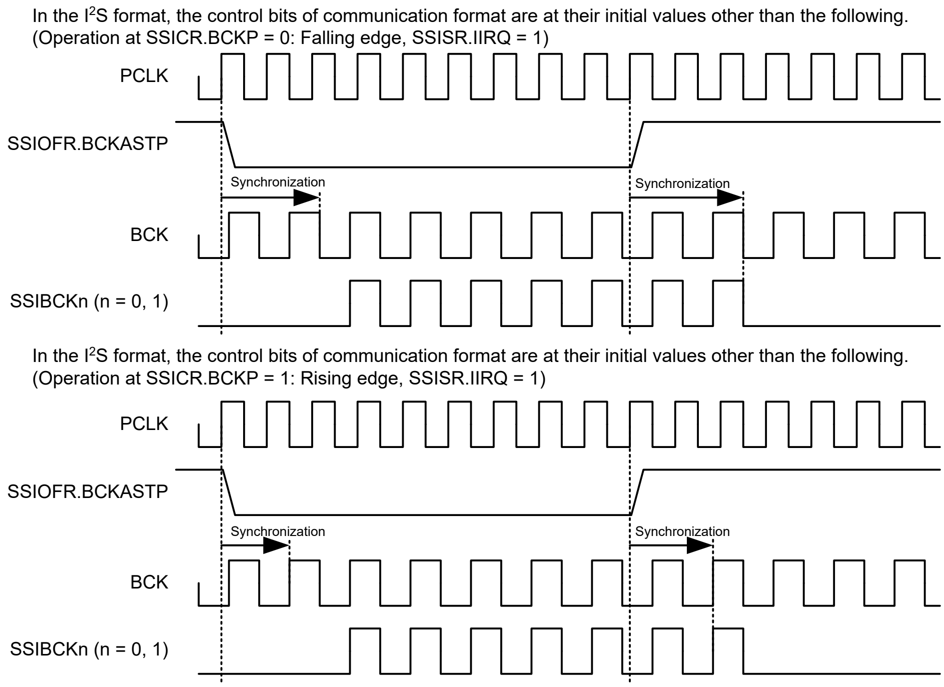
SSICR.MST	SSIOFR.BCKASTP	SSIOFR.LRCONT	Output from Pins		
			SSIBCKn (n = 0, 1)	SSILRCKn/SSIFSn (n = 0, 1)	SSITXD0/SSIDATA1
1	0	0	Supply	Stop	Stop
1	0	1	Supply	Supply	Stop
1	1	0	Stop	Stop	Stop
1	1	1	Stop	Supply	Stop



Note: PCLK = PCLKB

Figure 47.44 Example of disabling LR clock/frame synchronization continuation by SSIOFR.LRCONT

Note: To stop the output to the SSILRCKn/SSIFSn (n = 0, 1) pin with SSIOFR.LRCONT when SSIE is in the idle state in master-mode communication (SSICR.MST = 1), note the following: The output stops when the value of the SSIOFR.LRCONT bit is changed from 1 to 0. Make sure that the other-party device is not affected.



Note: PCLK = PCLKB

Figure 47.45 Example of stopping SSIBCKn (n = 0, 1) with SSIOFR.BCKASTP

Note: To stop the output to the SSIBCKn (n = 0, 1) pin with SSIOFR.BCKASTP in master-mode communication (SSICR.MST = 1) and while SSIE is in the idle state, note the following: The output stops when the value of the SSIOFR.BCKASTP bit is changed from 0 to 1. So, make sure that the other-party device is not affected.

47.5.2 Communication States

In this state, SSIE is during communication. Figure 47.46 shows transitions of communication states and Table 47.17 lists the conditions for transition. If the transition condition is not satisfied, the state does not transit.

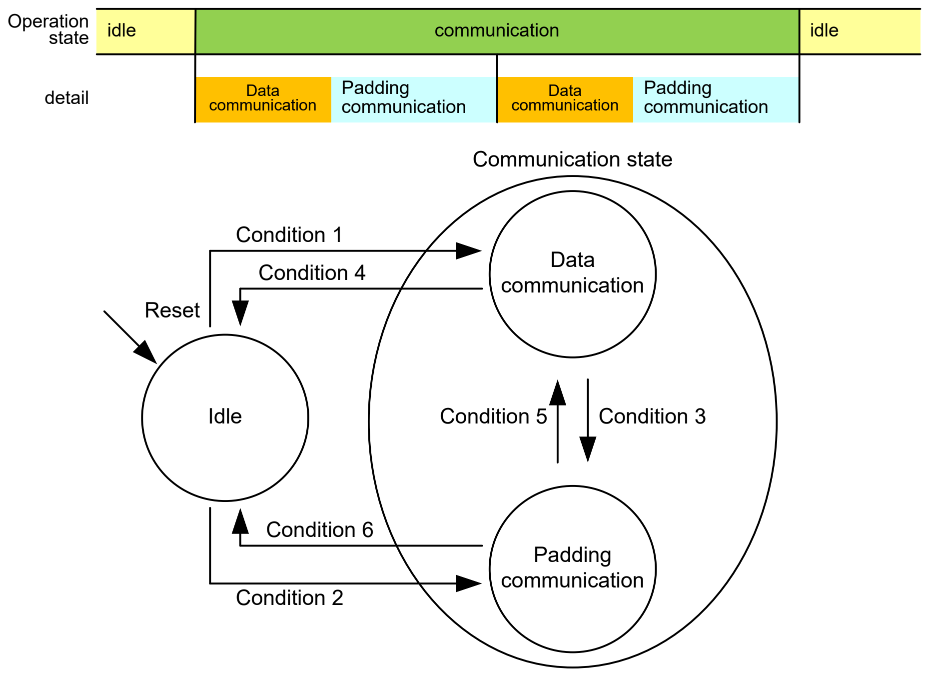


Figure 47.46 Communication state transition

Table 47.17 Condition for communication state transition

Condition Number	Condition for Transition
1	Writing SSICR.TEN = 1 or SSICR.REN = 1 while SSICR.SDTA = 0 or in the setting without padding bits.
2	Writing SSICR.TEN = 1 or SSICR.REN = 1 while SSICR.SDTA = 1 and in the setting with padding bits.
3	The following three conditions are all met: ● SSICR.TEN = 1 or SSICR.REN = 1 ● In the setting with padding bits ● The last bit of the data words has been transferred.
4	Both the following two conditions are met: ● SSICR.SDTA = 1 or without padding bits ● While SSICR.TEN = 0 and SSICR.REN = 0, the last bit of the data words in a frame has been transferred.
5	Transfer of the last padding bit is completed while SSICR.TEN = 1 or SSICR.REN = 1
6	Both the following two conditions are met: ● SSICR.SDTA = 0 and with padding bits ● While SSICR.TEN = 0 and SSICR.REN = 0, the last padding bit has been transferred.

See Table 47.13 for the setting with/without padding bits.

47.5.2.1 Data Communication State

In this state, SSIE is during communication. Data of the data word length set with the SSICR.DWL[2:0] bits is transmitted, received, or transmitted and received.

- State Transition in the Setting without Padding Bits

During communication (SSISR.IIRQ = 0), SSIE is during data communication for all the time. By disabling transmission and reception (SSICR.TEN = 0, SSICR.REN = 0), SSIE transits to the idle state. For details, see Figure 47.47 and Figure 47.48.

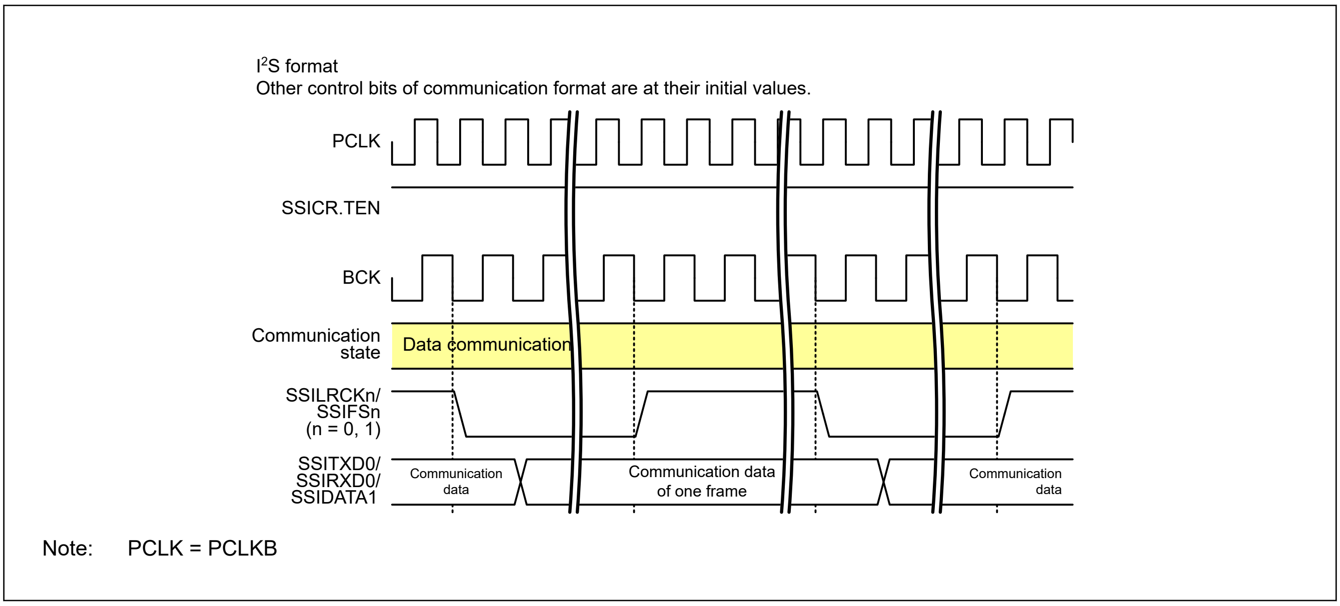


Figure 47.47 Continuation of the data communication

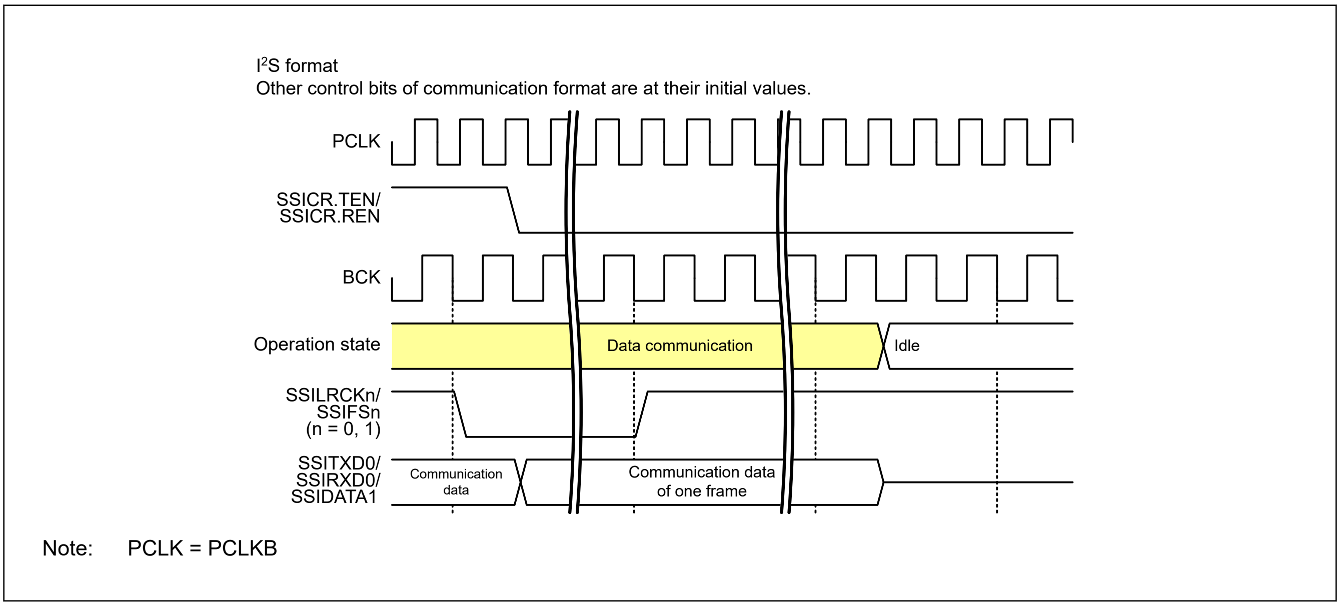
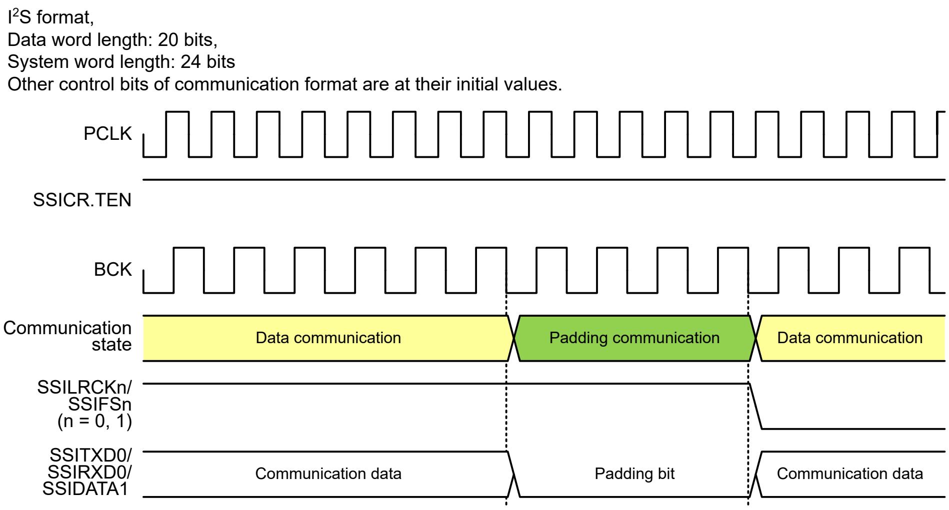


Figure 47.48 Halt from the data communication (without padding bits)

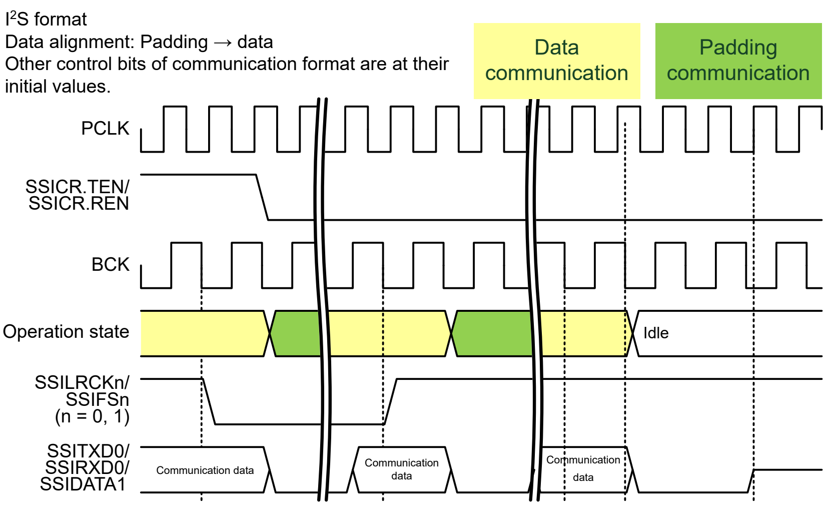
- State Transition in the Setting with Padding Bits

When SSIE ends transfer of the last bit of a data word during communication (SSISR.IIRQ = 0), SSIE transitions from the data communication state to the padding communication state in Figure 47.49. Except in the status with SSICR.SDTA = 1 and transmission and reception disabled (SSICR.TEN = 0 and SSICR.REN = 0), SSIE transitions from the data communication state to the idle state when it stops communication in Figure 47.51.



Note: PCLK = PCLKB

Figure 47.49 Transition from data communication to padding communication



Note: PCLK = PCLKB

Figure 47.50 Halt from data communication (with padding bits)

47.5.2.2 Padding Communication

In this state, SSIE is during communication. The padding bits set with the SSICR.SWL[2:0] bits and SSICR.DWL[2:0] bits are transmitted, received, or transmitted and received.

- State Transition in the Setting with Padding Bits

When SSIE ends transfer of the last padding bit during communication (SSISR.IIRQ = 0), SSIE transitions to the data communication state in Figure 47.49. If SSIE is in the status with SSICR.SDTA = 0 and transmission and reception disabled (SSICR.TEN = 0 and SSICR.REN = 0), SSIE transitions from the padding communication state to the idle state when it stops communication in Figure 47.51.

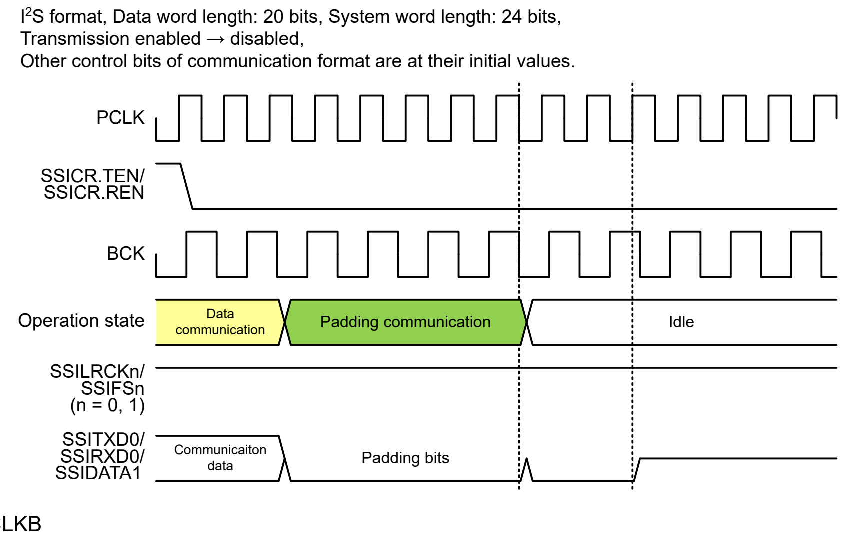


Figure 47.51 Halt from the padding communication

47.6 Communication Operation

Figure 47.52 shows the communication flow of SSIE.

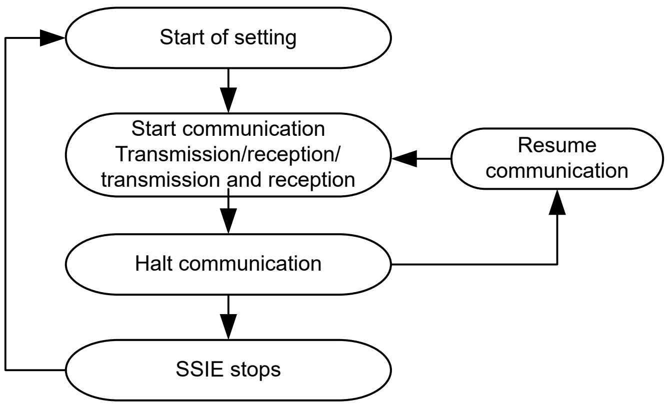
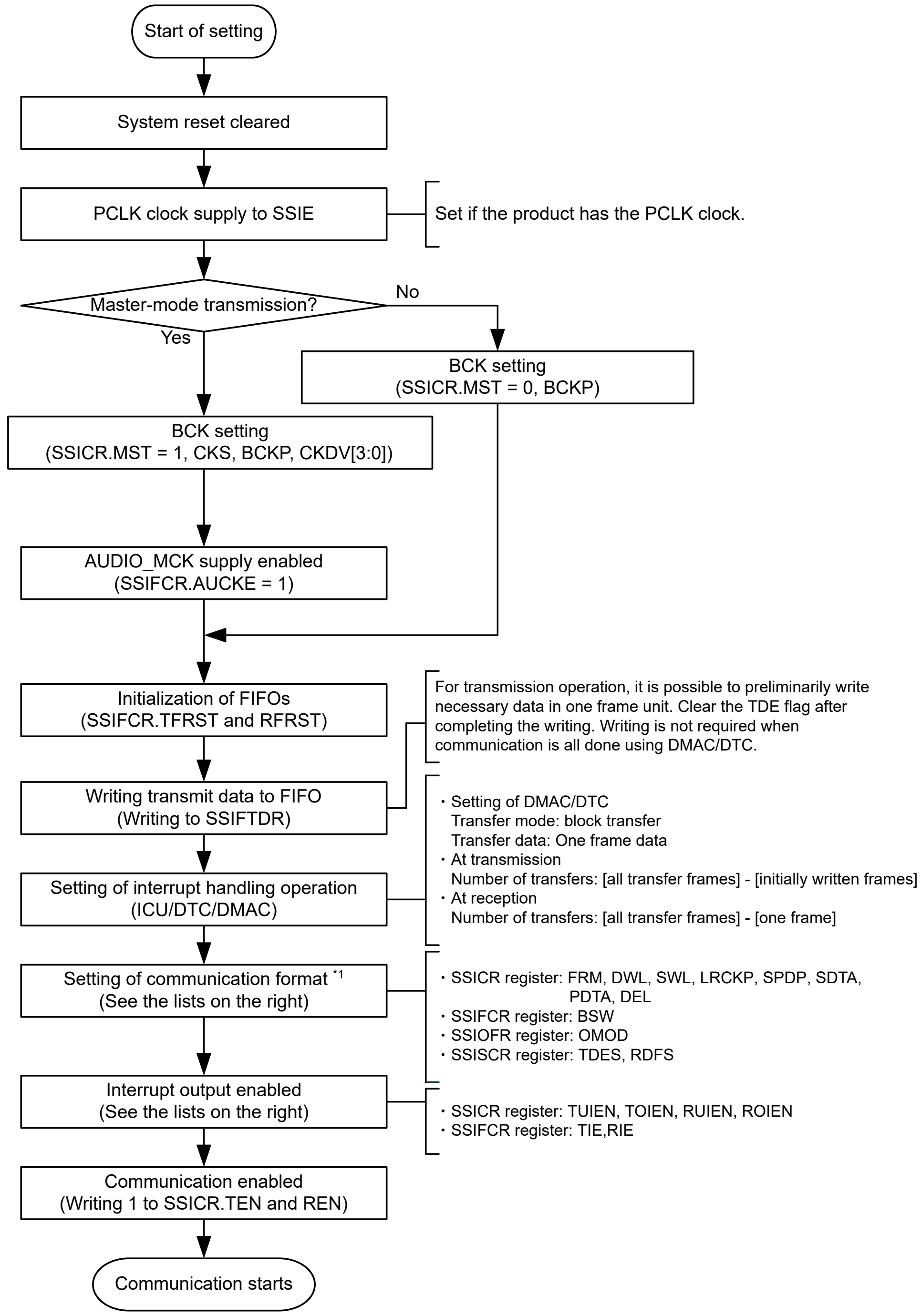


Figure 47.52 SSIE communication operation

The procedure of each operation is described from section 47.6.1. Start Communication to section 47.6.7. Resume Communication.

47.6.1 Start Communication

This section describes how to start communication of SSIE. Figure 47.53 shows the procedure to start communication. Be sure to follow the procedure. See section 47.6.2. Transmission for transmission operation and section 47.6.3. Reception for reception operation.



Note: PCLK = PCLKB

Note 1. Set SSICR.MUEN, SSIOFR.BCKASTP, and SSIOFR.LRCONT after completing the setting of the communication format.

Figure 47.53 Procedure to start communication (CPU operation procedure)

SSIE can perform continuous communication based on interrupts by the DTC/DMAC. For transmission, write 1 to SSIFCR.TIE, SSICR.TUIEN, and SSICR.TOIEN. For reception, write 1 to SSIFCR.RIE, SSICR.RUIEN, and SSICR.ROIEN.

47.6.2 Transmission

The transmission procedure in Figure 47.54 must be followed throughout a transmission operation.

After transmission is enabled (SSICR.TEN = 1 and SSICR.REN = 0), SSIE starts transmission when a start trigger is generated by SSILRCKn/SSIFSn (n = 0, 1) with the serial data for at least a frame contained in the transmit FIFO data register (SSIFTDR). SSIE outputs a transmit data empty interrupt to the DTC/DMAC according to the TDE setting condition (SSISCR.TDES) and the status of transmit data empty interrupt enable (SSIFCR.TIE) bit specified in the communication start procedure. This interrupt requests writing to the transmit FIFO data register (SSIFTDR). In the communication start procedure, specify writing to the transmit FIFO data register (SSIFTDR) as the DTC/DMAC operation in response to the transmit data empty interrupt. With this setting, SSIE can continuously transmit data not through the CPU. The transmit data empty interrupt is generated when the free space size of transmit FIFO data register reaches the value set in SSISCR.TDES. The number of times of writing must be specified in accordance with the free space size of the transmit FIFO data register indicated by the transmit data empty interrupt. If an error occurs, perform the error-handling procedure as instructed in the communication stop procedure.

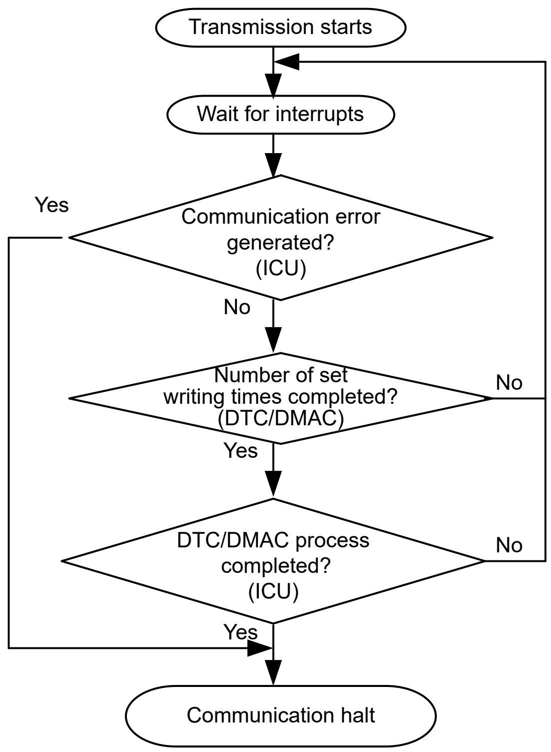


Figure 47.54 Transmission procedure

Note: The communication flow defined in SSIE uses the DTC/DMAC. If you do not use the DTC/DMAC, perform polling of the value 1 of SSIFSR.TDE to write data to SSIFTDR. The number of times of writing data to SSIFTDR by detecting the value 1 of SSIFSR.TDE must be in accordance with the free space size of the transmit FIFO data register specified by SSISCR.TDES. After as much transmit data as the free space size is written to SSIFTDR, the SSIFSR.TDE flag must be cleared. Continuous transmission is enabled by repeating data writing. If the SSIFSR.TDE flag is not cleared, the flag is not cleared automatically.

47.6.3 Reception

The reception procedure in Figure 47.55 must be followed throughout a reception operation.

After reception is enabled (SSICR.TEN = 0 and SSICR.REN = 1), SSIE starts reception when a start trigger is generated by SSILRCKn/SSIFSn (n = 0, 1). SSIE outputs a receive data full interrupt to the DTC/DMAC according to the RDF setting condition (SSISCR.RDFS) and the status of receive data full interrupt enable (SSIFCR.RIE) bit specified in the communication start procedure. This interrupt requests data reading from the receive FIFO data register (SSIFRDR). In the communication start procedure, specify reading from the receive FIFO data register (SSIFRDR) as the DTC/DMAC operation in response to the receive data full interrupt. With this setting, SSIE can continuously read data not through the CPU. The receive data full interrupt is generated when data as much as the capacity of receive FIFO data register has been stored. The number of times of reading must be specified in accordance with the data size of the receive FIFO data register indicated by the receive data full interrupt. If an error occurs, perform the error-handling procedure as instructed in the communication stop procedure.

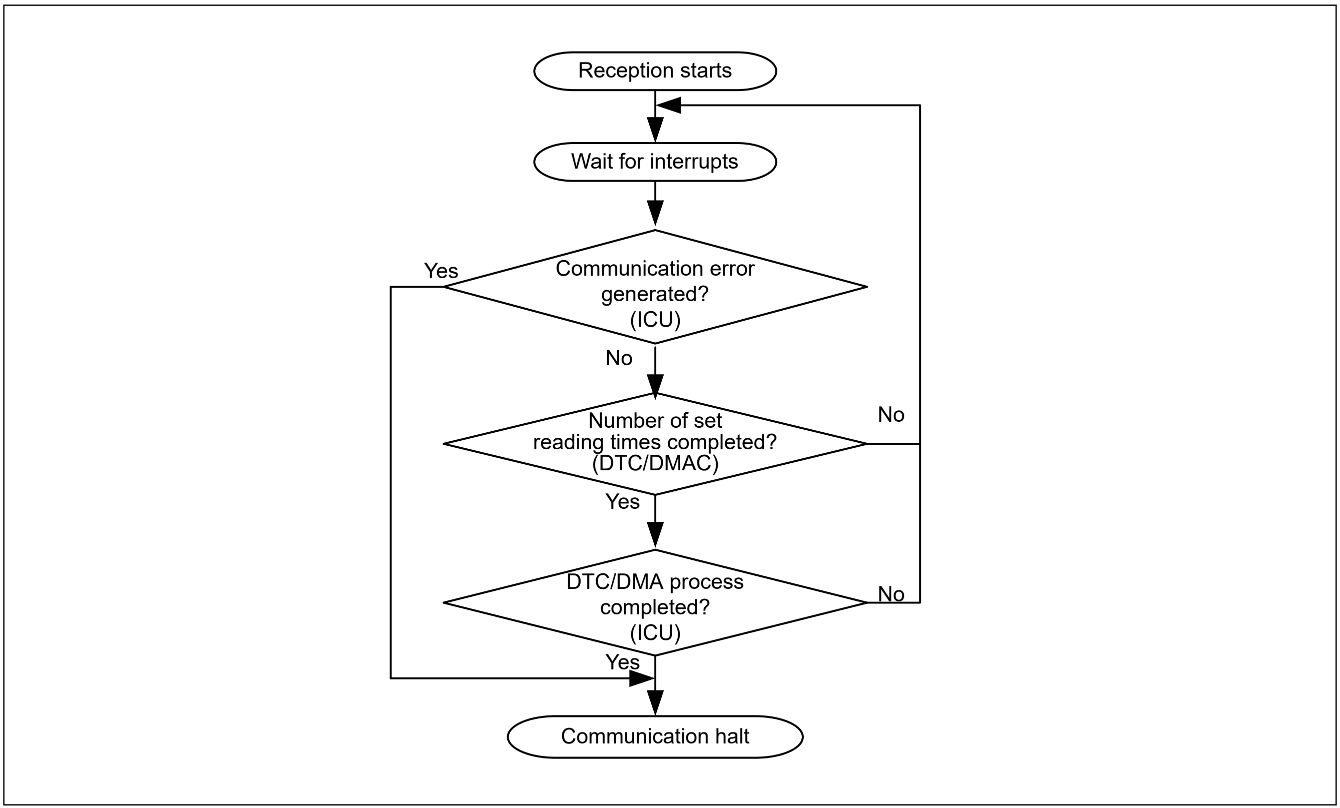


Figure 47.55 Reception procedure

Note: The communication flow defined in SSIE uses the DTC/DMAC. If you do not use the DTC/DMAC, perform polling of the value 1 of SSIFSR.RDF to read data from SSIFRDR. The number of times of reading data from SSIFRDR by detecting the value 1 of SSIFSR.RDF must be in accordance with the receive data storage capacity of the receive FIFO data register specified by SSISCR.RDFS. After received data is read from SSIFRDR, the SSIFSR.RDF flag must be cleared. Continuous reception is enabled by repeating data reading. If the SSIFSR.RDF flag is not cleared, the flag is not cleared automatically.

47.6.4 Transmission and Reception

After transmission and reception are enabled (SSICR.TEN = 1 and SSICR.REN = 1), SSIE starts transmission and reception when a start trigger is generated by SSILRCKn/SSIFSn (n = 0, 1) with the serial data for at least a frame contained in the transmit FIFO data register (SSIFTDR). SSIE can continuously transmit and receive data by performing the procedures described in section 47.6.2. Transmission and section 47.6.3. Reception, respectively. For how to stop transmission and reception, see section 47.6.5. Halt Communication.

47.6.5 Halt Communication

This section describes how to halt communication of SSIE. Figure 47.56 shows the procedure to halt communication. Be sure to follow the procedure.

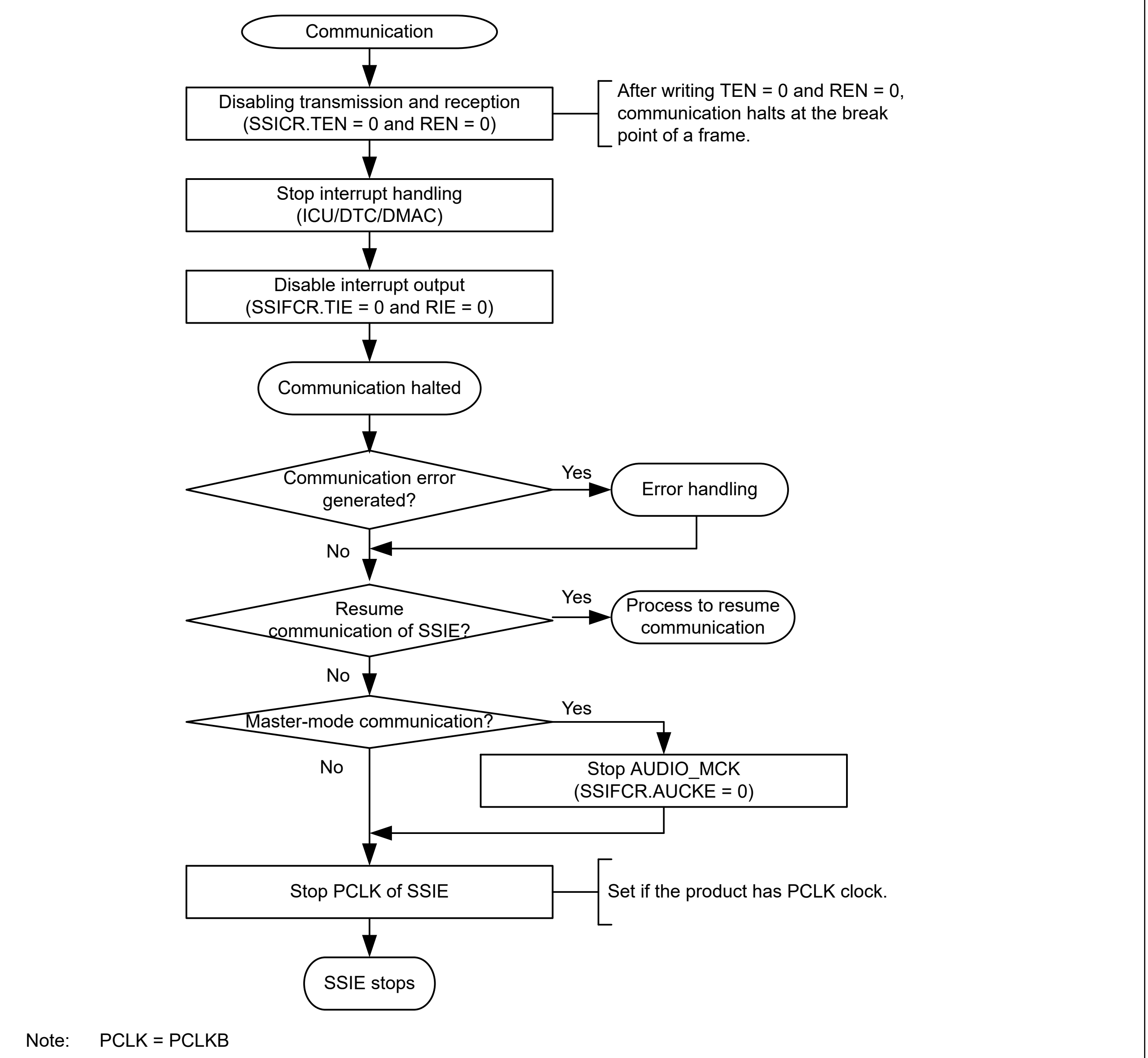


Figure 47.56 Procedure to halt communication (CPU operation procedure)

To halt the communication of SSIE, supply of the following clocks are required until the SSISR.IIRQ bit indicates an idle state.

- Input clock from the SSIBCKn (n = 0, 1) pin when SSICR.MST = 0
- AUDIO_MCK when SSICR.MST = 1
  To resume communication of SSIE in the previous setting, see section 47.6.7. Resume Communication.

Note: When communication of SSIE is halted according to the procedure to halt communication in Figure 47.56, resume communication according to the procedure to resume communication in Figure 47.58.

47.6.6 Error Handling

SSIE has the following four errors.

- Transmit underflow error
- Transmit overflow error
- Receive underflow error

- Receive overflow error.

When an underflow error or overflow error is generated, SSIE need to be restarted. Follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

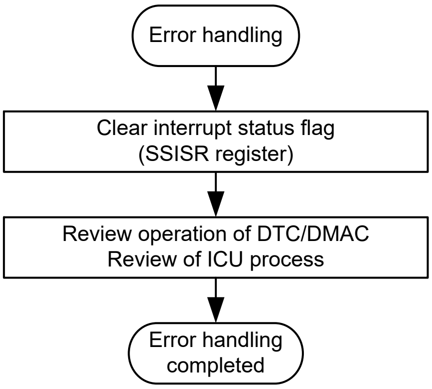


Figure 47.57 Error-handling procedure

Four error operations are described as follows. When the interrupt output enable bit of the SSICR register is enabled and error flags are set, an error interrupt is generated. See section 47.2.2. SSISR : Status Register for the setting conditions of error flags.

(1) Transmit Underflow Error

If a transmit underflow error occurs, review the number of times of writing data to the transmit FIFO data register (SSIFTDR) in response to a transmit data empty interrupt. After a transmit underflow error occurs, SSIE outputs 0s as data. To normally output the serial data written to the transmit FIFO data register (SSIFTDR) to the SSITXD0/SSIDATA1 pin, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57. After this error occurs, serial data is consumed as usual. If you resume communication, write the serial data from the beginning.

(2) Transmit Overflow Error

If a transmit overflow error occurs, review the number of times of writing data to the transmit FIFO data register (SSIFTDR) in response to transmit data empty interrupts. The serial data written to the transmit FIFO data register (SSIFTDR) that caused the transmit overflow error becomes invalid. This error can occur regardless of whether a transmission operation is being done. To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57. When you resume communication, deal with the invalid serial data appropriately.

(3) Receive Underflow Error

If a receive underflow error occurs, review the number of times of reading data from the receive FIFO data register (SSIFRDR) in response to receive data full interrupts. The values read from the receive FIFO data register (SSIFRDR) that caused the receive underflow error are undefined. This error can occur regardless of whether a reception operation is being done. To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

(4) Receive Overflow Error

If a receive overflow error occurs, review the number of times of reading data from the receive FIFO data register (SSIFRDR) in response to receive data full interrupts. The receive data that caused the receive overflow error cannot be stored in the receive FIFO data register (SSIFRDR). To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

47.6.7 Resume Communication

When you resume the communication using SSIE, follow the communication resume procedure in Figure 47.58. The communication resume procedure is designed on the assumption that you resume the communication stopped by the communication stop procedure without changing any settings. If you want to change clock and slave/master settings, use and follow the communication start procedure in Figure 47.53. For details about the transmission operation and reception operation after starting communication, see section 47.6.2. Transmission and section 47.6.3. Reception, respectively.

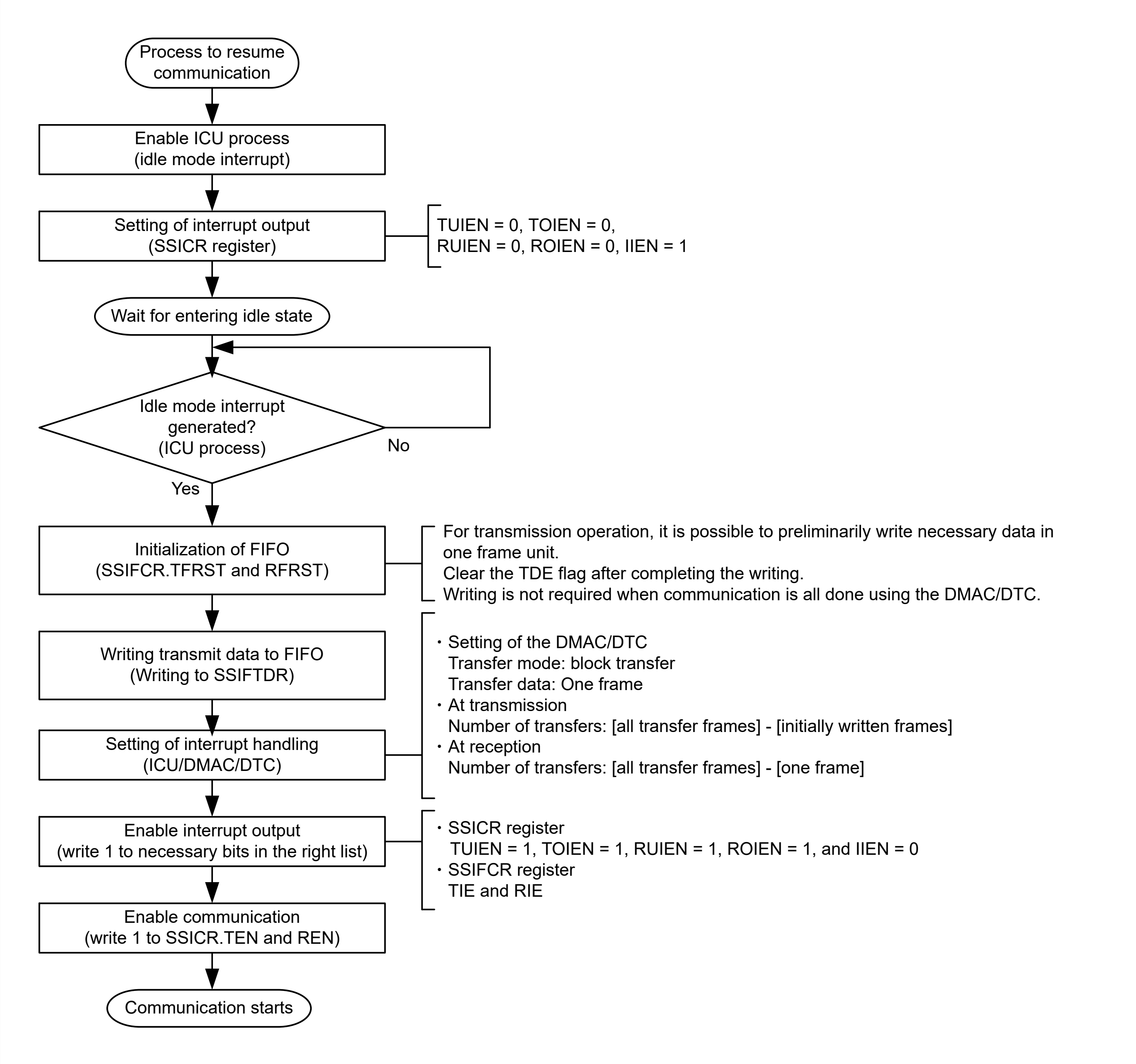


Figure 47.58 Procedure to resume communication (CPU operation procedure)

47.7 Interrupts

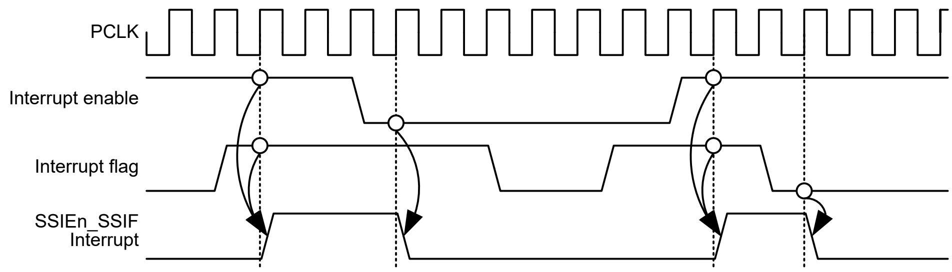
Table 47.18 lists the interrupt sources. Set enable/disable of interrupt output of each source with the TUIEN, TOIEN, RUIEN, ROIEN, and IIEN bits in the SSICR register and the TIE and RIE bits in the SSIFCR register.

Table 47.18 SSIE interrupt sources

Channel	Interrupt source	Description	Interrupt flag	DMAC/DTC activation
SSIE0	SSIE0_SSIF	• Transmit underflow interrupt • Transmit overflow interrupt • Receive underflow interrupt • Receive overflow interrupt • Idle interrupt	SSISR.TUIRQ SSISR.TOIRQ SSISR.RUIRQ SSISR.ROIRQ SSISR.IIRQ	Not possible
	SSIE0_SSIRXI	Receive data full interrupt	SSIFSR.RDF	Possible
	SSIE0_SSITXI	Transmit data empty interrupt	SSIFSR.TDE	Possible
SSIE1	SSIE1_SSIF	• Transmit underflow interrupt • Transmit overflow interrupt • Receive underflow interrupt • Receive overflow interrupt • Idle interrupt	SSISR.TUIRQ SSISR.TOIRQ SSISR.RUIRQ SSISR.ROIRQ SSISR.IIRQ	Not possible
	SSIE1_SSIRT	• Receive data full interrupt • Transmit data empty interrupt	SSIFSR.RDF SSIFSR.TDE	Possible

47.7.1 SSIEn_SSIF Interrupt (n = 0, 1)

This interrupt source combines five interrupts. Enable output of necessary interrupts before using SSIE. The five interrupts are operated by using the flags assigned to individual interrupts and interrupt output enable bits. To clear an interrupt, set the interrupt enable to 0 or clear the interrupt flag to 0.



Note: PCLK = PCLKB

Figure 47.59 Timing Diagram of the common interrupt source, SSIEn_SSIF

- Transmit underflow interrupt

As the transmit underflow interrupt, SSISR.TUIRQ is output while SSICR.TUIEN = 1. When you use SSIE for transmission, enable the output of this interrupt (SSICR.TUIRQ = 1). If this interrupt occurs, follow instructions in the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

- Transmit overflow interrupt

As the transmit overflow interrupt, SSISR.TOIRQ is output while SSICR.TOIRQ = 1. When you use SSIE for transmission, enable the output of this interrupt (SSICR.TOIRQ = 1). If this interrupt occurs, follow instructions in the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

- Receive underflow interrupt

As the receive underflow interrupt, SSISR.RUIRQ is output while SSICR.RUIRQ = 1. When you use SSIE for reception, enable the output of this interrupt (SSICR.RUIRQ = 1). If this interrupt occurs, follow instructions in the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

- Receive overflow interrupt

As the receive overflow interrupt, SSISR.ROIRQ is output while SSICR.ROIRQ = 1. When you use SSIE for reception, enable the output of this interrupt (SSICR.ROIRQ = 1). If this interrupt occurs, follow instructions in the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

- Idle mode interrupt

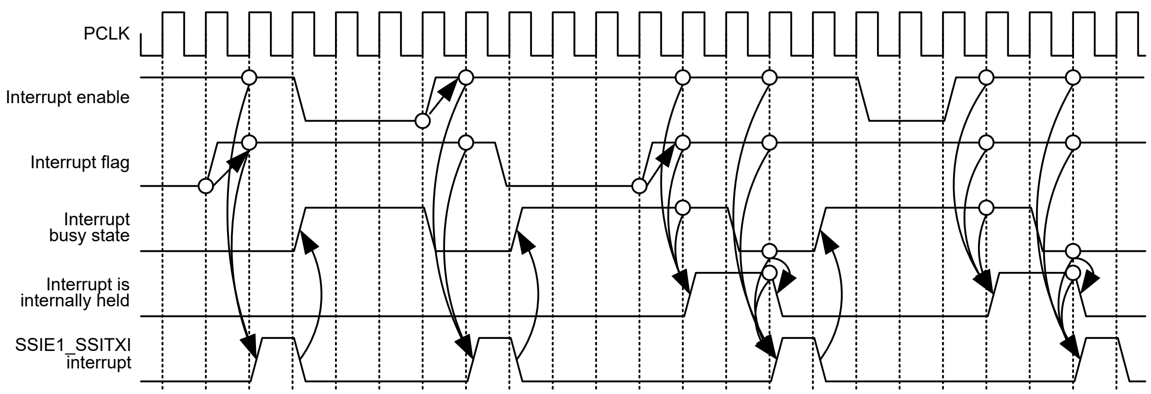
As the idle mode interrupt, SSISR.IIRQ is output while SSICR.IIEN = 1. This interrupt is used to make sure that communication has stopped fully.

47.7.2 SSIE0_SSITXI Interrupt [Full-duplex Communication]

The transmit data empty interrupt is a pulse interrupt that is output when the following condition is met:

- SSIFCR.TIE = 1 and SSIFSR.TDE = 1
  SSIE operation: When the value of SSIFSR.TDE changes from 0 to 1 while the value of SSIFCR.TIE is 1
  CPU instruction: When the value of SSIFCR.TIE changes from 0 to 1 while the value of SSIFSR.TDE is 1

This interrupt is subject to the interrupt suppression function. If an interrupt condition for this interrupt occurs when the DTC/DMAC is busy (when the DTC/DMAC cannot accept interrupts), the interrupt suppression function holds the output of this interrupt. The held interrupt will be output after the DTC/DMAC is enabled to accept interrupts. For details, see Figure 47.60.



Note: PCLK = PCLKB

Figure 47.60 SSIE1_SSITXI interrupt timing diagram

47.7.3 SSIE0_SSIRXI Interrupt [Full-duplex Communication]

The receive data full interrupt is a pulse interrupt that is output when the following condition is met:

- SSIFCR.RIE = 1 and SSIFSR.RDF = 1.
  SSIE operation: When the value of SSIFSR.RDF changes from 0 to 1 while the value of SSIFCR.RIE is 1
  CPU instruction: When the value of SSIFCR.RIE changes from 0 to 1 while the value of SSIFSR.RDE is 1

This interrupt is subject to the interrupt suppression function. If an interrupt condition for this interrupt occurs when the DTC/DMAC is busy (when the DTC/DMAC cannot accept interrupts), the interrupt suppression function holds the output of this interrupt. The held interrupt will be output after the DTC/DMAC is enabled to accept interrupts. The behavior of this interrupt is the same as the behavior shown in Figure 47.60.

47.7.4 SSIE1_SSIRT Interrupt [Half-duplex Communication]

This interrupt is output by two sources, transmit data empty interrupt and receive data full interrupt. When this interrupt is generated, read the interrupt flag and specify the interrupt source.

This interrupt is subject to the interrupt suppression function. If an interrupt condition for this interrupt occurs when the DTC/DMAC is busy (when the DTC/DMAC cannot accept interrupts), the interrupt suppression function holds the output of this interrupt. The held interrupt will be output after the DTC/DMAC is enabled to accept interrupts. For details, see Figure 47.61.

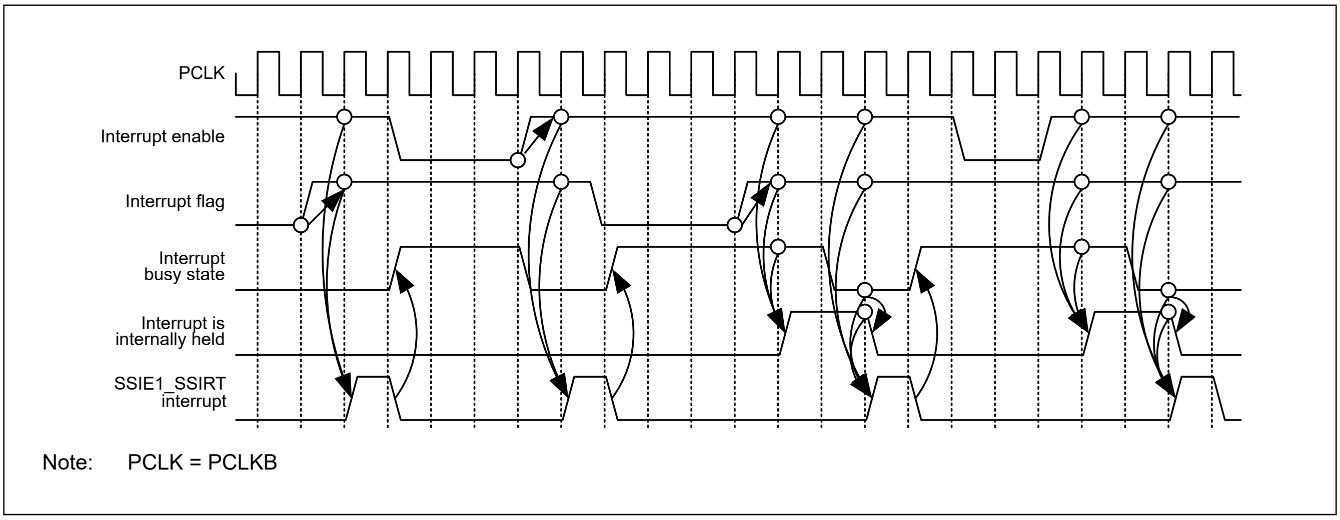


Figure 47.61 SSIE1_SSIRT interrupt timing diagram

47.8 Software Resets

SSIE has three software reset bits to reset its states.

- SSIE software reset (SSIFCR.SSIRST)
- Transmit FIFO data register reset (SSIFCR.TFRST)
- Receive FIFO data register reset (SSIFCR.RFRST).

This section describes the procedures for the three types of software resets.

47.8.1 Software Reset Procedure

(1) SSIE Software Reset

For the SSIE software reset bit (SSIFCR.SSIRST), follow the procedure shown in Figure 47.62. After a reset, the same setting is applied when it is resumed. To change the settings of clocks and slave/master mode, follow the procedure to start communication in Figure 47.53. See section 47.6.2. Transmission and section 47.6.3. Reception respectively for transmission and reception after communication is resumed.

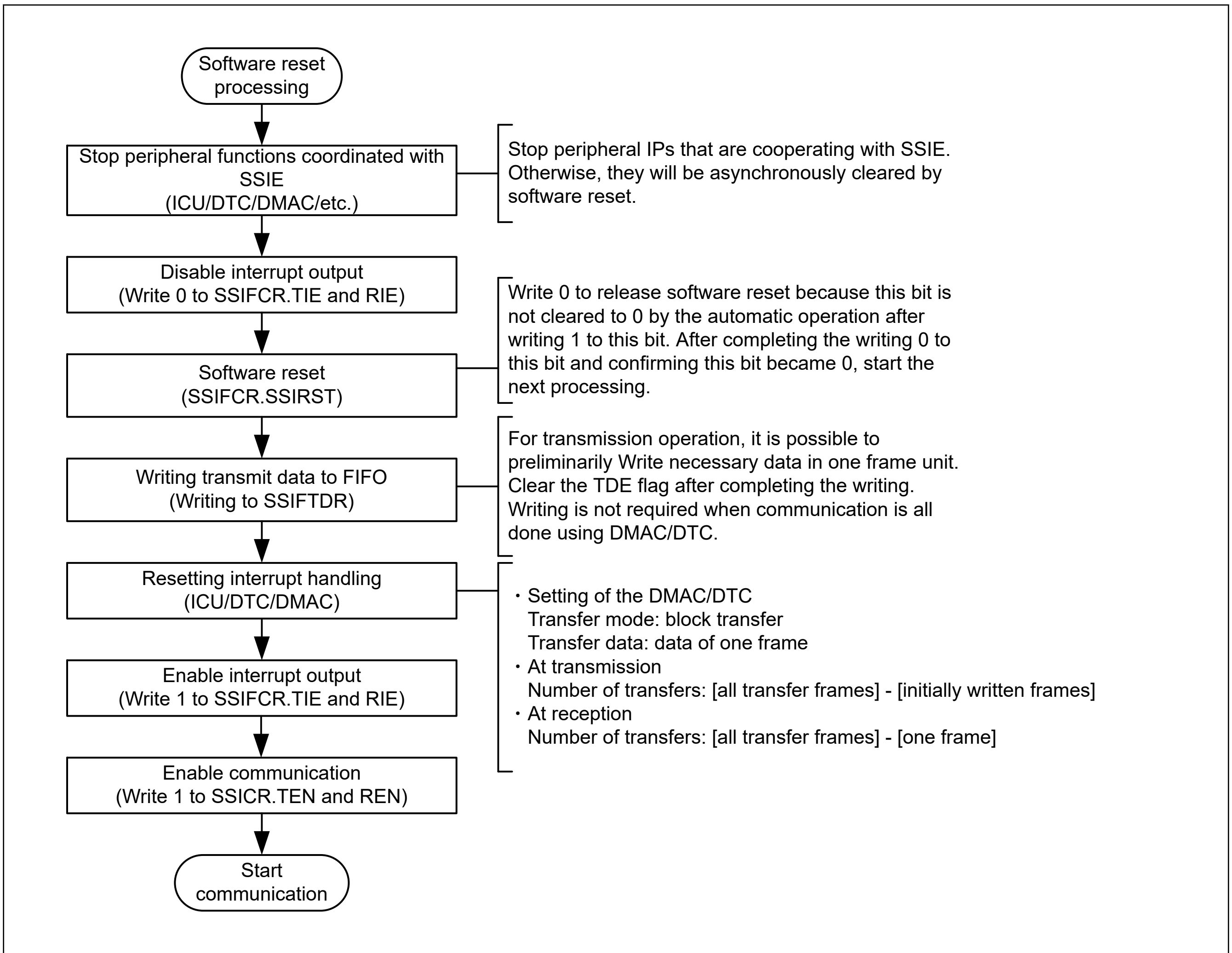


Figure 47.62 Software reset procedure (CPU operation procedure)

(2) Transmit FIFO data register reset

To perform a transmit FIFO data register reset, follow instructions in the procedure to start communication in Figure 47.53 and procedure to resume communication in Figure 47.58.

(3) Receive FIFO data register reset

To perform a receive FIFO data register reset, follow instructions in the procedure to start communication in Figure 47.53 and procedure to resume communication in Figure 47.58.

47.9 Usage Notes

47.9.1 Notes for Slave-mode Communication

47.9.1.1 SSIBCKn (n = 0, 1) Control

In slave-mode communication (SSICR.MST = 0), SSIE needs supply of SSIBCKn (n = 0, 1). To stop BCK on the master side, make sure that SSIE is in the idle state (SSISR.IIRQ = 1). If BCK is stopped before SSIE becomes idle, take the procedure to start communication in Figure 47.53 or wait for an idle state by taking the procedure to resume communication in Figure 47.58.

47.9.1.2 SSILRCKn/SSIFSn (n = 0, 1) pin

SSIE has the SSILRCKn/SSIFSn (n = 0, 1) pin, which indicates the synchronization of communication. When SSIE is in slave mode (SSICR.MST = 0), the communication format SSIE uses must match that of the other-party device to communicate. SSIE uses the signal input by the SSILRCKn/SSIFSn (n = 0, 1) pin only as a trigger to start communication.

47.9.2 Notes for Master-mode Communication

47.9.2.1 AUCKE Control

In master-mode communication (SSICR.MST = 1), SSIE operates with the audio clock (AUDIO_MCK). To stop SSIE completely, make sure that SSIE is in the idle state (SSISR.IIRQ = 1) and then write 0 to SSIFCR.AUCKE.

47.9.2.2 LRCONT Control

To stop the output to the SSILRCKn/SSIFSn (n = 0, 1) pin with SSIOFR.LRCONT when SSIE is in the idle state in master-mode communication (SSICR.MST = 1), note the following: The output stops when the value of the SSIOFR.LRCONT bit is changed from 1 to 0. Make sure that the other-party device is not affected. For details, see Figure 47.44.

47.9.2.3 BCKASTP Control

To stop the output to the SSIBCKn (n = 0, 1) pin with SSIOFR.BCKASTP in master-mode communication (SSICR.MST = 1) and while SSIE is in the idle state, note the following: The output stops when the value of the SSIOFR.BCKASTP bit is changed from 0 to 1. So, make sure that the other-party device is not affected. For details, see Figure 47.45.

The BCKASTP bit cannot be used when the other-party device (which is a slave) requires the clock output from the SSIBCKn (n = 0, 1) pin before and during communication.

47.9.3 Notes for Communication Flow

47.9.3.1 When an Error Interrupt is Generated

SSIE has the following four errors.

- Transmit underflow error
- Transmit overflow error
- Receive underflow error
- Receive overflow error

When an underflow error or overflow error is generated, SSIE need to be restarted. Follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

(1) Transmit Underflow Error

If a transmit underflow error occurs, review the number of times of writing data to the transmit FIFO data register (SSIFTDR) in response to a transmit data empty interrupt. After a transmit underflow error occurs, SSIE outputs 0s as data. To normally output the serial data written to the transmit FIFO data register (SSIFTDR) to the SSITXD0/SSIDATA1 pin, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57. After this error occurs, serial data is consumed as usual. If you resume communication, write the serial data from the beginning.

(2) Transmit Overflow Error

If a transmit overflow error occurs, review the number of times of writing data to the Transmit FIFO Data Register (SSIFTDR) in response to transmit data empty interrupts. The serial data written to the Transmit FIFO Data Register (SSIFTDR) that caused the transmit overflow error becomes invalid. This error can occur regardless of whether a transmission operation is being done. To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57. When you resume communication, deal with the invalid serial data appropriately.

(3) Receive Underflow Error

If a receive underflow error occurs, review the number of times of reading data from the receive FIFO data register (SSIFRDR) in response to receive data full interrupts. The values read from the receive FIFO data register (SSIFRDR) that caused the receive underflow error are undefined. This error can occur regardless of whether a reception operation is being done. To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

(4) Receive Overflow Error

If a receive overflow error occurs, review the number of times of reading data from the receive FIFO data register (SSIFRDR) in response to receive data full interrupts. The receive data that caused the receive overflow error cannot be stored in the receive FIFO data register (SSIFRDR). To recover from the error, follow the procedure to halt communication in Figure 47.56 and error-handling procedure in Figure 47.57.

47.9.3.2 Transmit Data Empty Interrupt

The communication flow defined in SSIE uses the DTC/DMAC. If you do not use the DTC/DMAC, perform polling of the value 1 of SSIFSR.TDE to write data to SSIFTDR. The number of times of writing data to SSIFTDR by detecting the value 1 of SSIFSR.TDE must be in accordance with the free space size of the transmit FIFO data register specified by SSISCR.TDES. After as much transmit data as the free space size is written to SSIFTDR, the SSIFSR.TDE flag must be cleared. Continuous transmission is enabled by repeating data writing. If the SSIFSR.TDE flag is not cleared, the flag is not cleared automatically.

47.9.3.3 Receive Data Full Interrupt

The communication flow defined in SSIE uses the DTC/DMAC. If you do not use the DTC/DMAC, perform polling of the value 1 of SSIFSR.RDF to read data from SSIFRDR. The number of times of reading data from SSIFRDR by detecting the value 1 of SSIFSR.RDF must be in accordance with the receive data storage capacity of the receive FIFO data register specified by SSISCR.RDFS. After received data is read from SSIFRDR, the SSIFSR.RDF flag must be cleared. Continuous reception is enabled by repeating data reading. If the SSIFSR.RDF flag is not cleared, the flag is not cleared automatically.

47.9.3.4 Switching Transfer Modes

1. For state transition from transmission, reception, and transmission and reception, disable transmission and reception (SSICR.TEN = 0, SSICR.REN = 0).
2. Confirm it is in the idle state (SSISR.IIRQ = 1).
3. In the idle state, set the SSICR.TEN bit or the SSICR.REN bit again and resume transfer.

47.9.3.5 Resume Communication After Halting SSIE

When communication of SSIE is halted according to the procedure to halt communication in Figure 47.56, resume communication according to the procedure to resume communication in Figure 47.58.

47.9.4 Write Access Restriction

47.9.4.1 SSICR Register

If the TEN bit or REN bit is rewritten, make sure that the SSISR.IIRQ bit is in the desired status. If the value of the TEN or REN bit is changed by rewriting, subsequent operation is unpredictable. For example, when transmission or reception is enabled, check that SSISR.IIRQ is 0; when transmission or reception is disabled, check that SSISR.IIRQ is 1.

(1) TEN Bit and REN Bit

These bits enable/disable transmission and reception. When 1 is written to one of these bits, the corresponding communication operation starts in synchronization with a start trigger by the SSILRCKn/SSIFSn (n = 0, 1) signal. For details, see section 47.6.2. Transmission, section 47.6.3. Reception, and section 47.6.4. Transmission and Reception. When 0 is written to this bit, the current communication operation stops at the next frame boundary. To use SSIE for both transmission and reception, always write 1 to these bits together. When stopping the communication using SSIE, always disable both transmission and reception (write 0 to the TEN and REN bits).

47.9.4.2 SSISR Register

(1) Clearing TUIRQ and TOIRQ

After communication is enabled (by changing the value of SSICR.TEN bit from 0 to 1), the transmission error flags (TOIRQ and TUIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the transmission error flags might be unable to be read.

(2) Clearing RUIRQ and ROIRQ

After communication is enabled (by changing the value of SSICR.REN bit from 0 to 1), the reception error flags (RUIRQ and ROIRQ in the SSISR register) are cleared. If, however, the SSISR register is read continuously, the cleared status of the reception error flags might be unable to be read.

47.9.4.3 Communication State

Writing to the bits with orange-shaded area in Table 47.19 is prohibited. If written, the operation performed immediately after writing is not guaranteed.

Table 47.19 Bits protected from writing during communication

Symbol	Address (BASE+)		+0								+1							
			31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
			15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SSICR	0x00	+0	—	CKS	TUIEN	TOIEN	RUIEN	ROIEN	IIEN	—	FRM[1:0]		DWL[2:0]			SWL[2:0]		
		+2	—	MST	BCKP	LRCKP	SPDP	SDTA	PDTA	DEL	CKDV[3:0]				MUEN	—	TEN	REN
SSISR	0x04	+0	—	—	TUIRQ	TOIRQ	RUIRQ	ROIRQ	IIRQ	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
SSIFCR	0x10	+0	AUCKE	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SSIRST
		+2	—	—	—	—	BSW	—	—	—	—	—	—	—	TIE	RIE	TFRST	RFRST
SSIFSR	0x14	+0	—	—	TDC[5:0]						—	—	—	—	—	—	—	TDE
		+2	—	—	RDC[5:0]						—	—	—	—	—	—	—	RDF
SSIFTDR	0x18	+0	FTDR[31:16]															
		+2	FTDR[15:0]															
SSIFRDR	0x1C	+0	FRDR[31:16]															
		+2	FRDR[15:0]															
SSIOFR	0x20	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	—	—	—	BCKASTP	LRCONT	—	—	—	—	—	—	OMOD[1:0]	
SSISCR	0x24	+0	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
		+2	—	—	—	TDES[4:0]					—	—	—	RDFS[4:0]				

47.9.5 Software Standby Mode

The SSIE must halt before transition to Software Standby mode.

Figure 47.63 shows an example flow before transition to Software Standby mode.

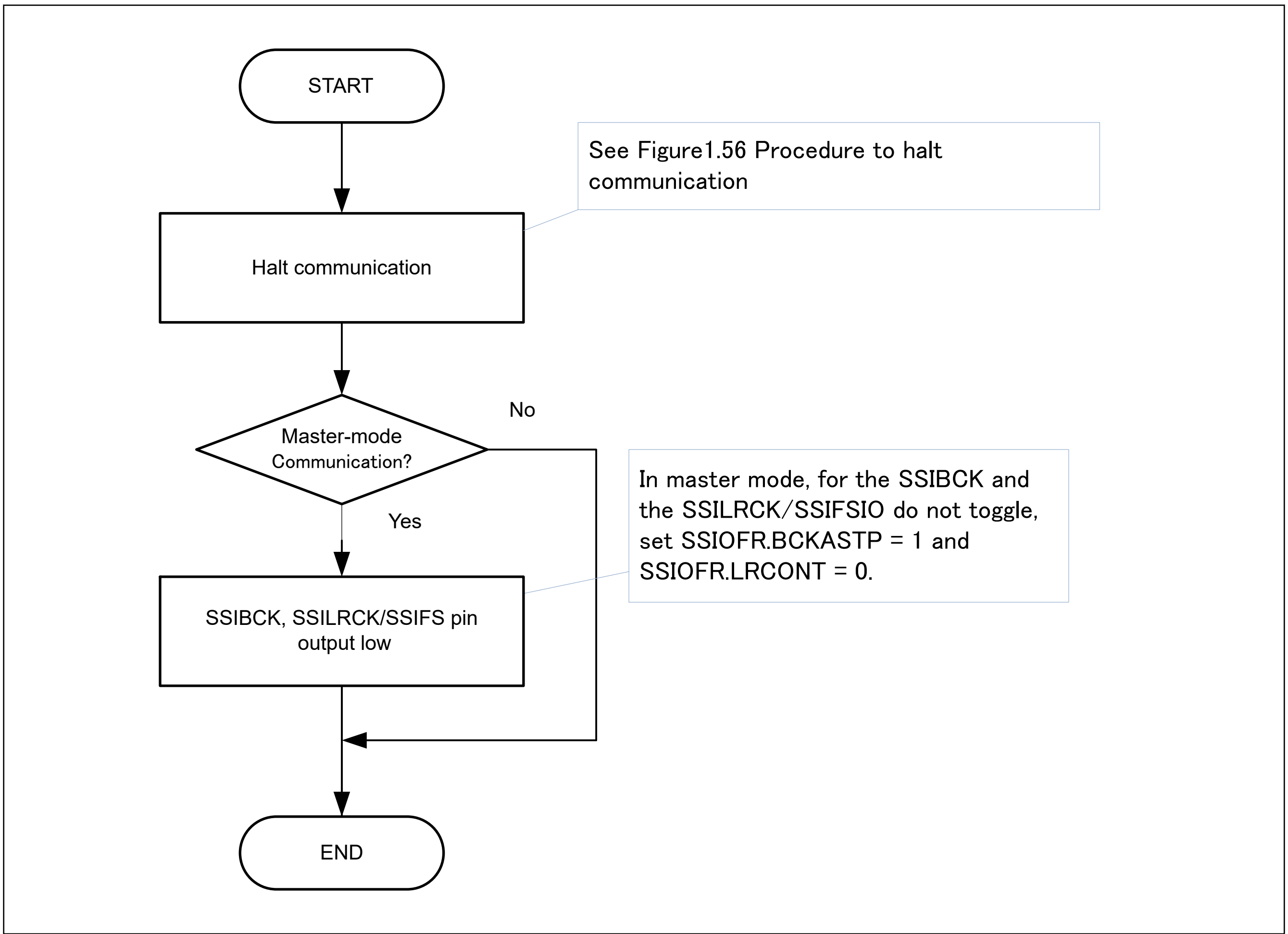


Figure 47.63 Example flow before transition to Software Standby mode

Figure 47.64 shows an example flow after canceling Software Standby mode.

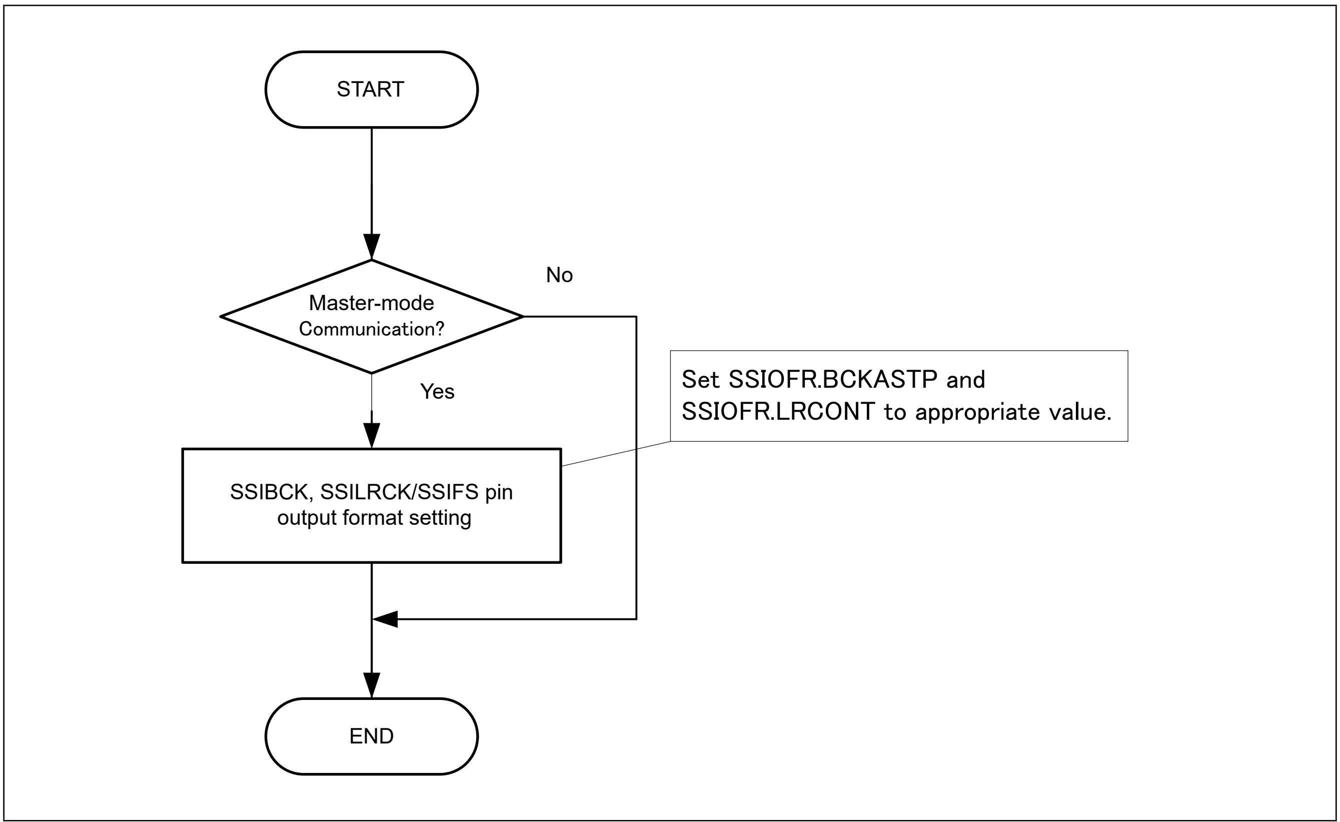


Figure 47.64 Example flow after canceling Software Standby mode